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Saudi Arabian Mining Company شركة التعدين العربية السعودية

Project Mansourah Massarah Gold Project

Customer: MA'ADEN GOLD AND BASE METALS (MA'ADEN)

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DEFINITIONS AND ABBREVIATIONS

Document	:	Information and the medium on which it is contained (ISO 9000:2015)
EMPLOYER	:	'EMPLOYER' means MA'ADEN GOLD AND BASE METALS (MGBM) and the PMC American Bechtel Inc. or their representatives
Contractor	:	Contractor' means consortium composed of LARSEN & TOUBRO LIMITED, Saudi Arabia AND OUTOTEC SAUDI ARABIA LLC, Saudi Arabia (for IK Contract) and LARSEN & TOUBRO LIMITED – ECC Construction Group – (SHJ BR) and OUTOTEC ENGINEERING DMCC, Dubai (for OOK Contract).
'PMT'	:	Ma'aden Project Management Team
'MMGP Project'	:	Mansourah Massarah Gold Processing Project
Attribute	:	Information element that describes a document or tag.
Supplier	:	An entity (supplier, consultant, sub-contractor) under the direct project management of Contractor
Sub-Supplier	:	A sub entity (supplier, consultant, sub-contractor) under the direct project management of Contractor.
QA	:	Quality Assurance
QC	:	Quality Control
Works	:	Full scope of the EPC Contract including the engineering, procurement, construction, commissioning and associated activities
STAAD	:	Commonly used analysis and design software package named STAAD.Pro
Sec.	:	Section
max.	:	Maximum
ref.	:	Reference
No.	:	Number
cl.	:	Clause
nos.	:	Numbers
Col.	:	Column
approx.	:	Approximate
eqn.	:	Equation
mkd.	:	Marked
Dia.	:	Diameter
Spac.	:	Spacing
Rebar	:	Reinforcement Bar

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1. INTENT

This document is to Prepare for design of **CRUSHED ORE STORAGE BIN (3210)** of approximate 18000T storage capacity. The analysis & design is carried out in accordance with project specifications along with codes & standards for the foundation raft and the dowel portion of vertical wall.

The design calculation ensures the safety and serviceability of the complete silo system.

2. EXECUTIVE SUMMARY

The Saudi Arabian Mining Company (Ma'aden) has announced that its wholly-owned subsidiary, Ma'aden Gold and Base Metals Company, have issued a Notice of Award to Outotec and Larsen & Toubro Consortium to complete the engineering, procurement, construction, pre-commissioning, commissioning and start-up assistance of the Mansourah & Massarah (MMGP) Gold Project.

MMGP project will become Ma'aden's largest ever gold project, producing an average of 250,000 ounces of gold per year over the life of mine. Mansourah-Massarah is situated within the Central Arabian Gold Region (CAGR), approximately 320km northeast of Taif towards Riyadh and about 35 km southeast of Hufairah, which is a small village situated on Jeddah - Riyadh highway. Mansourah - Massarah gold deposits are located 22°38'6.40"N; 42°44'22.90"E, and 22°42'1.40"N; 42°41'58.40"E respectively, approximately 537 km east of Jeddah and 77 km southeast of the town of Zalim.

The plant throughput capacity is 4000000t/a and is required to operate on a continuous basis at nominally 24 hours per day and 7 days a week.

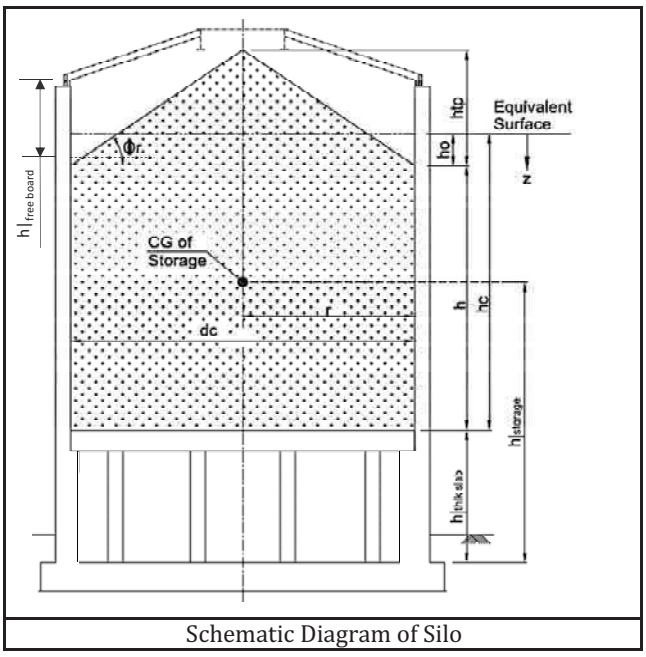
3. SCOPE

The analysis has been carried out in Excel and STAAD. The silo shell, thick slab and thick slab supporting columns are modeled in STAAD and possible load case are applied. The stresses due to all possible load case combinations are taken out from the analysis of STAAD and used directly as force values for member strength calculation.

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Schematic Diagram of Silo

4. INPUT DATA

4.1. Geometry of silo:

Capacity of silo	=	182600	KN	
Unloading Pattern	=	From Bottom		
Height of wall, h		20	m	
Inside diameter, dc	=	23	m	
Inside radius, r = dc/2	=	11.5	m	
ho = r/3.tan φ_r	=	2.89	m	[according to eqn. 5.77, EN 1991-4:2006]
hc = h + ho	=	22.89	m	
Height of Thick Slab from top of foundation, $h\|_{thick\ slab}$	=	9.55	m	
Height of free board, $h\|_{free\ board}$	=	4.2	m	
Total height of retaining wall	=	$h + h\|_{thick\ slab} + h\|_{freeboard}$	= 33.75 m	
Thickness of Thick slab	=	1.5	m	[assumed for design purpose]
Aerated bottom silo	=	No		
Slenderness, hc/dc	=	1.0		i.e, Squat Silo
Wall thickness, t_c	=	0.5	m	i.e, Thick-walled Circular Silo
Cross section area, A	=	415.48	sq.m	
Internal perimeter of wall, U	=	72.26	m	
Filling eccentricity, e_f	=	0	m	
Discharge eccentricity, e_o	=	6.52	m	
Maximum eccentricity, e	=	6.52	m	[according to eqn. 5.32, EN 1991-4:2006]
Action Assessment Class	=	Class-3		[ref. table 2.1, EN 1991-4:2006]
Wall Category	=	D2		Moderate friction classed as`Smooth'

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4.2. Bulk material parameters:

Type of Particulate Solid = Crushed Gold Ore

		Lower Value	Upper Value		
Density of bulk material, γ	=	16	19.2	KN/m^3	[according to Annex E, EN 1991-4:2006]
Angle of repose, ϕ_r	=	37		degrees	
Angle of internal friction, ϕ_i	=	26.72	35.96	degrees	[according to Annex E, EN 1991-4:2006]
Coefficient of wall friction, μ	=	0.53	0.66		[according to Annex E, EN 1991-4:2006]
Lateral pressure ratio, K	=	0.45	0.6		[according to Annex E, EN 1991-4:2006]
Patch load solid, C_{op}	=	0.5			[according to Annex E, EN 1991-4:2006]

• The unit weight of the solid γ_u is the upper characteristic value, to be used for all calculations of actions. The lower characteristic value γ_l is to assist in estimating the required volume of silo that will have a defiend capacity.

[according to Note: b, Annex E, EN 1991-4:2006]

		Characteristic Values of stored material to be adopted			
		Coefficient of Wall friction		Lateral Pressure Ratio	
	Purpose	Type	μ	Type	K
Case-I	Max. Normal pressure on vertical wall	Lower Value	0.53	Upper Value	0.6
Case-II	Max. Frictional traction on vertical wall	Upper Value	0.66	Upper Value	0.6
Case-III	Max. Vertical load on silo bottom	Lower Value	0.53	Lower Value	0.45

4.3. Concrete Parameters:

Type of concrete used	=	C35		
f_c'	=	35	Mpa	[i.e., 28 days compressive strength]
Unit weight of RCC	=	24	KN/m^3	[as per document number: MD-216-1000-EG-ST-CRT-1001]

4.4. Reinforcement Parameters:

Type of Steel used	=	ASTM A 615/ A 615M Grade 60
f_y	=	420 Mpa

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5. LOAD

5.1. Dead load:

5.1.1. Self-weight of Silo:

Assigned directly using self-weight command in STAAD.

5.1.2. Surcharge Load:

5.1.2.1. Due to Soil Filling:

Depth of soil filling	=	1.0	m		
Unit weight of soil filling	=	20	KN/m^3		[as per document number:
Pressure due to soil filling	=	20	KN/m^2	[vertical]	MD-216-1000-EG-ST-CRT-1001]

5.1.2.2. Due to spilled material heap on ground:

Depth of Material Stack	=	5.0	m	[average height]
Pressure due to	=	96.0	KN/m^2	[vertical]
stacked material	=	57.6	KN/m^2	[horizontal = Vertical x K]

5.1.3. Self-weight of Roof structure:

The analysis & design of structural roof is carried out by separate STAAD model.

For the design of Silo the self-weight of roof structure, feed conveyor & bag filter assumed = 2.0 KN/m^2

Total self-weight	=	π.(23^2)/4 x 2	=	831	KN
Number of node in STAAD	=	72	nos.		
Self-weight per STAAD node	=	831/ 72	=	11.5	KN

5.1.4. Self-weight of Structural Floor:

The analysis of structural Floor is carried out by separate STAAD model.

For design of Silo the self-weight of structural floor including bin reclaim hooper assumed = 2.5 KN/m^2

Total self-weight	=	π.(23^2)/4 x 2.5	=	1038.7	KN
Number of node in STAAD	=	72	nos.		
Self-weight per STAAD node	=	1038.7/ 72	=	14.4	KN

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5.1.5. Self-weight of Liner Plate:

Thickness of liner plate	=	10	mm	
Unit weight of liner	=	77.5	KN/m^3	[unit weight of liner considered same as steel]
Weight of liner	=	775	KN/m^2	

5.2. Live load:

5.2.1. Live load from Roof structure:

For the design of Silo the live load of roof structure, feed conveyor & bag filter assumed = 2.0 KN/m^2

Total load	=	π.(23^2)/4 x 2	=	831 KN	[as per document number:
Number of node in STAAD	=	72 nos.			MD-216-1000-EG-ST-CRT-1001]
Live load per STAAD node	=	831/ 72	=	11.5 KN	

5.2.2. Live load from Structural Floor:

For the design of Silo the live load of roof structure, feed conveyor & bag filter assumed = 4.0 KN/m^2

Total load	=	π.(23^2)/4 x 4	=	1661.9 KN	[as per document number:
Number of node in STAAD	=	72 nos.			MD-216-1000-EG-ST-CRT-1001]
Live load per STAAD node	=	1661.9/ 72	=	23.1 KN	

5.2.3. Load of discharge chute:

Load of discharge chute (below thick slab)	=	240 KN per chute			[ref. mechanical drawing number: MD-216-3210-EG-ME-ELV-1009]
Load applied in STAAD	=	240/ 4	=	60 KN/ side per chute	

5.3. Material load:

5.3.1. Filling load:

Horizontal pressure, P_{hf}	=	Pho.Yr	[according to eqn. 5.71, EN 1991-4:2006]
Wall friction pressure, P_{wf}	=	μ.Phf	[according to eqn. 5.72, EN 1991-4:2006]
Vertical pressure, $P_{vf} = \gamma.Z_V$	=	γ.Zv	[according to eqn. 5.79, EN 1991-4:2006]
Cumulative wall friction, n_zSk	=	μ Pho (Z - Zv)	[according to eqn. 5.81, EN 1991-4:2006]

Where,		Case-I	Case-II	Case-III	
$P_{ho} = \gamma.(1/\mu).(A/U)$	=	208.3	167.3	208.3	[according to eqn. 5.73, EN 1991-4:2006]
$Z_o = \{1/(K\mu)\}.(A/U)$	=	18.1	14.5	24.1	[according to eqn. 5.75, EN 1991-4:2006]
Yr = [1 - {(z-ho)/(zo-ho)+1}^n]					[according to eqn. 5.74, EN 1991-4:2006]
n = -(1+tan φr).(1-ho/zo)	=	-1.47	-1.40	-1.54	[according to eqn. 5.76, EN 1991-4:2006]
Zv = ho - {1/(n+1)}*[zo - ho - {(z+zo-2ho)^(n+1) / (zo-ho)^n }]					[according to eqn. 5.80, EN 1991-4:2006]

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5.3.2. Discharge load:

Horizontal pressure, P_{he} = Phf [according to cl no.5.3.2.1.1, EN 1991-4:2006]

Wall friction pressure, P_{we} = Pwf [according to cl no.5.3.2.1.1, EN 1991-4:2006]

n_{zSk} = μ Pho (Z - Zv) [according to eqn. 5.91, EN 1991-4:2006]

Table-1 : Wall pressure for Case-I				Case-I						
				Filling Condition				Emptying Condition		
Sec.	z*	$z=z^*+h_o$	Yr	P_{hf}	P_{wf}	P_{vf}	n_{zSk}	P_{he}	P_{we}	n_{zSk}
-	m	m	-	KN/m^2	KN/m^2	KN/m^2	KN/m	KN/m^2	KN/m^2	KN/m
1	0	2.89	0.00	0.0	0.0	55.5	0.0	0.0	0.0	0.0
2	1	3.89	0.09	18.7	9.9	73.8	5.1	18.7	9.9	5.1
3	2	4.89	0.17	34.7	18.4	90.5	19.3	34.7	18.4	19.3
4	3	5.89	0.23	48.6	25.7	105.9	41.5	48.6	25.7	41.5
5	4	6.89	0.29	60.7	32.2	120.0	70.5	60.7	32.2	70.5
6	5	7.89	0.34	71.3	37.8	133.1	105.5	71.3	37.8	105.5
7	6	8.89	0.39	80.7	42.8	145.3	145.9	80.7	42.8	145.9
8	7	9.89	0.43	89.1	47.2	156.7	190.9	89.1	47.2	190.9
9	8	10.89	0.46	96.6	51.2	167.3	240.2	96.6	51.2	240.2
10	9	11.89	0.50	103.3	54.8	177.3	293.2	103.3	54.8	293.2
11	10	12.89	0.53	109.4	58.0	186.7	349.6	109.4	58.0	349.6
12	11	13.89	0.55	114.9	60.9	195.5	409.1	114.9	60.9	409.1
13	12	14.89	0.58	120.0	63.6	203.9	471.4	120.0	63.6	471.4
14	13	15.89	0.60	124.5	66.0	211.8	536.2	124.5	66.0	536.2
15	14	16.89	0.62	128.7	68.2	219.4	603.3	128.7	68.2	603.3
16	15	17.89	0.64	132.6	70.3	226.5	672.6	132.6	70.3	672.6
17	16	18.89	0.65	136.1	72.1	233.3	743.8	136.1	72.1	743.8
18	17	19.89	0.67	139.4	73.9	239.8	816.8	139.4	73.9	816.8
19	18	20.89	0.68	142.4	75.5	246.0	891.5	142.4	75.5	891.5
20	19	21.89	0.70	145.3	77.0	252.0	967.7	145.3	77.0	967.7
21	20	22.89	0.71	147.9	78.4	257.7	1045.4	147.9	78.4	1045.4

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Table-2 : Wall pressure for Case-II				Case-II						
				Filling Condition				Emptying Condition		
Sec.	z*	$z=z^*+h_o$	Yr	P_{hf}	P_{wf}	P_{vf}	n_{zSk}	P_{he}	P_{we}	n_{zSk}
-	m	m	-	KN/m²	KN/m²	KN/m²	KN/m	KN/m²	KN/m²	KN/m
1	0	2.89	0.00	0.0	0.0	55.5	0.0	0.0	0.0	0.0
2	1	3.89	0.11	22.8	15.0	73.6	6.2	22.8	15.0	6.2
3	2	4.89	0.20	41.6	27.5	89.8	23.5	41.6	27.5	23.5
4	3	5.89	0.28	57.4	37.9	104.4	49.8	57.4	37.9	49.8
5	4	6.89	0.34	70.8	46.7	117.7	83.9	70.8	46.7	83.9
6	5	7.89	0.39	82.2	54.3	129.8	124.5	82.2	54.3	124.5
7	6	8.89	0.44	92.2	60.8	141.0	170.8	92.2	60.8	170.8
8	7	9.89	0.48	100.8	66.5	151.3	222.0	100.8	66.5	222.0
9	8	10.89	0.52	108.4	71.6	160.8	277.5	108.4	71.6	277.5
10	9	11.89	0.55	115.2	76.0	169.7	336.8	115.2	76.0	336.8
11	10	12.89	0.58	121.2	80.0	178.0	399.4	121.2	80.0	399.4
12	11	13.89	0.61	126.5	83.5	185.8	465.1	126.5	83.5	465.1
13	12	14.89	0.63	131.3	86.7	193.1	533.4	131.3	86.7	533.4
14	13	15.89	0.65	135.7	89.6	200.0	604.2	135.7	89.6	604.2
15	14	16.89	0.67	139.6	92.2	206.5	677.2	139.6	92.2	677.2
16	15	17.89	0.69	143.2	94.5	212.7	752.2	143.2	94.5	752.2
17	16	18.89	0.70	146.5	96.7	218.5	829.0	146.5	96.7	829.0
18	17	19.89	0.72	149.5	98.7	224.1	907.4	149.5	98.7	907.4
19	18	20.89	0.73	152.3	100.5	229.4	987.4	152.3	100.5	987.4
20	19	21.89	0.74	154.8	102.2	234.4	1068.8	154.8	102.2	1068.8
21	20	22.89	0.75	157.2	103.8	239.2	1151.5	157.2	103.8	1151.5

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Table-3 : Wall pressure for Case-III				Case-III						
				Filling Condition				Emptying Condition		
Sec.	z*	$z=z^*+h_o$	Yr	P_{hf}	P_{wf}	P_{vf}	n_{zSk}	P_{he}	P_{we}	n_{zSk}
-	m	m	-	KN/m²	KN/m²	KN/m²	KN/m	KN/m²	KN/m²	KN/m
1	0	2.89	0.00	0.0	0.0	55.5	0.0	0.0	0.0	0.0
2	1	3.89	0.07	14.3	9.4	74.0	3.9	14.3	9.4	3.9
3	2	4.89	0.13	27.0	17.8	91.3	14.9	27.0	17.8	14.9
4	3	5.89	0.18	38.5	25.4	107.5	32.3	38.5	25.4	32.3
5	4	6.89	0.23	48.7	32.2	122.6	55.4	48.7	32.2	55.4
6	5	7.89	0.28	58.0	38.3	136.9	83.8	58.0	38.3	83.8
7	6	8.89	0.32	66.5	43.9	150.4	116.8	66.5	43.9	116.8
8	7	9.89	0.36	74.1	48.9	163.1	154.1	74.1	48.9	154.1
9	8	10.89	0.39	81.2	53.6	175.1	195.3	81.2	53.6	195.3
10	9	11.89	0.42	87.6	57.8	186.5	240.0	87.6	57.8	240.0
11	10	12.89	0.45	93.5	61.7	197.4	288.0	93.5	61.7	288.0
12	11	13.89	0.48	99.0	65.3	207.7	339.1	99.0	65.3	339.1
13	12	14.89	0.50	104.0	68.6	217.6	392.9	104.0	68.6	392.9
14	13	15.89	0.52	108.7	71.7	227.0	449.3	108.7	71.7	449.3
15	14	16.89	0.54	113.0	74.6	235.9	508.0	113.0	74.6	508.0
16	15	17.89	0.56	117.0	77.2	244.5	569.0	117.0	77.2	569.0
17	16	18.89	0.58	120.8	79.7	252.8	632.0	120.8	79.7	632.0
18	17	19.89	0.60	124.3	82.0	260.7	697.0	124.3	82.0	697.0
19	18	20.89	0.61	127.6	84.2	268.3	763.7	127.6	84.2	763.7
20	19	21.89	0.63	130.7	86.2	275.6	832.2	130.7	86.2	832.2
21	20	22.89	0.64	133.5	88.1	282.6	902.2	133.5	88.1	902.2

5.3.3. Patch load:

- **Filling Patch Load:**

$P_{pf} = C_{pf} P_{hf}$ (outward direction) [according to eqn. 5.8, EN 1991-4:2006]

$P_{pfi} = P_{pf} / 7$ (inward direction) [according to eqn. 5.13, EN 1991-4:2006]

C_{pf} = 0.21 Cop (1+2E^2).[1-exp{-1.5(h_c/d_c-1)}] [according to eqn. 5.9, EN 1991-4:2006]

= 0.00

E = $2e_f / d_c$ = 0.00 [according to eqn. 5.10, EN 1991-4:2006]

- **Discharge Patch Load:**

$P_{pe} = C_{pe} P_{he}$ (outward direction) [according to eqn. 5.27, EN 1991-4:2006]

$P_{pei} = P_{pe} / 7$ (inward direction) [according to eqn. 5.33, EN 1991-4:2006]

C_{pe} = max [0.42Cop(1+2E^2)[1-exp{-1.5(hc/dc-1)}], 0.272Cop(hc/dc-1+E), 0]

= 0.08 [for, hc/dc≤1.2] [according to eqn. 5.28, 5.29, 5.30, EN 1991-4:2006]

E = $2e / d_c$ = 0.57 [according to eqn. 5.31, EN 1991-4:2006]

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- **Substitute uniform pressure increase for Patch load:**

  Filling: $p_{hf,u} = p_{hf}\,(1 + \zeta.C_{pf})$

  Discharge: $p_{he,u} = p_{he}\,(1 + \zeta.C_{pe})$

where, $\zeta = 0.5 + 0.01(d_c/t) = 1.00$ [with, $\zeta \geq 1.0$] [according to eqn. 5.31, EN 1991-4:2006]

Table-4		Case-I							
		Filling Condition				Emptying Condition			
Sec.	z*	p_{hf}	p_{pf}	p_{hfi}	$p_{hf,u}$	p_{he}	p_{pe}	p_{pei}	$p_{he,u}$
-	m	KN/m^2	KN/m^2	KN/m^2	KN/m^2	KN/m^2	KN/m^2	KN/m^2	KN/m^2
1	0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
2	1	18.67	0.00	0.00	18.67	18.67	1.44	-0.21	20.11
3	2	34.70	0.00	0.00	34.70	34.70	2.68	-0.38	37.37
4	3	48.57	0.00	0.00	48.57	48.57	3.75	-0.54	52.32
5	4	60.68	0.00	0.00	60.68	60.68	4.68	-0.67	65.36
6	5	71.32	0.00	0.00	71.32	71.32	5.50	-0.79	76.82
7	6	80.74	0.00	0.00	80.74	80.74	6.23	-0.89	86.97
8	7	89.12	0.00	0.00	89.12	89.12	6.87	-0.98	95.99
9	8	96.61	0.00	0.00	96.61	96.61	7.45	-1.06	104.06
10	9	103.35	0.00	0.00	103.35	103.35	7.97	-1.14	111.31
11	10	109.43	0.00	0.00	109.43	109.43	8.44	-1.21	117.86
12	11	114.94	0.00	0.00	114.94	114.94	8.86	-1.27	123.80
13	12	119.95	0.00	0.00	119.95	119.95	9.25	-1.32	129.20
14	13	124.53	0.00	0.00	124.53	124.53	9.60	-1.37	134.13
15	14	128.72	0.00	0.00	128.72	128.72	9.93	-1.42	138.65
16	15	132.57	0.00	0.00	132.57	132.57	10.22	-1.46	142.80
17	16	136.12	0.00	0.00	136.12	136.12	10.50	-1.50	146.62
18	17	139.40	0.00	0.00	139.40	139.40	10.75	-1.54	150.15
19	18	142.44	0.00	0.00	142.44	142.44	10.98	-1.57	153.42
20	19	145.26	0.00	0.00	145.26	145.26	11.20	-1.60	156.46
21	20	147.88	0.00	0.00	147.88	147.88	11.40	-1.63	159.28

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Table-5		Case-II							
		Filling Condition				Emptying Condition			
Sec.	z*	p_{hf}	p_{pf}	p_{hfi}	$p_{hf,u}$	p_{he}	p_{pe}	p_{pei}	$p_{he,u}$
-	m	KN/m^2	KN/m^2	KN/m^2	KN/m^2	KN/m^2	KN/m^2	KN/m^2	KN/m^2
1	0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
2	1	22.79	0.00	0.00	22.79	22.79	1.76	-0.25	24.54
3	2	41.62	0.00	0.00	41.62	41.62	3.21	-0.46	44.82
4	3	57.39	0.00	0.00	57.39	57.39	4.43	-0.63	61.82
5	4	70.78	0.00	0.00	70.78	70.78	5.46	-0.78	76.23
6	5	82.25	0.00	0.00	82.25	82.25	6.34	-0.91	88.59
7	6	92.17	0.00	0.00	92.17	92.17	7.11	-1.02	99.28
8	7	100.83	0.00	0.00	100.83	100.83	7.77	-1.11	108.61
9	8	108.44	0.00	0.00	108.44	108.44	8.36	-1.19	116.80
10	9	115.17	0.00	0.00	115.17	115.17	8.88	-1.27	124.05
11	10	121.16	0.00	0.00	121.16	121.16	9.34	-1.33	130.50
12	11	126.52	0.00	0.00	126.52	126.52	9.76	-1.39	136.28
13	12	131.34	0.00	0.00	131.34	131.34	10.13	-1.45	141.47
14	13	135.69	0.00	0.00	135.69	135.69	10.46	-1.49	146.15
15	14	139.64	0.00	0.00	139.64	139.64	10.77	-1.54	150.41
16	15	143.23	0.00	0.00	143.23	143.23	11.04	-1.58	154.28
17	16	146.51	0.00	0.00	146.51	146.51	11.30	-1.61	157.81
18	17	149.52	0.00	0.00	149.52	149.52	11.53	-1.65	161.05
19	18	152.29	0.00	0.00	152.29	152.29	11.74	-1.68	164.03
20	19	154.84	0.00	0.00	154.84	154.84	11.94	-1.71	166.78
21	20	157.20	0.00	0.00	157.20	157.20	12.12	-1.73	169.32

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Table-6		Case-III							
		Filling Condition				Emptying Condition			
Sec.	z*	p_{hf}	p_{pf}	p_{hfi}	$p_{hf,u}$	p_{he}	p_{pe}	p_{pei}	$p_{he,u}$
-	m	KN/m^2	KN/m^2	KN/m^2	KN/m^2	KN/m^2	KN/m^2	KN/m^2	KN/m^2
1	0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
2	1	14.29	0.00	0.00	14.29	14.29	1.10	-0.16	15.39
3	2	27.03	0.00	0.00	27.03	27.03	2.08	-0.30	29.12
4	3	38.46	0.00	0.00	38.46	38.46	2.97	-0.42	41.42
5	4	48.74	0.00	0.00	48.74	48.74	3.76	-0.54	52.49
6	5	58.03	0.00	0.00	58.03	58.03	4.47	-0.64	62.51
7	6	66.47	0.00	0.00	66.47	66.47	5.12	-0.73	71.59
8	7	74.15	0.00	0.00	74.15	74.15	5.72	-0.82	79.87
9	8	81.17	0.00	0.00	81.17	81.17	6.26	-0.89	87.43
10	9	87.60	0.00	0.00	87.60	87.60	6.75	-0.96	94.36
11	10	93.52	0.00	0.00	93.52	93.52	7.21	-1.03	100.73
12	11	98.97	0.00	0.00	98.97	98.97	7.63	-1.09	106.60
13	12	104.01	0.00	0.00	104.01	104.01	8.02	-1.15	112.02
14	13	108.67	0.00	0.00	108.67	108.67	8.38	-1.20	117.05
15	14	113.00	0.00	0.00	113.00	113.00	8.71	-1.24	121.72
16	15	117.03	0.00	0.00	117.03	117.03	9.02	-1.29	126.06
17	16	120.79	0.00	0.00	120.79	120.79	9.31	-1.33	130.10
18	17	124.30	0.00	0.00	124.30	124.30	9.58	-1.37	133.88
19	18	127.58	0.00	0.00	127.58	127.58	9.84	-1.41	137.42
20	19	130.66	0.00	0.00	130.66	130.66	10.07	-1.44	140.73
21	20	133.55	0.00	0.00	133.55	133.55	10.30	-1.47	143.84

5.3.4. Vertical Pressure on Flat Bottom (i.e., on top of Thick Slab):

Pv = Pvsq = Pvb + ΔPsq.(2.0-hc/dc)/(2.0-htp/dc) [according to cl. 6.2.2, EN 1991-4:2006]

where, ΔPsq = Pvtp - Pvho = 166.464 - 55.488 = 110.98 KN/m^2

Pvtp = γ.htp = 19.2 x 8.67 = 166.46 KN/m^2

			Case-I	Case-II	Case-III	
Pvf	=	Filling value of the vertical pressure (for z = hc)	= 257.7	239.2	282.6	KN/m^2
Pvb	=	Uniform component of vertical pressure = Cb.Pvf	= 257.7	239.2	282.6	KN/m^2

Cb = Bottom Load magnifier = 1.0 [for Action Assessment Class-3]

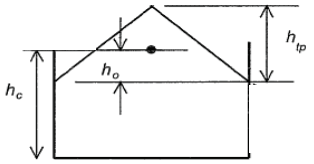
Pvho = Janssen vertical pressure at base of top pile = γ.Z = 19.2 x 2.89 KN/m^2

[for z= ho] = 55.488 KN/m^2

ho = Depth below equivalent surface to base of top pile = 2.89 m

htp = Total height of top pile = 8.67 m

hc = Depth of the silo base below equivalent surface = 22.9 m



			Case-I	Case-II	Case-III	
so, Pv	=	The vertical pressure acting on a flat bottom	= 326.4	307.9	351.3	KN/m^2

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5.4. Wind Load:

Basic wind speed, V = 155 km/h = **43.06** m/sec [as per design criteria, for Process Plant Area]
Exposure category = C [as per design criteria, for Process Plant Area]
Building Category = II [ref. Table 1.6-1 : SBC 301-2007]
Wind directionality factor, K_d = 0.95 [ref. Table 6.4-1 : SBC 301-2007]
Topography Factor, K_{zt} = 1 [as not located in isolated hills/ escarpments]
Importance Factor, I = 1 [ref. Table 6.5-1 : SBC 301-2007]
Velocity Pressure Exposure Coefficient, Kz = [ref. Table 7.2-2 : SBC 301-2007]

Height m	Velocity Pressure Exposure Coefficient, Kz
0	0.86
5	0.86
6	0.95
8	1.00
10	1.04
12	1.07
14	1.10
16	1.13
18	1.16
20	1.18
22	1.21
25	1.26
30	1.30
35	1.34

Wind Velocity Pressure, qz = $0.0473 \times 10^{-3} K_z . K_{zt} . K_d . V^2 . I$ KN/m^2 [ref. Eq. 7.2-12 : SBC 301-2007]
= 1.080 Kz KN/m^2

Height	Kz	Wind Velocity Pressure, qz
m	-	KN/m^2
0.0	0.86	0.928
5.0	0.86	0.928
6.0	0.95	1.026
8.0	1.00	1.080
10.0	1.04	1.123
12.0	1.07	1.155
14.0	1.10	1.188
16.0	1.13	1.220
18.0	1.16	1.252
20.0	1.18	1.274
22.0	1.21	1.306
25.0	1.26	1.360
30.0	1.30	1.403
35.0	1.34	1.447

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Gust Effect Factor, G = 0.85 [For Rigid Structure, ref. Section 7.2.7.1 : SBC 301-2007]

Force co-efficient, C_f = 0.71 [ref. Figure 7.2-15 : SBC 301-2007]

[For Round Cross-section, $D\sqrt{qz} > 0.17$, Type of Surface = Rough, $h/D \approx 1.6$]

where, D = Diameter of circular cross-section

qz = Velocity pressure evaluated at height z above ground in KN/m^2

h = Height of structure in meters

Design Wind Load, F = $qz.G.C_f.A_f$ [ref. section 7.2.11 : SBC 301-2007]

where, A_f = Projected area normal to the wind

Height	Design Wind Pressure, $qz.G.C_f$	Design Wind Load, $qz.G.C_f.A_f$
m	KN/m^2	KN
0.0	0.560	0.560.Af
5.0	0.560	0.560.Af
6.0	0.619	0.619.Af
8.0	0.652	0.652.Af
10.0	0.678	0.678.Af
12.0	0.697	0.697.Af
14.0	0.717	0.717.Af
16.0	0.736	0.736.Af
18.0	0.756	0.756.Af
20.0	0.769	0.769.Af
22.0	0.788	0.788.Af
25.0	0.821	0.821.Af
30.0	0.847	0.847.Af
35.0	0.873	0.873.Af
0.0	0.000	0.000.Af

5.5. Earthquake Load:

Location of plant = Saudi Arabia

Seismic zone = 2A [ref. Appendix of Chapter 16, Section 1653 : UBC 97]

Seismic zone facor, Z = 0.15 [ref. table 16I : UBC 97]

Numerical coefficient representative of the inherent over strength and global ductility capacity of lateral force resisting systems, R = 3.6 [ref. table 16P : UBC 97]

Importance factor, I = 1 [ref. table 16K : UBC 97]

Time period, T = $C_t \times (h_n)^{3/4}$ [ref. eqn. 30-8 : UBC 97]

= 1.11 sec

Where,

C_t = 0.0731

h_n = Height of building above ground level = 37.5 m

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Soil profile type	=	S_C	[ref. table 16J : UBC 97]
Seismic coefficient, C_a	=	0.18	[ref. table 16Q : UBC 97]
Seismic coefficient, C_v	=	0.25	[ref. table 16R : UBC 97]
The total design base shear, V	=	C_v x I x W/ (R x T)	[ref. eqn. 30-4 : UBC 97]
	=	0.06 x W KN	
The total design base shear shall not exceed the following, $V|_{max}$	=	2.5 x Ca x I x W/ R	[ref. eqn. 30-5 : UBC 97]
	=	0.13 x W KN	
The total design base shear shall not less than the following, $V|_{min}$	=	0.11 x Ca x I x W	[ref. eqn. 30-6 : UBC 97]
	=	0.02 x W KN	
Hence, the design seismic base shear	=	0.06 x W	[where, W = vertical load considered for seismic load]
	=	6.0 % of W	

During earthquake only 80% of bulk solids are assumed to be stored inside the silo. [refer clause 4.4.8 of ACI 313-97]

Height of material storage considered for seismic = 80% of hc = 18.31 m

Total lever arm available, $h|_{storage}$ = $h|_{thick\ slab}$ + (Height of material storage considered)/2
= 18.71 m

Total Load due to stored material, W = 146061 KN [with stored material height as hc]

- Total Seismic Load due to self weight of silo, Roof structure & structual floor is considered seperately in STAAD Command.

Accordingly seismic joint weight per STAAD node for 18.31m high stored material calculated below.

Number of nodes in STAAD for silo shell = 72 nos.

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Sec.	z	Height of Stored Material above joint level	Influence Height of Stored Material Considered in joint level	Weight of Stored Material Considered in joint level	Weight of Stored Material Considered per joint
-	m	m	m	KN	KN
1	2.89	-	-	-	-
2	3.89	-	0.155	1236.5	17.2
3	4.89	0.3	0.655	5225.0	72.6
4	5.89	1.0	1	7977.1	110.8
5	6.89	1.0	1	7977.1	110.8
6	7.89	1.0	1	7977.1	110.8
7	8.89	1.0	1	7977.1	110.8
8	9.89	1.0	1	7977.1	110.8
9	10.89	1.0	1	7977.1	110.8
10	11.89	1.0	1	7977.1	110.8
11	12.89	1.0	1	7977.1	110.8
12	13.89	1.0	1	7977.1	110.8
13	14.89	1.0	1	7977.1	110.8
14	15.89	1.0	1	7977.1	110.8
15	16.89	1.0	1	7977.1	110.8
16	17.89	1.0	1	7977.1	110.8
17	18.89	1.0	1	7977.1	110.8
18	19.89	1.0	1	7977.1	110.8
19	20.89	1.0	1	7977.1	110.8
20	21.89	1.0	1	7977.1	110.8
21	22.89	1.0	0.5	3988.6	55.4
			Sum=18.31		

5.6. Temperature Load:

The thermal effects of the hot stored materials on to the silo walls are considered in accordance with ACI 313, clause 4.4.9 with moment formula given below.

$$M_t = E_c h^2 \alpha_c \Delta T / 12(1-\nu)$$

[ref. Design and Construction of Silos and Bunkers - Safarian, Harris, equation 4.15, in which h is in inches, ΔT in °F, E_c in kips/in^2, M_t in kips-in/ft]

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Where, E_c = Modulus of elasticity of concrete = $4700\sqrt{f_c'}$ = 27805.6 MPa = 278056 kg/cm^2 [as per cl. 19.2.2.1, SBC 304-2018]

h = Wall thickness = 1000 mm = 100 cm

α_c = Thermal coefficient of expansion of concrete = 1.1 x 10^{-5} per °C

ΔT = Temperature difference between inside face and outside face of wall = 25 °C [assumed] [as per Civil & Structural design criteria]

ν = Poisson's ratio of concrete = 0.2

M_t = 8.328 x 278056 x 100^2 x (1.1x10^-5) x 25/ {12(1-0.2)} = 663337 Kg-cm/m = 66.3 KN-m/m

$M_{t|f}$ = Factored Moment = 1.4 x M_t = 1.4 x 66.3 KN-m/m = 92.82 KN-m/m

[where, 1.4 is load factor & 8.328 is unit conversion factor as per "Design and Construction of Silos and Bunkers" by Sagis S. Safarian, Ernest C. Harris]

6. CORRECTION FACTORS FOR LOAD APPLICATION IN STAAD

The pressure due to bulk solids calculated is to be reduced as the STAAD model the centre line dimension of silo is differ from the actual inner face dimension of silo wall. In reality the material pressures are only in the inner face of the silo.

Actual inner face diameter of silo = 23.0 m

Centre Line dimension of silo in STAAD = 23.225 m

6.1. Correction Factor for variance in dimension:

6.1.1. Correction Factor for Thick Slab:

Area of Thick Slab in actual = 415.5 sq.m

Area of Thick Slab in STAAD = 423.6 sq.m

Correction Factor, CF1 = Ratio of Areas = 0.981

6.1.2. Correction Factor for Silo Wall:

The pressure calculated in wall will be acting on the inner surface of silo wall with internal diameter 23 m, whereas in STAAD it will be applied on the centre-line diameter of the silo i.e., 23.225 m.

Correction Factor to be applied to calculated pressure values, CF2 = 23/ 23.225 = 0.990

6.2. Correction Factors for variance between Calculated and Actual Vertical Load:

Total load of stored material inside silo = $\{\pi.(dc)^2/4\}.(hc).\gamma$ = 182597 KN

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6.2.1. Correction Factors considering maximum load on Thick Slab:

				Case-I	Case-II	Case-III	
Normalised pressure on Thick Slab, $Pv\|_{Norm.}$	=	CF1 x Pv	=	320.1	302.0	344.5	KN/m^2
				Case-I	Case-II	Case-III	
Load taken by thick slab in STAAD	=	(Area of thick slab in STAAD) x $Pv\|_{Norm.}$	=	135600	127930	145949	KN
				Case-I	Case-II	Case-III	
Remaining load to be taken by silo wall	=	(Total load of stored material) – (Load taken by thick slab)	=	46997	54667	36647	KN
				Case-I	Case-II	Case-III	
Calculated load taken by silo wall	=	For Filling, $\sum P_{wf}.\Delta z$	=	78315	107272	84315	KN
	=	For Emptying, $\sum P_{we}.\Delta z$	=	78315	107272	84315	KN
				Case-I	Case-II	Case-III	
Correction Factor for vertical pressure on wall, CF3	=	For Filling		0.60	0.44	0.56	
	=	For Emptying		0.60	0.44	0.56	

6.2.2. Correction Factors considering maximum load on Silo Wall:

				Case-I	Case-II	Case-III	
Calculated Load taken by silo wall	=	For Filling, $\sum P_{wf}.\Delta z$	=	78315	107272	84315	KN
	=	For Emptying, $\sum P_{we}.\Delta z$	=	78315	107272	84315	KN
				Case-I	Case-II	Case-III	
Calculated Load taken by silo wall in STAAD (multiplied with CF2)	=	For Filling, $\sum P_{wf}.\Delta z$	=	77556	106233	83498	KN
	=	For Emptying, $\sum P_{we}.\Delta z$	=	77556	106233	83498	KN
				Case-I	Case-II	Case-III	
Remaining load taken by thick slab	=	(Total load of stored material) – (Load taken by silo wall)	=	105040	76364	99098	KN
				105040	76364	99098	KN
				Case-I	Case-II	Case-III	
Calculated Load on thick slab	=	(Area of thick slab in STAAD) x $Pv\|_{Norm.}$	=	135600	127930	145949	KN
				Case-I	Case-II	Case-III	
Correction Factor for vertical pressure on thick slab, CF4	=	For Filling		0.77	0.60	0.68	
	=	For Emptying		0.77	0.60	0.68	

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7. SUMMARY OF MATERIAL LOADS APPLIED IN STAAD (CONSIDERING CORRECTION FACTORS)

7.1. Vertical pressure on Silo (Considering Maximum pressure on Thick Slab):

7.1.1. Vertical pressure on Silo Wall:

Table-7			Filling Condition			Emptying Condition		
			Case-I	Case-II	Case-III	Case-I	Case-II	Case-III
Sec.	z*	$z=z^*+h_o$	P_{wf}	P_{wf}	P_{wf}	P_{we}	P_{we}	P_{we}
-	m	m	KN/m²	KN/m²	KN/m²	KN/m²	KN/m²	KN/m²
1	0	2.89	0.0	0.0	0.0	0.0	0.0	0.0
2	1	3.89	5.9	6.5	5.2	5.9	6.5	5.2
3	2	4.89	10.9	11.9	9.8	10.9	11.9	9.8
4	3	5.89	15.3	16.4	14.0	15.3	16.4	14.0
5	4	6.89	19.1	20.3	17.8	19.1	20.3	17.8
6	5	7.89	22.5	23.6	21.1	22.5	23.6	21.1
7	6	8.89	25.4	26.4	24.2	25.4	26.4	24.2
8	7	9.89	28.1	28.9	27.0	28.1	28.9	27.0
9	8	10.89	30.4	31.1	29.6	30.4	31.1	29.6
10	9	11.89	32.6	33.0	31.9	32.6	33.0	31.9
11	10	12.89	34.5	34.7	34.1	34.5	34.7	34.1
12	11	13.89	36.2	36.2	36.1	36.2	36.2	36.1
13	12	14.89	37.8	37.6	37.9	37.8	37.6	37.9
14	13	15.89	39.2	38.9	39.6	39.2	38.9	39.6
15	14	16.89	40.5	40.0	41.2	40.5	40.0	41.2
16	15	17.89	41.8	41.0	42.6	41.8	41.0	42.6
17	16	18.89	42.9	42.0	44.0	42.9	42.0	44.0
18	17	19.89	43.9	42.8	45.3	43.9	42.8	45.3
19	18	20.89	44.9	43.6	46.5	44.9	43.6	46.5
20	19	21.89	45.8	44.3	47.6	45.8	44.3	47.6
21	20	22.89	46.6	45.0	48.7	46.6	45.0	48.7

7.1.2. Vertical pressure on Thick Slab:

Table-8	Condition	Case-I	Case-II	Case-III
		P_v	P_v	P_v
		KN/m²	KN/m²	KN/m²
Pressure on Thick Slab	Filling Condition	320.1	302.0	344.5
	Emptying Condition	320.1	302.0	344.5

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7.2. Vertical pressure on Silo (Considering Maximum pressure on Silo Wall):

7.2.1. Vertical pressure on Silo Wall:

Table-9			Filling Condition			Emptying Condition		
			Case-I	Case-II	Case-III	Case-I	Case-II	Case-III
Sec.	z*	$z=z^*+h_o$	P_{wf}	P_{wf}	P_{wf}	P_{we}	P_{we}	P_{we}
-	m	m	KN/m²	KN/m²	KN/m²	KN/m²	KN/m²	KN/m²
1	0	2.89	0.0	0.0	0.0	0.0	0.0	0.0
2	1	3.89	9.8	14.9	9.3	9.8	14.9	9.3
3	2	4.89	18.2	27.2	17.7	18.2	27.2	17.7
4	3	5.89	25.5	37.5	25.1	25.5	37.5	25.1
5	4	6.89	31.8	46.3	31.9	31.8	46.3	31.9
6	5	7.89	37.4	53.8	37.9	37.4	53.8	37.9
7	6	8.89	42.4	60.2	43.4	42.4	60.2	43.4
8	7	9.89	46.8	65.9	48.5	46.8	65.9	48.5
9	8	10.89	50.7	70.9	53.1	50.7	70.9	53.1
10	9	11.89	54.2	75.3	57.3	54.2	75.3	57.3
11	10	12.89	57.4	79.2	61.1	57.4	79.2	61.1
12	11	13.89	60.3	82.7	64.7	60.3	82.7	64.7
13	12	14.89	63.0	85.8	68.0	63.0	85.8	68.0
14	13	15.89	65.4	88.7	71.0	65.4	88.7	71.0
15	14	16.89	67.6	91.3	73.9	67.6	91.3	73.9
16	15	17.89	69.6	93.6	76.5	69.6	93.6	76.5
17	16	18.89	71.4	95.8	78.9	71.4	95.8	78.9
18	17	19.89	73.2	97.7	81.2	73.2	97.7	81.2
19	18	20.89	74.8	99.5	83.4	74.8	99.5	83.4
20	19	21.89	76.2	101.2	85.4	76.2	101.2	85.4
21	20	22.89	77.6	102.7	87.3	77.6	102.7	87.3

7.2.2. Vertical pressure on Thick Slab:

Table-10	Condition	Case-I	Case-II	Case-III
		P_v	P_v	P_v
		KN/m²	KN/m²	KN/m²
Pressure on Thick Slab	Filling Condition	247.9	180.3	233.9
	Emptying Condition	247.9	180.3	233.9

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7.3. Horizontal pressure on Silo Wall:

Table-11			Filling Condition			Emptying Condition		
			Case-I	Case-II	Case-III	Case-I	Case-II	Case-III
Sec.	z*	$z=z^*+h_o$	$p_{hf,u}$	$p_{hf,u}$	$p_{hf,u}$	$p_{he,u}$	$p_{he,u}$	$p_{he,u}$
-	m	m	KN/m²	KN/m²	KN/m²	KN/m²	KN/m²	KN/m²
1	0	2.89	0.0	0.0	0.0	0.0	0.0	0.0
2	1	3.89	18.5	22.6	14.2	19.9	24.3	15.2
3	2	4.89	34.4	41.2	26.8	37.0	44.4	28.8
4	3	5.89	48.1	56.8	38.1	51.8	61.2	41.0
5	4	6.89	60.1	70.1	48.3	64.7	75.5	52.0
6	5	7.89	70.6	81.5	57.5	76.1	87.7	61.9
7	6	8.89	80.0	91.3	65.8	86.1	98.3	70.9
8	7	9.89	88.3	99.9	73.4	95.1	107.6	79.1
9	8	10.89	95.7	107.4	80.4	103.1	115.7	86.6
10	9	11.89	102.3	114.1	86.8	110.2	122.9	93.4
11	10	12.89	108.4	120.0	92.6	116.7	129.2	99.8
12	11	13.89	113.8	125.3	98.0	122.6	135.0	105.6
13	12	14.89	118.8	130.1	103.0	127.9	140.1	110.9
14	13	15.89	123.3	134.4	107.6	132.8	144.7	115.9
15	14	16.89	127.5	138.3	111.9	137.3	148.9	120.5
16	15	17.89	131.3	141.8	115.9	141.4	152.8	124.8
17	16	18.89	134.8	145.1	119.6	145.2	156.3	128.8
18	17	19.89	138.1	148.1	123.1	148.7	159.5	132.6
19	18	20.89	141.1	150.8	126.3	151.9	162.4	136.1
20	19	21.89	143.8	153.3	129.4	154.9	165.2	139.4
21	20	22.89	146.4	155.7	132.3	157.7	167.7	142.4

8. DESIGN REQUIREMENTS

8.1. Design Basis:

Design of concrete structures and foundations shall be in accordance with SBC-304, ACI 318M & ACI 313-97.

8.1.1. Minimum Concrete Cover:

Concrete Exposure	Member		Minimum
Exposed to weather or contact with ground	Wall		60 mm
	Column & Pedestal		60 mm
	Slab		40 mm
	Foundation Raft	Side & Bottom	75 mm
		Top	50 mm

8.1.2. Allowable Soil Bearing Capacity:

The net safe bearing capacity of soil for foundations design has been considered as 600 KN/m^2 at 2.85m (i.e., RL 909.400m) below FGL (i.e., RL 912.250m) for BH-17.

Vertical Modulus of Subgrade Reaction (applied in STAAD) = 54000 KN/m^2/m
[ref. Table 3 & 4 of Geotechnical Report No. MD-216-3000-EG-CV-RPT-1008]

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9. PRIMARY LOAD CASES CONSIDERED IN STAAD:

Primary Load Case	Description	Direction	Remarks	STAAD Nomenclature	Load Case Number
Dead Load (D)	-	Downward	-	D	1
Live Load (L)	-	Downward	-	L	2
Wind Load (W)	Along X-Direction	Along +X	-	W1A	3
	Along Z-Direction	Along +Z	-	W1B	4
Earthquake Load (E)	Along X-Direction	Along +X	-	E1A	5
	Along Z-Direction	Along +Z	-	E1B	6
Bulk Material Load (H)	Load on Silo Wall	Horizontal	Filling Load for Case-I	H1A	11
			Filling Load for Case-II	H1B	12
			Filling Load for Case-III	H1C	13
			Emptying Load for Case-I	H2A	14
			Emptying Load for Case-II	H2B	15
			Emptying Load for Case-III	H2C	16
		Downward (for max. load on Thick Slab)	Filling & Emptying, Case-I	H3A	21
			Filling & Emptying, Case-II	H3B	22
			Filling & Emptying, Case-III	H3C	23
		Downward (for max. load on Silo Wall)	Filling & Emptying, Case-I	H4A	24
			Filling & Emptying, Case-II	H4B	25
			Filling & Emptying, Case-III	H4C	26
	Load on Thick Slab	Downward (for max. load on Thick Slab)	Filling & Emptying, Case-I	H5A	31
			Filling & Emptying, Case-II	H5B	32
			Filling & Emptying, Case-III	H5C	33
		Downward (for max. load on Silo Wall)	Filling & Emptying, Case-I	H6A	34
			Filling & Emptying, Case-II	H6B	35
			Filling & Emptying, Case-III	H6C	36

• The Load due to Temperature not considered in STAAD analysis. Alternatively reinforement required due to temperature calculated seperately and provided on the wall.

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10. LOAD COMBINATIONS CONSIDERED IN STAAD:

Load combinations are determined accordance with SBC 301, SBC 304 and Civil, Structural Design Criteria document number MD-216-1000-EG-ST-CRT-1001.

10.1. Factored Load Combinations for strength design:

- 1.4D [ref. cl. 2.3.2 : SBC 301-2007]
- 1.4D + 1.7 (L + H)
- 1.2D + 1.0L
- 1.2D + 0.8W
- 1.2D + 1.6W + 1.0L
- 1.2D + 1.0E + 1.0L
- 1.2D + 1.0E + 1.0L
- 0.9D + 1.6W + 1.6H
- 0.9D + 1.0E + 1.6H

Serial No.	STAAD Load Combination Number	Description
1	101	1.4x[D]
2	102	1.4x[D] + 1.7x[L + H1A + H3A + H5A]
3	103	1.4x[D] + 1.7x[L + H1B + H3B + H5B]
4	104	1.4x[D] + 1.7x[L + H1C + H3C + H5C]
5	105	1.4x[D] + 1.7x[L + H1A + H4A + H6A]
6	106	1.4x[D] + 1.7x[L + H1B + H4B + H6B]
7	107	1.4x[D] + 1.7x[L + H1C + H4C + H6C]
8	108	1.4x[D] + 1.7x[L + H2A + H3A + H5A]
9	109	1.4x[D] + 1.7x[L + H2B + H3B + H5B]
10	110	1.4x[D] + 1.7x[L + H2C + H3C + H5C]
11	111	1.4x[D] + 1.7x[L + H2A + H4A + H6A]
12	112	1.4x[D] + 1.7x[L + H2B + H4B + H6B]
13	113	1.4x[D] + 1.7x[L + H2C + H4C + H6C]
14	114	1.2x[D] + 1.0x[L]
15	115	1.2x[D] + 0.8x[W1A]
16	116	1.2x[D] + 0.8x[W1B]
17	117	1.2x[D] + 1.6x[W1A] + 1.0x[L]
18	118	1.2x[D] + 1.6x[W1B] + 1.0x[L]
19	119	1.2x[D] + 1.0x[E1A] + 1.0x[L]
20	120	1.2x[D] + 1.0x[E1B] + 1.0x[L]
21	121	0.9x[D] + 1.6x[W1A + H1A + H3A + H5A]
22	122	0.9x[D] + 1.6x[W1A + H1B + H3B + H5B]
23	123	0.9x[D] + 1.6x[W1A + H1C + H3C + H5C]
24	124	0.9x[D] + 1.6x[W1A + H1A + H4A + H6A]

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Serial No.	STAAD Load Combination Number	Description
25	125	0.9x[D] + 1.6x[W1A + H1B + H4B + H6B]
26	126	0.9x[D] + 1.6x[W1A + H1C + H4C + H6C]
27	127	0.9x[D] + 1.6x[W1A + H2A + H3A + H5A]
28	128	0.9x[D] + 1.6x[W1A + H2B + H3B + H5B]
29	129	0.9x[D] + 1.6x[W1A + H2C + H3C + H5C]
30	130	0.9x[D] + 1.6x[W1A + H2A + H4A + H6A]
31	131	0.9x[D] + 1.6x[W1A + H2B + H4B + H6B]
32	132	0.9x[D] + 1.6x[W1A + H2C + H4C + H6C]
33	133	0.9x[D] + 1.6x[W1B + H1A + H3A + H5A]
34	134	0.9x[D] + 1.6x[W1B + H1B + H3B + H5B]
35	135	0.9x[D] + 1.6x[W1B + H1C + H3C + H5C]
36	136	0.9x[D] + 1.6x[W1B + H1A + H4A + H6A]
37	137	0.9x[D] + 1.6x[W1B + H1B + H4B + H6B]
38	138	0.9x[D] + 1.6x[W1B + H1C + H4C + H6C]
39	139	0.9x[D] + 1.6x[W1B + H2A + H3A + H5A]
40	140	0.9x[D] + 1.6x[W1B + H2B + H3B + H5B]
41	141	0.9x[D] + 1.6x[W1B + H2C + H3C + H5C]
42	142	0.9x[D] + 1.6x[W1B + H2A + H4A + H6A]
43	143	0.9x[D] + 1.6x[W1B + H2B + H4B + H6B]
44	144	0.9x[D] + 1.6x[W1B + H2C + H4C + H6C]
45	145	0.9x[D] + 1.0x[E1A] + 1.6x[H1A + H3A + H5A]
46	146	0.9x[D] + 1.0x[E1A] + 1.6x[H1B + H3B + H5B]
47	147	0.9x[D] + 1.0x[E1A] + 1.6x[H1C + H3C + H5C]
48	148	0.9x[D] + 1.0x[E1A] + 1.6x[H1A + H4A + H6A]
49	149	0.9x[D] + 1.0x[E1A] + 1.6x[H1B + H4B + H6B]
50	150	0.9x[D] + 1.0x[E1A] + 1.6x[H1C + H4C + H6C]
51	151	0.9x[D] + 1.0x[E1A] + 1.6x[H2A + H3A + H5A]
52	152	0.9x[D] + 1.0x[E1A] + 1.6x[H2B + H3B + H5B]
53	153	0.9x[D] + 1.0x[E1A] + 1.6x[H2C + H3C + H5C]
54	154	0.9x[D] + 1.0x[E1A] + 1.6x[H2A + H4A + H6A]
55	155	0.9x[D] + 1.0x[E1A] + 1.6x[H2B + H4B + H6B]
56	156	0.9x[D] + 1.0x[E1A] + 1.6x[H2C + H4C + H6C]
57	157	0.9x[D] + 1.0x[E1B] + 1.6x[H1A + H3A + H5A]
58	158	0.9x[D] + 1.0x[E1B] + 1.6x[H1B + H3B + H5B]
59	159	0.9x[D] + 1.0x[E1B] + 1.6x[H1C + H3C + H5C]
60	160	0.9x[D] + 1.0x[E1B] + 1.6x[H1A + H4A + H6A]
61	161	0.9x[D] + 1.0x[E1B] + 1.6x[H1B + H4B + H6B]

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Serial No.	STAAD Load Combination Number	Description
62	162	0.9x[D] + 1.0x[E1B] + 1.6x[H1C + H4C + H6C]
63	163	0.9x[D] + 1.0x[E1B] + 1.6x[H2A + H3A + H5A]
64	164	0.9x[D] + 1.0x[E1B] + 1.6x[H2B + H3B + H5B]
65	165	0.9x[D] + 1.0x[E1B] + 1.6x[H2C + H3C + H5C]
66	166	0.9x[D] + 1.0x[E1B] + 1.6x[H2A + H4A + H6A]
67	167	0.9x[D] + 1.0x[E1B] + 1.6x[H2B + H4B + H6B]
68	168	0.9x[D] + 1.0x[E1B] + 1.6x[H2C + H4C + H6C]

10.2. Load Combinations for serviceability and allowable stress design:

- D + H + L [ref. cl. 2.4.1 : SBC 301-2007]
- D + H + 0.75L
- D + H + W
- D + H + 0.7E
- D + H + 0.75W + 0.75L
- D + H + 0.75(0.7E) + 0.75L
- 0.6D + W+ H
- 0.6D + 0.7E+ H

Serial No.	STAAD Load Combination Number	Description
1	501	1x[D + H1A + H3A + H5A + L]
2	502	1x[D + H1B + H3B + H5B + L]
3	503	1x[D + H1C + H3C + H5C + L]
4	504	1x[D + H1A + H4A + H6A + L]
5	505	1x[D + H1B + H4B + H6B + L]
6	506	1x[D + H1C + H4C + H6C + L]
7	507	1x[D + H2A + H3A + H5A + L]
8	508	1x[D + H2B + H3B + H5B + L]
9	509	1x[D + H2C + H3C + H5C + L]
10	510	1x[D + H2A + H4A + H6A + L]
11	511	1x[D + H2B + H4B + H6B + L]
12	512	1x[D + H2C + H4C + H6C + L]
13	513	1x[D + H1A + H3A + H5A]
14	514	1x[D + H1B + H3B + H5B]
15	515	1x[D + H1C + H3C + H5C]
16	516	1x[D + H1A + H4A + H6A]
17	517	1x[D + H1B + H4B + H6B]
18	518	1x[D + H1C + H4C + H6C]
19	519	1x[D + H2A + H3A + H5A]
20	520	1x[D + H2B + H3B + H5B]

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Serial No.	STAAD Load Combination Number	Description
21	521	1x[D + H2C + H3C + H5C]
22	522	1x[D + H2A + H4A + H6A]
23	523	1x[D + H2B + H4B + H6B]
24	524	1x[D + H2C + H4C + H6C]
25	525	1x[D + H1A + H3A + H5A] + 0.75x[L]
26	526	1x[D + H1B + H3B + H5B] + 0.75x[L]
27	527	1x[D + H1C + H3C + H5C] + 0.75x[L]
28	528	1x[D + H1A + H4A + H6A] + 0.75x[L]
29	529	1x[D + H1B + H4B + H6B] + 0.75x[L]
30	530	1x[D + H1C + H4C + H6C] + 0.75x[L]
31	531	1x[D + H2A + H3A + H5A] + 0.75x[L]
32	532	1x[D + H2B + H3B + H5B] + 0.75x[L]
33	533	1x[D + H2C + H3C + H5C] + 0.75x[L]
34	534	1x[D + H2A + H4A + H6A] + 0.75x[L]
35	535	1x[D + H2B + H4B + H6B] + 0.75x[L]
36	536	1x[D + H2C + H4C + H6C] + 0.75x[L]
37	537	1x[D + H1A + H3A + H5A + W1A]
38	538	1x[D + H1B + H3B + H5B + W1A]
39	539	1x[D + H1C + H3C + H5C + W1A]
40	540	1x[D + H1A + H4A + H6A + W1A]
41	541	1x[D + H1B + H4B + H6B + W1A]
42	542	1x[D + H1C + H4C + H6C + W1A]
43	543	1x[D + H2A + H3A + H5A + W1A]
44	544	1x[D + H2B + H3B + H5B + W1A]
45	545	1x[D + H2C + H3C + H5C + W1A]
46	546	1x[D + H2A + H4A + H6A + W1A]
47	547	1x[D + H2B + H4B + H6B + W1A]
48	548	1x[D + H2C + H4C + H6C + W1A]
49	549	1x[D + H1A + H3A + H5A + W1B]
50	550	1x[D + H1B + H3B + H5B + W1B]
51	551	1x[D + H1C + H3C + H5C + W1B]
52	552	1x[D + H1A + H4A + H6A + W1B]
53	553	1x[D + H1B + H4B + H6B + W1B]
54	554	1x[D + H1C + H4C + H6C + W1B]
55	555	1x[D + H2A + H3A + H5A + W1B]
56	556	1x[D + H2B + H3B + H5B + W1B]
57	557	1x[D + H2C + H3C + H5C + W1B]
58	558	1x[D + H2A + H4A + H6A + W1B]
59	559	1x[D + H2B + H4B + H6B + W1B]
60	560	1x[D + H2C + H4C + H6C + W1B]
61	561	1x[D + H1A + H3A + H5A] + 0.7x[E1A]
62	562	1x[D + H1B + H3B + H5B] + 0.7x[E1A]
63	563	1x[D + H1C + H3C + H5C] + 0.7x[E1A]
64	564	1x[D + H1A + H4A + H6A] + 0.7x[E1A]

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Serial No.	STAAD Load Combination Number	Description
65	565	1x[D + H1B + H4B + H6B] + 0.7x[E1A]
66	566	1x[D + H1C + H4C + H6C] + 0.7x[E1A]
67	567	1x[D + H2A + H3A + H5A] + 0.7x[E1A]
68	568	1x[D + H2B + H3B + H5B] + 0.7x[E1A]
69	569	1x[D + H2C + H3C + H5C] + 0.7x[E1A]
70	570	1x[D + H2A + H4A + H6A] + 0.7x[E1A]
71	571	1x[D + H2B + H4B + H6B] + 0.7x[E1A]
72	572	1x[D + H2C + H4C + H6C] + 0.7x[E1A]
73	573	1x[D + H1A + H3A + H5A] + 0.7x[E1B]
74	574	1x[D + H1B + H3B + H5B] + 0.7x[E1B]
75	575	1x[D + H1C + H3C + H5C] + 0.7x[E1B]
76	576	1x[D + H1A + H4A + H6A] + 0.7x[E1B]
77	577	1x[D + H1B + H4B + H6B] + 0.7x[E1B]
78	578	1x[D + H1C + H4C + H6C] + 0.7x[E1B]
79	579	1x[D + H2A + H3A + H5A] + 0.7x[E1B]
80	580	1x[D + H2B + H3B + H5B] + 0.7x[E1B]
81	581	1x[D + H2C + H3C + H5C] + 0.7x[E1B]
82	582	1x[D + H2A + H4A + H6A] + 0.7x[E1B]
83	583	1x[D + H2B + H4B + H6B] + 0.7x[E1B]
84	584	1x[D + H2C + H4C + H6C] + 0.7x[E1B]
85	585	1x[D + H1A + H3A + H5A] + 0.75x[W1A + L]
86	586	1x[D + H1B + H3B + H5B] + 0.75x[W1A + L]
87	587	1x[D + H1C + H3C + H5C] + 0.75x[W1A + L]
88	588	1x[D + H1A + H4A + H6A] + 0.75x[W1A + L]
89	589	1x[D + H1B + H4B + H6B] + 0.75x[W1A + L]
90	590	1x[D + H1C + H4C + H6C] + 0.75x[W1A + L]
91	591	1x[D + H2A + H3A + H5A] + 0.75x[W1A + L]
92	592	1x[D + H2B + H3B + H5B] + 0.75x[W1A + L]
93	593	1x[D + H2C + H3C + H5C] + 0.75x[W1A + L]
94	594	1x[D + H2A + H4A + H6A] + 0.75x[W1A + L]
95	595	1x[D + H2B + H4B + H6B] + 0.75x[W1A + L]
96	596	1x[D + H2C + H4C + H6C] + 0.75x[W1A + L]
97	597	1x[D + H1A + H3A + H5A] + 0.75x[W1B + L]
98	598	1x[D + H1B + H3B + H5B] + 0.75x[W1B + L]
99	599	1x[D + H1C + H3C + H5C] + 0.75x[W1B + L]
100	600	1x[D + H1A + H4A + H6A] + 0.75x[W1B + L]
101	601	1x[D + H1B + H4B + H6B] + 0.75x[W1B + L]
102	602	1x[D + H1C + H4C + H6C] + 0.75x[W1B + L]
103	603	1x[D + H2A + H3A + H5A] + 0.75x[W1B + L]
104	604	1x[D + H2B + H3B + H5B] + 0.75x[W1B + L]
105	605	1x[D + H2C + H3C + H5C] + 0.75x[W1B + L]
106	606	1x[D + H2A + H4A + H6A] + 0.75x[W1B + L]
107	607	1x[D + H2B + H4B + H6B] + 0.75x[W1B + L]
108	608	1x[D + H2C + H4C + H6C] + 0.75x[W1B + L]

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Serial No.	STAAD Load Combination Number	Description
109	609	1x[D + H1A + H3A + H5A] + 0.75x0.7x[E1A] + 0.75x[L]
110	610	1x[D + H1B + H3B + H5B] + 0.75x0.7x[E1A] + 0.75x[L]
111	611	1x[D + H1C + H3C + H5C] + 0.75x0.7x[E1A] + 0.75x[L]
112	612	1x[D + H1A + H4A + H6A] + 0.75x0.7x[E1A] + 0.75x[L]
113	613	1x[D + H1B + H4B + H6B] + 0.75x0.7x[E1A] + 0.75x[L]
114	614	1x[D + H1C + H4C + H6C] + 0.75x0.7x[E1A] + 0.75x[L]
115	615	1x[D + H2A + H3A + H5A] + 0.75x0.7x[E1A] + 0.75x[L]
116	616	1x[D + H2B + H3B + H5B] + 0.75x0.7x[E1A] + 0.75x[L]
117	617	1x[D + H2C + H3C + H5C] + 0.75x0.7x[E1A] + 0.75x[L]
118	618	1x[D + H2A + H4A + H6A] + 0.75x0.7x[E1A] + 0.75x[L]
119	619	1x[D + H2B + H4B + H6B] + 0.75x0.7x[E1A] + 0.75x[L]
120	620	1x[D + H2C + H4C + H6C] + 0.75x0.7x[E1A] + 0.75x[L]
121	621	1x[D + H1A + H3A + H5A] + 0.75x0.7x[E1B] + 0.75x[L]
122	622	1x[D + H1B + H3B + H5B] + 0.75x0.7x[E1B] + 0.75x[L]
123	623	1x[D + H1C + H3C + H5C] + 0.75x0.7x[E1B] + 0.75x[L]
124	624	1x[D + H1A + H4A + H6A] + 0.75x0.7x[E1B] + 0.75x[L]
125	625	1x[D + H1B + H4B + H6B] + 0.75x0.7x[E1B] + 0.75x[L]
126	626	1x[D + H1C + H4C + H6C] + 0.75x0.7x[E1B] + 0.75x[L]
127	627	1x[D + H2A + H3A + H5A] + 0.75x0.7x[E1B] + 0.75x[L]
128	628	1x[D + H2B + H3B + H5B] + 0.75x0.7x[E1B] + 0.75x[L]
129	629	1x[D + H2C + H3C + H5C] + 0.75x0.7x[E1B] + 0.75x[L]
130	630	1x[D + H2A + H4A + H6A] + 0.75x0.7x[E1B] + 0.75x[L]
131	631	1x[D + H2B + H4B + H6B] + 0.75x0.7x[E1B] + 0.75x[L]
132	632	1x[D + H2C + H4C + H6C] + 0.75x0.7x[E1B] + 0.75x[L]
133	633	0.6x[D] + [W1A + H1A + H3A + H5A]
134	634	0.6x[D] + [W1A + H1B + H3B + H5B]
135	635	0.6x[D] + [W1A + H1C + H3C + H5C]
136	636	0.6x[D] + [W1A + H1A + H4A + H6A]
137	637	0.6x[D] + [W1A + H1B + H4B + H6B]
138	638	0.6x[D] + [W1A + H1C + H4C + H6C]
139	639	0.6x[D] + [W1A + H2A + H3A + H5A]
140	640	0.6x[D] + [W1A + H2B + H3B + H5B]
141	641	0.6x[D] + [W1A + H2C + H3C + H5C]
142	642	0.6x[D] + [W1A + H2A + H4A + H6A]
143	643	0.6x[D] + [W1A + H2B + H4B + H6B]
144	644	0.6x[D] + [W1A + H2C + H4C + H6C]
145	645	0.6x[D] + [W1B + H1A + H3A + H5A]
146	646	0.6x[D] + [W1B + H1B + H3B + H5B]
147	647	0.6x[D] + [W1B + H1C + H3C + H5C]
148	648	0.6x[D] + [W1B + H1A + H4A + H6A]
149	649	0.6x[D] + [W1B + H1B + H4B + H6B]
150	650	0.6x[D] + [W1B + H1C + H4C + H6C]
151	651	0.6x[D] + [W1B + H2A + H3A + H5A]
152	652	0.6x[D] + [W1B + H2B + H3B + H5B]

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Serial No.	STAAD Load Combination Number	Description
153	653	0.6x[D] + [W1B + H2C + H3C + H5C]
154	654	0.6x[D] + [W1B + H2A + H4A + H6A]
155	655	0.6x[D] + [W1B + H2B + H4B + H6B]
156	656	0.6x[D] + [W1B + H2C + H4C + H6C]
157	657	0.6x[D] + 0.7x[E1A] + [H1A + H3A + H5A]
158	658	0.6x[D] + 0.7x[E1A] + [H1B + H3B + H5B]
159	659	0.6x[D] + 0.7x[E1A] + [H1C + H3C + H5C]
160	660	0.6x[D] + 0.7x[E1A] + [H1A + H4A + H6A]
161	661	0.6x[D] + 0.7x[E1A] + [H1B + H4B + H6B]
162	662	0.6x[D] + 0.7x[E1A] + [H1C + H4C + H6C]
163	663	0.6x[D] + 0.7x[E1A] + [H2A + H3A + H5A]
164	664	0.6x[D] + 0.7x[E1A] + [H2B + H3B + H5B]
165	665	0.6x[D] + 0.7x[E1A] + [H2C + H3C + H5C]
166	666	0.6x[D] + 0.7x[E1A] + [H2A + H4A + H6A]
167	667	0.6x[D] + 0.7x[E1A] + [H2B + H4B + H6B]
168	668	0.6x[D] + 0.7x[E1A] + [H2C + H4C + H6C]
169	669	0.6x[D] + 0.7x[E1B] + [H1A + H3A + H5A]
170	670	0.6x[D] + 0.7x[E1B] + [H1B + H3B + H5B]
171	671	0.6x[D] + 0.7x[E1B] + [H1C + H3C + H5C]
172	672	0.6x[D] + 0.7x[E1B] + [H1A + H4A + H6A]
173	673	0.6x[D] + 0.7x[E1B] + [H1B + H4B + H6B]
174	674	0.6x[D] + 0.7x[E1B] + [H1C + H4C + H6C]
175	675	0.6x[D] + 0.7x[E1B] + [H2A + H3A + H5A]
176	676	0.6x[D] + 0.7x[E1B] + [H2B + H3B + H5B]
177	677	0.6x[D] + 0.7x[E1B] + [H2C + H3C + H5C]
178	678	0.6x[D] + 0.7x[E1B] + [H2A + H4A + H6A]
179	679	0.6x[D] + 0.7x[E1B] + [H2B + H4B + H6B]
180	680	0.6x[D] + 0.7x[E1B] + [H2C + H4C + H6C]

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11. DESIGN OF FOUNDATION

11.1. Bearing Capacity Check:

	Node	L/C	Horizontal	Vertical	Horizontal
			Fx in kN/m2	Fy in kN/m2	Fz in kN/m2
Max Px	5001	501	0	471.789	0
Min Px	5001	501	0	471.789	0
Max Py	13294	583	0	553.292	0
Min Py	13276	658	0	197.673	0
Max Pz	5001	501	0	471.789	0
Min Pz	5001	501	0	471.789	0

[from STAAD analysis]

Gross Maximum Base pressure, p_{max} = 553.3 KN/m^2

Net Allowable Safe Bearing Capacity = 600 KN/m^2 [for Bore hole 17]

Depth of soil above footing = 1 m

Unit weight of soil = 18 KN/m^3

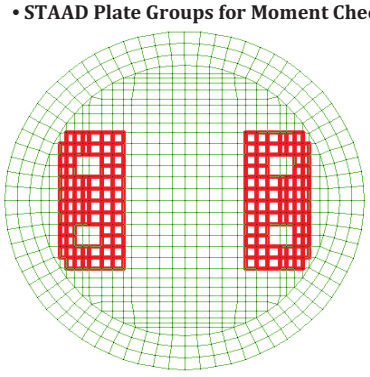
Gross Safe Bearing Capacity = 618 KN/m^2 > 553.3 KN/m^2 **Hence, OK**

11.2. Design of Foundation Raft:

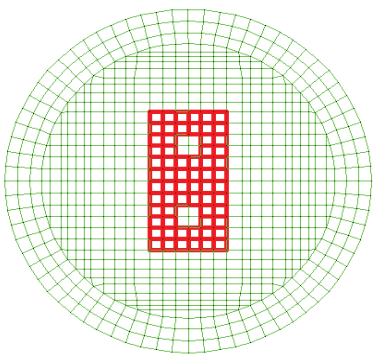
Based on the maximum moments and shear from the STAAD analysis, the design of raft is carried out for both directions X & Y and top & bottom reinforcements.

The plate elements of STAAD model devided in several groups and individually for each group the reinforements calculated for maximum moment and shear of that group.

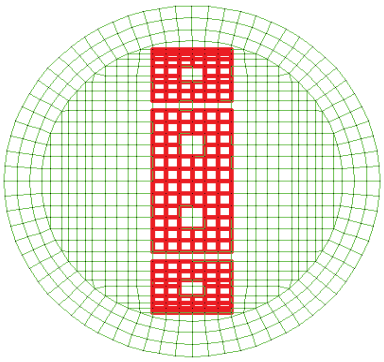
• STAAD Plate Groups for Moment Check:



RAFT_COL1_MOM



RAFT_COL2_MOM

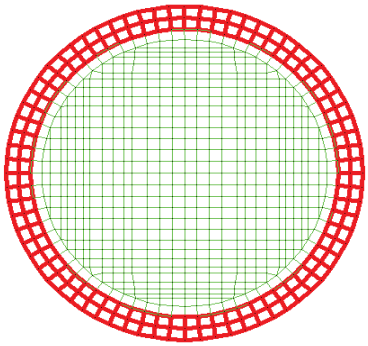


RAFT_COL3_MOM

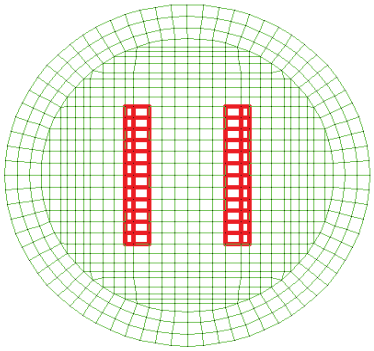
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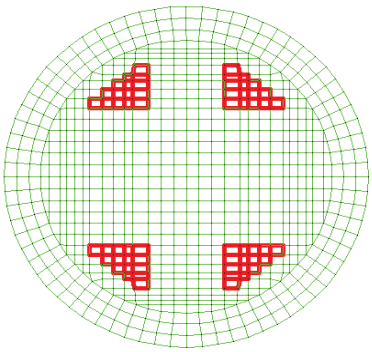
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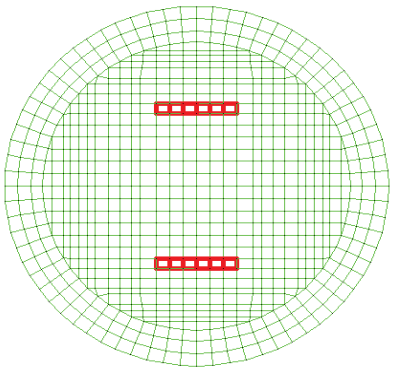
RAFT_CANT_MOM



RAFT_OTHER_1

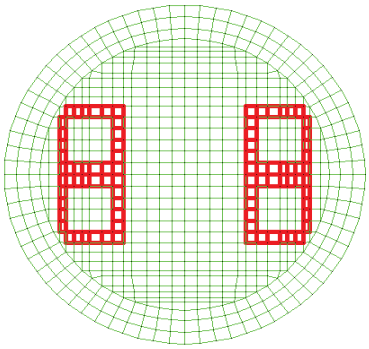


RAFT_OTHER_2

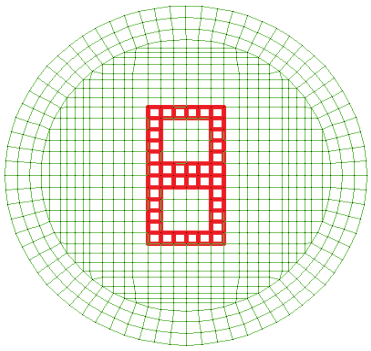


RAFT_OTHER_3

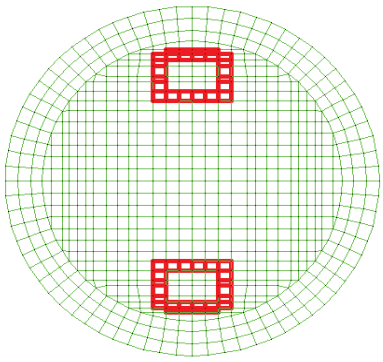
• **STAAD Plate Groups for Shear Check:**



RAFT_COL1_SHEAR



RAFT_COL2_SHEAR

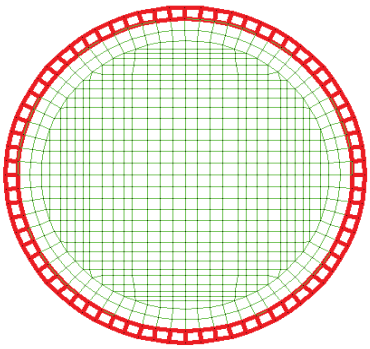


RAFT_COL3_SHEAR

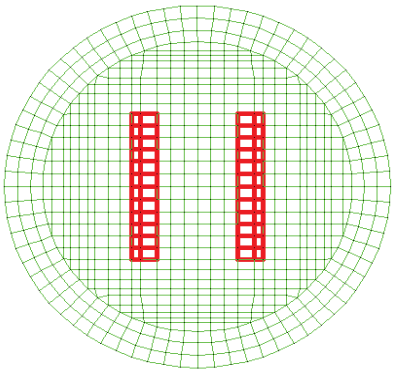
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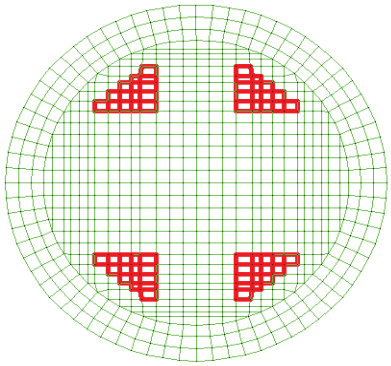
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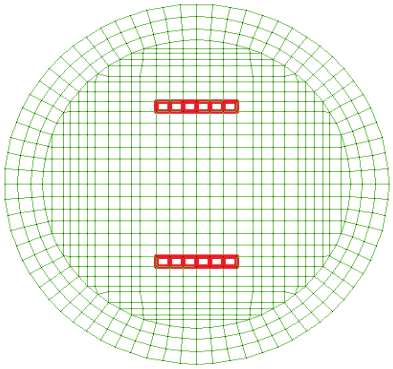
RAFT_CANT_SHEAR



RAFT_OTHER_1



RAFT_OTHER_2



RAFT_OTHER_3

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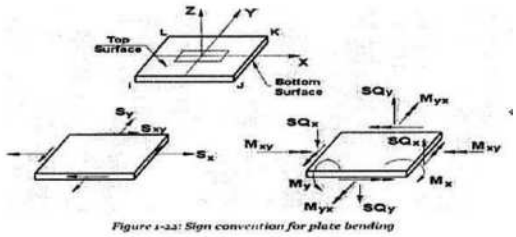
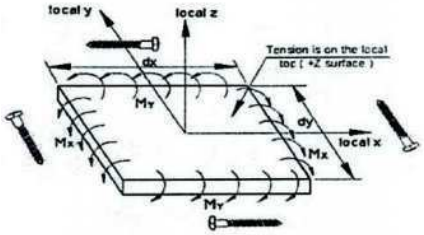


Figure 1-22: Sign convention for plate bending



Sign convention :
Local Z axis of plate is perpendicular to surface & positive towards upward direction.

Moment :
Mx : BM along X direction, +ve means tension on +Z surface of plate
My : BM along Y direction, +ve means tension on +Z surface of plate
For -ve Mx & My it is vice-virsa

Shear force :
SQx : Shear at face perpendicular to X axis
SQy : Shear at face perpendicular to Y axis

Axial force :
+ve Sx means tension along local X axis of plate & -ve Sx means compression along local X axis. for Sy this is similar.

Reinforcement in Foundation:

PLATE CENTER STRESS STAAD GROUP : RAFT_COL1_MOMENT At the face of the column

Load condition	Plate	Shear		Membrane			Bending Moment			M + I M_{xy} I		L/C
		SQX (local) Mpa	SQY(local)MPa	SX (local) MPa	SY (local) MPa	SXY (local) MPa	Mx (KN-m/m)	My (KN-m/m)	Mxy (KN-m/m)	Mx (KN-m/m)	My (KN-m/m)	
Max Qx	11948	0.935	0.028	0	0	0	864.5	-405.1	-128.4	992.9	-533.5	104 1.4X[D] + 1.7X[L + H1C + H3C + H5C]
Min Qx	10897	-0.935	0.028	0	0	0	864.5	-404.7	0	864.5	-404.7	104 1.4X[D] + 1.7X[L + H1C + H3C + H5C]
Max Qy	11964	0.438	0.798	0	0	0	-1008	724.7	-28.93	-1037	753.6	104 1.4X[D] + 1.7X[L + H1C + H3C + H5C]
Min Qy	11253	-0.436	-0.78	0	0	0	-996.7	757.6	-55.48	-1052	813.1	104 1.4X[D] + 1.7X[L + H1C + H3C + H5C]
Max Sx	12041	0.238	-0.128	0	0	0	-625.1	-214.6	255.8	-880.9	-470.3	101 1.4X[D]
Min Sx	12041	0.238	-0.128	0	0	0	-625.1	-214.6	255.8	-880.9	-470.3	101 1.4X[D]
Max Sy	12041	0.238	-0.128	0	0	0	-625.1	-214.6	255.8	-880.9	-470.3	101 1.4X[D]
Min Sy	12041	0.238	-0.128	0	0	0	-625.1	-214.6	255.8	-880.9	-470.3	101 1.4X[D]
Max Sxy	12041	0.238	-0.128	0	0	0	-625.1	-214.6	255.8	-880.9	-470.3	101 1.4X[D]
Min Sxy	12041	0.238	-0.128	0	0	0	-625.1	-214.6	255.8	-880.9	-470.3	101 1.4X[D]
Max Mx	11952	0.446	0.143	0	0	0	1776	1046	72.25	1848	1118	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Min Mx	11214	-0.71	0.371	0	0	0	-2206	-700.2	387.1	-2593	-1087	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Max My	10888	-0.401	-0.159	0	0	0	170.3	1791	27.08	197.4	1818	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Min My	11944	-0.166	0.066	0	0	0	-1184	-872.9	202.9	-1386	-1076	104 1.4X[D] + 1.7X[L + H1C + H3C + H5C]
Max Mxy	12041	0.778	-0.375	0	0	0	-2081	-577.9	644.8	-2726	-1223	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Min Mxy	11716	-0.778	-0.375	0	0	0	-2081	-577.9	-644.8	-2726	-1223	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]

PLATE CENTER STRESS STAAD GROUP : RAFT_COL1_SHEAR At a distance "d" from the face of the column

Load condition	Plate	Shear		Membrane			Bending Moment			M + I M_{xy} I		L/C
		SQX (local) Mpa	SQY(local)MPa	SX (local) MPa	SY (local) MPa	SXY (local) MPa	Mx (KN-m/m)	My (KN-m/m)	Mxy (KN-m/m)	Mx (KN-m/m)	My (KN-m/m)	
Max Qx	11906	0.802	0.376	0	0	0	-2080	-643.7	-572.7	-2653	-1216	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Min Qx	11210	-0.805	0.376	0	0	0	-2082	-641.5	0	-2082	-641.5	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Max Qy	11958	0.396	0.379	0	0	0	-1528	107.8	-268.3	-1796	376.1	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Min Qy	11716	-0.778	-0.375	0	0	0	-2081	-577.9	-644.8	-2726	-1223	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Max Sx	12041	0.238	-0.128	0	0	0	-625.1	-214.6	255.8	-880.9	-470.3	101 1.4X[D]
Min Sx	12041	0.238	-0.128	0	0	0	-625.1	-214.6	255.8	-880.9	-470.3	101 1.4X[D]
Max Sy	12041	0.238	-0.128	0	0	0	-625.1	-214.6	255.8	-880.9	-470.3	101 1.4X[D]
Min Sy	12041	0.238	-0.128	0	0	0	-625.1	-214.6	255.8	-880.9	-470.3	101 1.4X[D]
Max Sxy	12041	0.238	-0.128	0	0	0	-625.1	-214.6	255.8	-880.9	-470.3	101 1.4X[D]
Min Sxy	12041	0.238	-0.128	0	0	0	-625.1	-214.6	255.8	-880.9	-470.3	101 1.4X[D]
Max Mx	11952	0.446	0.143	0	0	0	1776	1046	72.25	1848	1118	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Min Mx	11214	-0.71	0.371	0	0	0	-2206	-700.2	387.1	-2593	-1087	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Max My	10888	-0.401	-0.159	0	0	0	170.3	1791	27.08	197.4	1818	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Min My	11818	0.404	0.192	0	0	0	-1654	-760.1	62.02	-1716	-822.1	110 1.4X[D] + 1.7X[L + H2C + H3C + H5C]
Max Mxy	12041	0.778	-0.375	0	0	0	-2081	-577.9	644.8	-2726	-1223	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Min Mxy	11716	-0.778	-0.375	0	0	0	-2081	-577.9	-644.8	-2726	-1223	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]

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Material Property :

Grade of concrete =		fc'35	
Grade of reinf. =		fy420	
Clear cover =	bottom face	75	mm
	top face	50	mm
Thicknes of slab =		1850	mm

Design Constants :

φ for Moment =	0.9
φ for shear =	0.75
β_1	0.8
ε_t limit (for tension controlled section) =	0.005
Concrete compressive strain, ε_{cu} =	0.003

Reinforcement Along X:

Force Values			Depth		Reinforcement Calculation								
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	$a=f_y/0.85f_c' bA_s$	A_s required	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,}$ provided	
	KN-m/m		mm	mm	KN-m/m	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	
Max Mx	1848	top face	50	1784	1848		2771		32	241.61	200	4022.9	
Min Mx	45	top face	50	1784	45	0.014	66.7	3330	32	241.61	200	4022.9	
Max Mxy	-2726	bottom face	75	1757	2726		4174		35.7	239.88	200	5007	[i.e., 32-200 + 16-200 (additional)]
Min Mxy	-2726	bottom face	75	1757	2726	*As	4175		35.7	239.88	200	5007	[i.e., 32-200 + 16-200 (additional)]

Shear Check		Qx	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{ca}	$V_{c,design}$	$\varphi(V_c+0.66\sqrt{f_c'}bd)\geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa	KN	KN-m/m		m^2	KN	KN	KN	KN	KN			KN		
Load condition	Max Qx	0.80	1483.7	2653	0.002	1.784	1757	1757	fc'3061	1756.9	6542	> 1484	ok	1318	φVc<Vu	required
	Min Qx	0.81	1489.3	2082			1776			1757.1	6542	> 1489	ok	1318	φVc<Vu	required

Nominal shear strength required by Reinf. Vs= 228.8 KN SECTION 11.2

Spacing Required of Strrups, s= 324 mm SECTION 11.5.6.2
or 879 mm SECTION 11.5.4.1
or 1388 mm

Hence, spacing required = 324 mm

Diameter of Shear Link considered = 16 mm
Spacing : Along X-X = 200 mm
Along Y-Y = 200 mm OK (Provided D20 Shear Links in drawing)

Reinforcement Along Y:

Force Values			Depth		Reinforcement Calculation								
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	$a=f_y/0.85f_c' bA_s$	A_s required	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,}$ provided	
	KN-m/m		mm	mm	KN-m/m	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	
Max My	1818	top face	50	1784	1818		2726		32	241.61	200	4022.9	
Min My	-1076	bottom face	75	1759	1076	0.014	1629	3330	32	241.61	200	4022.9	
Max Mxy	-1223	bottom face	75	1757	1223		1855		35.7	300.72	200	5007	[i.e., 32-200 + 16-200 (additional)]
Min Mxy	-1223	bottom face	75	1757	1223	*As	1855		35.7	300.72	200	5007	[i.e., 32-200 + 16-200 (additional)]

Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	$\varphi(V_c+0.66\sqrt{f_c'}bd)\geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa		KN	KN-m/m		m^2	KN	KN	KN	KN	KN			KN		
Load condition	Max Qy	0.38	1	701	376.1	0.002	1.784	1916	1757	3061	1757.1	1840	> 701	ok	1318	φVc>Vu	not required
	Min Qy	0.38	1	694	1223			1758			1757.1	1840	>694	ok	1318	φVc>Vu	not required

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• **Area Below Column C2:**

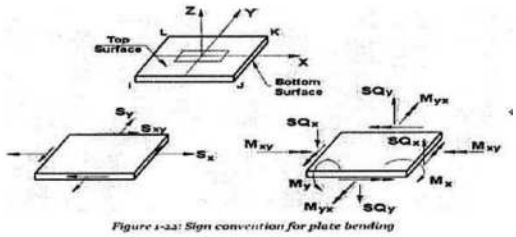
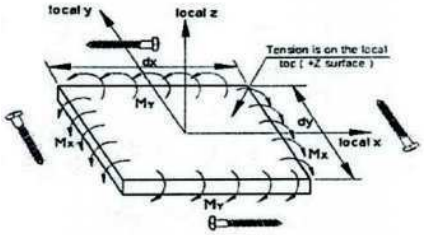


Figure 1-22: Sign convention for plate bending



Sign convention :
Local Z axis of plate is perpendicular to surface & positive towards upward direction.

Moment :
Mx : BM along X direction, +ve means tension on +Z surface of plate
My : BM along Y direction, +ve means tension on +Z surface of plate
For -ve Mx & My it is vice-virsa

Shear force :
SQx : Shear at face perpendicular to X axis
SQy : Shear at face perpendicular to Y axis

Axial force :
+ve Sx means tension along local X axis of plate & -ve Sx means compression along local X axis. for Sy this is similar.

Reinforcement in Foundation:

PLATE CENTER STRESS STAAD GROUP : RAFT_COL2_MOMENT At the face of the column

Load condition	Plate	Shear		Membrane			Bending Moment			M + I M_{xy} I		L/C
		SQX (local) Mpa	SQY(local)MPa	SX (local) MPa	SY (local) MPa	SXY (local) MPa	Mx (KN-m/m)	My (KN-m/m)	Mxy (KN-m/m)	Mx (KN-m/m)	My (KN-m/m)	
Max Qx	10955	1.532	0.188	0	0	0	-324.2	-1302	344.6	-668.8	-1646	104 1.4X[D] + 1.7X[L + H1C + H3C + H5C]
Min Qx	11821	-1.532	0.188	0	0	0	-324.3	-1302	0	-324.3	-1302	104 1.4X[D] + 1.7X[L + H1C + H3C + H5C]
Max Qy	11702	-0.321	1.422	0	0	0	-2177	268.7	-520.4	-2698	789.1	104 1.4X[D] + 1.7X[L + H1C + H3C + H5C]
Min Qy	11683	-0.329	-1.446	0	0	0	-2270	239	515.9	-2786	754.9	104 1.4X[D] + 1.7X[L + H1C + H3C + H5C]
Max Sx	11919	-0.016	-0.041	0	0	0	468	398.2	-26.31	494.4	424.6	101 1.4X[D]
Min Sx	11919	-0.016	-0.041	0	0	0	468	398.2	-26.31	494.4	424.6	101 1.4X[D]
Max Sy	11919	-0.016	-0.041	0	0	0	468	398.2	-26.31	494.4	424.6	101 1.4X[D]
Min Sy	11919	-0.016	-0.041	0	0	0	468	398.2	-26.31	494.4	424.6	101 1.4X[D]
Max Sxy	11919	-0.016	-0.041	0	0	0	468	398.2	-26.31	494.4	424.6	101 1.4X[D]
Min Sxy	11919	-0.016	-0.041	0	0	0	468	398.2	-26.31	494.4	424.6	101 1.4X[D]
Max Mx	10954	0.464	0.12	0	0	0	1837	961.2	71.37	1908	1033	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Min Mx	11686	-0.321	1.308	0	0	0	-2453	415.3	-421.4	-2875	836.8	104 1.4X[D] + 1.7X[L + H1C + H3C + H5C]
Max My	10979	0.144	-0.244	0	0	0	150.5	2100	-66.56	217	2166	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Min My	11820	-1.371	-0.186	0	0	0	-399.3	-1352	483.1	-882.4	-1835	135 0.9X[D] + 1.6X[W1B + H1C + H3C + H5C]
Max Mxy	11706	0.912	0.732	0	0	0	-242.6	237.3	957.4	-1200	1195	104 1.4X[D] + 1.7X[L + H1C + H3C + H5C]
Min Mxy	11916	-0.912	0.732	0	0	0	-242.6	237.3	-957.4	-1200	1195	104 1.4X[D] + 1.7X[L + H1C + H3C + H5C]

PLATE CENTER STRESS STAAD GROUP : RAFT_COL2_SHEAR At a distance "d" from the face of the column

Load condition	Plate	Shear		Membrane			Bending Moment			M + I M_{xy} I		L/C
		SQX (local) Mpa	SQY(local)MPa	SX (local) MPa	SY (local) MPa	SXY (local) MPa	Mx (KN-m/m)	My (KN-m/m)	Mxy (KN-m/m)	Mx (KN-m/m)	My (KN-m/m)	
Max Qx	10954	0.798	0.137	0	0	0	1179	-192.5	75.13	1254	-267.6	104 1.4X[D] + 1.7X[L + H1C + H3C + H5C]
Min Qx	11823	-0.798	0.137	0	0	0	1179	-192.5	0	1179	-192.5	104 1.4X[D] + 1.7X[L + H1C + H3C + H5C]
Max Qy	11703	-0.204	0.584	0	0	0	-1106	1304	-253.9	-1359	1558	104 1.4X[D] + 1.7X[L + H1C + H3C + H5C]
Min Qy	10945	0.211	-0.592	0	0	0	-1207	1316	-242.7	-1449	1559	104 1.4X[D] + 1.7X[L + H1C + H3C + H5C]
Max Sx	11919	-0.016	-0.041	0	0	0	468	398.2	-26.31	494.4	424.6	101 1.4X[D]
Min Sx	11919	-0.016	-0.041	0	0	0	468	398.2	-26.31	494.4	424.6	101 1.4X[D]
Max Sy	11919	-0.016	-0.041	0	0	0	468	398.2	-26.31	494.4	424.6	101 1.4X[D]
Min Sy	11919	-0.016	-0.041	0	0	0	468	398.2	-26.31	494.4	424.6	101 1.4X[D]
Max Sxy	11919	-0.016	-0.041	0	0	0	468	398.2	-26.31	494.4	424.6	101 1.4X[D]
Min Sxy	11919	-0.016	-0.041	0	0	0	468	398.2	-26.31	494.4	424.6	101 1.4X[D]
Max Mx	10954	0.464	0.12	0	0	0	1837	961.2	71.37	1908	1033	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Min Mx	11687	-0.232	0.382	0	0	0	-1591	1465	-112.3	-1704	1577	104 1.4X[D] + 1.7X[L + H1C + H3C + H5C]
Max My	10979	0.144	-0.244	0	0	0	150.5	2100	-66.56	217	2166	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Min My	11822	-0.575	-0.071	0	0	0	728.3	-389.5	226.7	955.1	-616.3	159 0.9X[D] + 1.0X[E1B] + 1.6X 0.8X[H1C + H3C + H5C]
Max Mxy	11707	0.56	0.206	0	0	0	1056	320.9	621.2	1677	942.1	104 1.4X[D] + 1.7X[L + H1C + H3C + H5C]
Min Mxy	11917	-0.56	0.206	0	0	0	1056	320.9	-621.2	1677	942.1	104 1.4X[D] + 1.7X[L + H1C + H3C + H5C]

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Material Property :

Grade of concrete =		fc'35	
Grade of reinf. =		fy420	
Clear cover =	bottom face	75	mm
	top face	50	mm
Thicknes of slab =		1850	mm

Design Constants :

φ for Moment =	0.9
φ for shear =	0.75
β_1	0.8
ε_t limit (for tension controlled section) =	0.005
Concrete compressive strain, ε_{cu} =	0.003

Reinforcement Along X:

Force Values			Depth		Reinforcement Calculation								
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	$a=f_y/0.85f_c' bA_s$	A_s required	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,}$ provided	
	KN-m/m		mm	mm	KN-m/m	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	
Max Mx	1908	top face	50	1784	1908		2862		32	241.61	200	4022.9	
Min Mx	45	top face	50	1784	45	0.014	66.7	3330	32	241.61	200	4022.9	
Max Mxy	-1200	bottom face	75	1757	1200		1820		35.7	300.72	200	5007	[i.e., 32-200 + 16-200 (additional)]
Min Mxy	-1200	bottom face	75	1757	1200	*As	1820		35.7	300.72	200	5007	[i.e., 32-200 + 16-200 (additional)]

Shear Check		Qx	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{ca}	$V_{c,design}$	$\varphi(V_c+0.66\sqrt{f_c'}bd)\geq V_u$		Rema rks	φV_c	Remar ks	Remarks :Min shear reinf
		Mpa	KN	KN-m/m		m^2	KN	KN	KN	KN	KN			KN		
Load condition	Max Qx	0.80	1476.3	1254	0.002	1.784	1832	1757	fc'3061	1757.1	6542	> 1476	ok	1318	φVc<Vu	required
	Min Qx	0.80	1476.3	1179			1841			1757.1	6542	> 1476	ok	1318	φVc<Vu	required

Nominal shear strength required by Reinf. Vs= 211.3 KN SECTION 11.2

Spacing Required of Strrups, s= 351 mm SECTION 11.5.6.2
or 879 mm SECTION 11.5.4.1
or 1388 mm

Hence, spacing required = 351 mm

Diameter of Shear Link considered = 16 mm
Spacing : Along X-X = 200 mm
Along Y-Y = 200 mm OK [Provided D20 Shear Links in drawing]

Reinforcement Along Y:

Force Values			Depth		Reinforcement Calculation								
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	$a=f_y/0.85f_c' bA_s$	A_s required	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,}$ provided	
	KN-m/m		mm	mm	KN-m/m	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	
Max My	2166	top face	50	1784	2166		3254		32	241.61	200	4022.9	
Min My	-1835	bottom face	75	1759	1835	0.014	2791	3330	32	241.61	200	4022.9	
Max Mxy	1195	top face	50	1782	1195		1786		35.7	300.72	200	5007	[i.e., 32-200 + 16-200 (additional)]
Min Mxy	1195	top face	50	1782	1195	*As	1786		35.7	300.72	200	5007	[i.e., 32-200 + 16-200 (additional)]

Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	$\varphi(V_c+0.66\sqrt{f_c'}bd)\geq V_u$		Rema rks	φV_c	Remar ks	Remarks :Min shear reinf
		Mpa		KN	KN-m/m		m^2	KN	KN	KN	KN	KN			KN		
Load condition	Max Qy	0.58	1	1080	1558	0.002	1.784	1773	1757	3061	1757.1	1840	> 1080	ok	1318	φVc>Vu	not required
	Min Qy	0.59	1	1095	1559			1774			1757.1	1840	>1095	ok	1318	φVc>Vu	not required

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• **Area Below Column C3:**

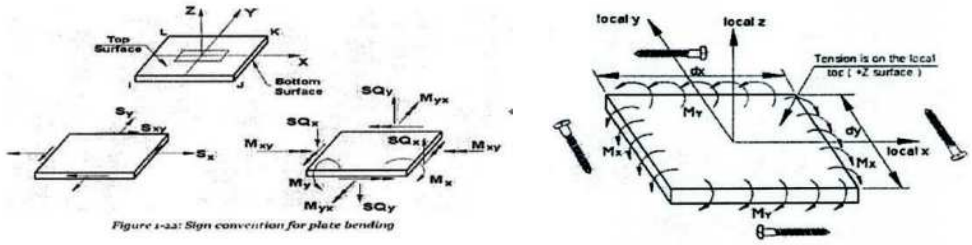


Figure 1-22: Sign convention for plate bending

Sign convention :
Local Z axis of plate is perpendicular to surface & positive towards upward direction.

Moment :
Mx : BM along X direction, +ve means tension on +Z surface of plate
My : BM along Y direction, +ve means tension on +Z surface of plate
For -ve Mx & My it is vice-virsa

Shear force :
SQx : Shear at face perpendicular to X axis
SQy : Shear at face perpendicular to Y axis

Axial force :
+ve Sx means tension along local X axis of plate & -ve Sx means compression along local X axis. for Sy this is similar.

Reinforcement in Foundation:

PLATE CENTER STRESS STAAD GROUP : RAFT_COL3_MOMENT At the face of the column

Load condition	Plate	Shear		Membrane			Bending Moment			M + I M_{xy} I		L/C
		SQX (local) Mpa	SQY(local)MPa	SX (local) MPa	SY (local) MPa	SXY (local) MPa	Mx (KN-m/m)	My (KN-m/m)	Mxy (KN-m/m)	Mx (KN-m/m)	My (KN-m/m)	
Max Qx	10920	0.628	0.226	0	0	0	6.256	-359.1	-38.71	44.97	-397.8	104 1.4X[D] + 1.7X[L + H1C + H3C + H5C]
Min Qx	11801	-0.628	0.226	0	0	0	5.759	-359.1	0	5.759	-359.1	104 1.4X[D] + 1.7X[L + H1C + H3C + H5C]
Max Qy	11679	-0.575	1.167	0	0	0	-1267	192.2	-233.5	-1500	425.7	104 1.4X[D] + 1.7X[L + H1C + H3C + H5C]
Min Qy	11738	0.528	-1.104	0	0	0	-1046	160.1	-194.8	-1241	354.9	104 1.4X[D] + 1.7X[L + H1C + H3C + H5C]
Max Sx	11939	0.004	-0.155	0	0	0	319.9	16.41	74.35	394.2	90.76	101 1.4X[D]
Min Sx	11939	0.004	-0.155	0	0	0	319.9	16.41	74.35	394.2	90.76	101 1.4X[D]
Max Sy	11939	0.004	-0.155	0	0	0	319.9	16.41	74.35	394.2	90.76	101 1.4X[D]
Min Sy	11939	0.004	-0.155	0	0	0	319.9	16.41	74.35	394.2	90.76	101 1.4X[D]
Max Sxy	11939	0.004	-0.155	0	0	0	319.9	16.41	74.35	394.2	90.76	101 1.4X[D]
Min Sxy	11939	0.004	-0.155	0	0	0	319.9	16.41	74.35	394.2	90.76	101 1.4X[D]
Max Mx	11755	0.061	-0.366	0	0	0	1241	863.7	-85.07	1326	948.8	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Min Mx	11679	-0.575	1.167	0	0	0	-1267	192.2	-233.5	-1500	425.7	104 1.4X[D] + 1.7X[L + H1C + H3C + H5C]
Max My	11680	-0.182	0.496	0	0	0	-884.3	1169	-73.22	-957.5	1242	110 1.4X[D] + 1.7X[L + H2C + H3C + H5C]
Min My	11862	0.196	-0.958	0	0	0	-157.6	-2798	379.3	-536.8	-3177	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Max Mxy	11384	-0.107	0.981	0	0	0	-231	-2616	399.2	-630.2	-3015	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Min Mxy	11700	0.107	0.979	0	0	0	-233.3	-2615	-400.3	-633.6	-3016	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]

PLATE CENTER STRESS STAAD GROUP : RAFT_COL3_SHEAR At a distance "d" from the face of the column

Load condition	Plate	Shear		Membrane			Bending Moment			M + I M_{xy} I		L/C
		SQX (local) Mpa	SQY(local)MPa	SX (local) MPa	SY (local) MPa	SXY (local) MPa	Mx (KN-m/m)	My (KN-m/m)	Mxy (KN-m/m)	Mx (KN-m/m)	My (KN-m/m)	
Max Qx	10938	0.348	0.322	0	0	0	-53.08	1004	53.56	-106.6	1057	104 1.4X[D] + 1.7X[L + H1C + H3C + H5C]
Min Qx	11803	-0.348	0.322	0	0	0	-53.33	1004	0	-53.33	1004	104 1.4X[D] + 1.7X[L + H1C + H3C + H5C]
Max Qy	11699	0.006	0.843	0	0	0	-345.2	-2377	-109	-454.2	-2486	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Min Qy	11793	-0.062	-0.821	0	0	0	-159.5	-2517	-102.9	-262.5	-2620	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Max Sx	11939	0.004	-0.155	0	0	0	319.9	16.41	74.35	394.2	90.76	101 1.4X[D]
Min Sx	11939	0.004	-0.155	0	0	0	319.9	16.41	74.35	394.2	90.76	101 1.4X[D]
Max Sy	11939	0.004	-0.155	0	0	0	319.9	16.41	74.35	394.2	90.76	101 1.4X[D]
Min Sy	11939	0.004	-0.155	0	0	0	319.9	16.41	74.35	394.2	90.76	101 1.4X[D]
Max Sxy	11939	0.004	-0.155	0	0	0	319.9	16.41	74.35	394.2	90.76	101 1.4X[D]
Min Sxy	11939	0.004	-0.155	0	0	0	319.9	16.41	74.35	394.2	90.76	101 1.4X[D]
Max Mx	11755	0.061	-0.366	0	0	0	1241	863.7	-85.07	1326	948.8	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Min Mx	11675	-0.115	0.118	0	0	0	-968.7	-1109	161.2	-1130	-1270	104 1.4X[D] + 1.7X[L + H1C + H3C + H5C]
Max My	11680	-0.182	0.496	0	0	0	-884.3	1169	-73.22	-957.5	1242	110 1.4X[D] + 1.7X[L + H2C + H3C + H5C]
Min My	11793	-0.062	-0.821	0	0	0	-159.5	-2517	-102.9	-262.5	-2620	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Max Mxy	11386	0.002	0.763	0	0	0	333.3	-1467	285.8	619.1	-1753	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Min Mxy	11872	-0.002	0.763	0	0	0	332.2	-1467	-286.4	618.6	-1754	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]

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Material Property :

Grade of concrete =		fc'35	
Grade of reinf. =		fy420	
Clear cover =	bottom face	75	mm
	top face	50	mm
Thicknes of slab =		1850	mm

Design Constants :

φ for Moment =	0.9
φ for shear =	0.75
β_1	0.8
ε_t limit (for tension controlled section) =	0.005
Concrete compressive strain, ε_{cu} =	0.003

Reinforcement Along X:

Force Values			Depth		Reinforcement Calculation								
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	$a=f_y/0.85f_c' bA_s$	A_s required	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,}$ provided	
	KN-m/m		mm	mm	KN-m/m	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	
Max Mx	1326	top face	50	1784	1326		1982		32	241.61	200	4022.9	
Min Mx	45	top face	50	1784	45	0.014	66.7	3330	32	241.61	200	4022.9	
Max Mxy	-630	bottom face	75	1757	630.2		952		35.7	300.72	200	5007	[i.e., 32-200 + 16-200 (additional)]
Min Mxy	-634	bottom face	75	1757	633.6	*As	958		35.7	300.72	200	5007	[i.e., 32-200 + 16-200 (additional)]

Shear Check		Qx	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{ca}	$V_{c,design}$	$\varphi(V_c+0.66\sqrt{f_c'}bd)\geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa	KN	KN-m/m		m^2	KN	KN	KN	KN	KN			KN		
Load condition	Max Qx	0.35	643.8	106.6	0.002	1.784	2425	1757	fc'3061	1757.1	6542	> 644	ok	1318	φVc>Vu	not required
	Min Qx	0.35	643.8	53.33			3161			1757.1	6542	> 644	ok	1318	φVc>Vu	not required

Reinforcement Along Y:

Force Values			Depth		Reinforcement Calculation								
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	$a=f_y/0.85f_c' bA_s$	A_s required	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,}$ provided	
	KN-m/m		mm	mm	KN-m/m	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	
Max My	1242	top face	50	1784	1242		1856		32	241.61	200	4022.9	
Min My	-3177	bottom face	75	1757	3177	0.014	4879	3330	35.7	205.25	200	5007	
Max Mxy	-3015	bottom face	75	1757	3015		4625		35.7	216.5	200	5007	[i.e., 32-200 + 16-200 (additional)]
Min Mxy	-3016	bottom face	75	1757	3016	*As	4626		35.7	216.47	200	5007	[i.e., 32-200 + 16-200 (additional)]

Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	$\varphi(V_c+0.66\sqrt{f_c'}bd)\geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa		KN	KN-m/m		m^2	KN	KN	KN	KN	KN			KN		
Load condition	Max Qy	0.84	1	1560	2486	0.002	1.784	1765	1757	3061	1757.1	1840	> 1560	ok	1318	φVc<Vu	required
	Min Qy	0.82	1	1519	2620			1759			1757.1	1840	>1519	ok	1318	φVc<Vu	required

Nominal shear strength required by Reinf. Vs= 322.3 KN SECTION 11.2

Spacing Required of Strrups, s= 230 mm SECTION 11.5.6.2

or 879 mm SECTION 11.5.4.1

or 1388 mm

Hence, spacing required = 230 mm

Diameter of Shear Link Provided = 16 mm

Spacing Provided : Along X-X = 200 mm

Along Y-Y = 200 mm OK [Provided D20 Shear Links in drawing]

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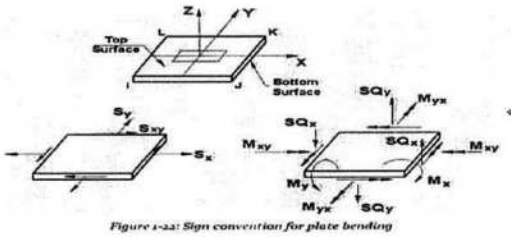
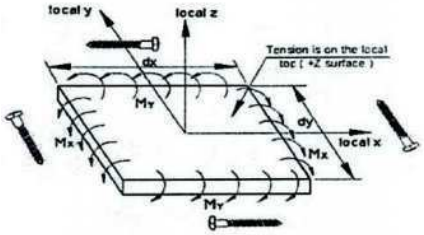


Figure 1-22: Sign convention for plate bending



Sign convention :
Local Z axis of plate is perpendicular to surface & positive towards upward direction.

Moment :
Mx : BM along X direction, +ve means tension on +Z surface of plate
My : BM along Y direction, +ve means tension on +Z surface of plate
For -ve Mx & My it is vice-virsa

Shear force :
SQx : Shear at face perpendicular to X axis
SQy : Shear at face perpendicular to Y axis

Axial force :
+ve Sx means tension along local X axis of plate & -ve Sx means compression along local X axis. for Sy this is similar.

Reinforcement in Foundation:

PLATE CENTER STRESS STAAD GROUP : RAFT_CANT_MOM At the face of the column

Load condition	Plate	Shear		Membrane			Bending Moment			M + I M_{xy} I		L/C
		SQX (local) Mpa	SQY(local)MPa	SX (local) MPa	SY (local) MPa	SXY (local) MPa	Mx (KN-m/m)	My (KN-m/m)	Mxy (KN-m/m)	Mx (KN-m/m)	My (KN-m/m)	
Max Qx	11604	0.706	-0.004	0	0	0	-1042	-756.7	122.3	-1164	-878.9	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Min Qx	12923	-0.168	-0.186	0	0	0	75.55	-177.7	0	75.55	-177.7	106 1.4X[D] + 1.7X[L + H1B + H4B + H6B]
Max Qy	11660	0.573	0.204	0	0	0	-895.4	-20.17	-76.16	-971.5	-96.33	106 1.4X[D] + 1.7X[L + H1B + H4B + H6B]
Min Qy	12950	0.024	-0.228	0	0	0	-602.7	-193.7	58.63	-661.3	-252.3	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Max Sx	12956	0.01	-0.049	0	0	0	159.8	-56.35	2.818	162.6	-59.17	101 1.4X[D]
Min Sx	12956	0.01	-0.049	0	0	0	159.8	-56.35	2.818	162.6	-59.17	101 1.4X[D]
Max Sy	12956	0.01	-0.049	0	0	0	159.8	-56.35	2.818	162.6	-59.17	101 1.4X[D]
Min Sy	12956	0.01	-0.049	0	0	0	159.8	-56.35	2.818	162.6	-59.17	101 1.4X[D]
Max Sxy	12956	0.01	-0.049	0	0	0	159.8	-56.35	2.818	162.6	-59.17	101 1.4X[D]
Min Sxy	12956	0.01	-0.049	0	0	0	159.8	-56.35	2.818	162.6	-59.17	101 1.4X[D]
Max Mx	12885	-0.033	-0.153	0	0	0	589.8	-178.9	5.274	595.1	-184.2	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Min Mx	11608	0.686	0.086	0	0	0	-1069	-622	77.68	-1146	-699.7	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Max My	11592	0.466	0.066	0	0	0	-827.4	579.1	-45.29	-872.7	624.4	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Min My	11654	0.658	0.065	0	0	0	-920.6	-941.6	-174.1	-1095	-1116	104 1.4X[D] + 1.7X[L + H1C + H3C + H5C]
Max Mxy	11604	0.661	-0.008	0	0	0	-951.1	-911.2	173	-1124	-1084	104 1.4X[D] + 1.7X[L + H1C + H3C + H5C]
Min Mxy	11580	0.632	0.022	0	0	0	-925.4	-789.5	-177.3	-1103	-966.8	110 1.4X[D] + 1.7X[L + H2C + H3C + H5C]

PLATE CENTER STRESS STAAD GROUP : RAFT_CANT_SHEAR At a distance "d" from the face of the column

Load condition	Plate	Shear		Membrane			Bending Moment			M + I M_{xy} I		L/C
		SQX (local) Mpa	SQY(local)MPa	SX (local) MPa	SY (local) MPa	SXY (local) MPa	Mx (KN-m/m)	My (KN-m/m)	Mxy (KN-m/m)	Mx (KN-m/m)	My (KN-m/m)	
Max Qx	12954	0.168	-0.185	0	0	0	75.73	-177.7	31.1	106.8	-208.8	106 1.4X[D] + 1.7X[L + H1B + H4B + H6B]
Min Qx	12923	-0.168	-0.186	0	0	0	75.55	-177.7	0	75.55	-177.7	106 1.4X[D] + 1.7X[L + H1B + H4B + H6B]
Max Qy	12956	0	-0.011	0	0	0	77.53	-14.83	2.862	80.39	-17.69	119 1.2X[D] + 1.0X[E1A] + 1.0X[L]
Min Qy	12950	0.024	-0.228	0	0	0	-602.7	-193.7	58.63	-661.3	-252.3	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Max Sx	12956	0.01	-0.049	0	0	0	159.8	-56.35	2.818	162.6	-59.17	101 1.4X[D]
Min Sx	12956	0.01	-0.049	0	0	0	159.8	-56.35	2.818	162.6	-59.17	101 1.4X[D]
Max Sy	12956	0.01	-0.049	0	0	0	159.8	-56.35	2.818	162.6	-59.17	101 1.4X[D]
Min Sy	12956	0.01	-0.049	0	0	0	159.8	-56.35	2.818	162.6	-59.17	101 1.4X[D]
Max Sxy	12956	0.01	-0.049	0	0	0	159.8	-56.35	2.818	162.6	-59.17	101 1.4X[D]
Min Sxy	12956	0.01	-0.049	0	0	0	159.8	-56.35	2.818	162.6	-59.17	101 1.4X[D]
Max Mx	12885	-0.033	-0.153	0	0	0	589.8	-178.9	5.274	595.1	-184.2	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Min Mx	12926	-0.09	-0.21	0	0	0	-759.9	-166.7	-85.14	-845.1	-251.9	104 1.4X[D] + 1.7X[L + H1C + H3C + H5C]
Max My	12886	-0.003	-0.013	0	0	0	46.45	-14.82	7.369	53.82	-22.19	119 1.2X[D] + 1.0X[E1A] + 1.0X[L]
Min My	12946	-0.068	-0.202	0	0	0	7.482	-201.5	-6.821	14.3	-208.3	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Max Mxy	12951	0.09	-0.21	0	0	0	-759.4	-166.7	85.4	-844.8	-252.1	110 1.4X[D] + 1.7X[L + H2C + H3C + H5C]
Min Mxy	12926	-0.09	-0.21	0	0	0	-759.4	-166.8	-85.14	-844.6	-251.9	110 1.4X[D] + 1.7X[L + H2C + H3C + H5C]

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Material Property :

Grade of concrete =		fc'35	
Grade of reinf. =		fy420	
Clear cover =	bottom face	75	mm
	top face	50	mm
Thicknes of slab =		1850	mm

Design Constants :

φ for Moment =	0.9
φ for shear =	0.75
β_1	0.8
ε_t limit (for tension controlled section) =	0.005
Concrete compressive strain, ε_{cu} =	0.003

Reinforcement Along X:

Force Values			Depth		Reinforcement Calculation							
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	$a=f_y/0.85f_c' bA_s$	A_s required	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,}$ provided
	KN-m/m		mm	mm	KN-m/m	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2
Max Mx	595	top face	50	1784	595.1	0.014	886	3330	32	241.61	200	4022.9
Min Mx	45	top face	50	1784	45		66.7		32	241.61	200	4022.9
Max Mxy	-1124	bottom face	75	1759	1124		1702		32	241.61	200	4022.9
Min Mxy	-1103	bottom face	75	1759	1103	*As	1670		32	241.61	200	4022.9

Shear Check		Qx	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{ca}	$V_{c,design}$	$\varphi(V_c+0.66\sqrt{f_c'}bd)\geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa	KN	KN-m/m		m^2	KN	KN	KN	KN	KN			KN		
Load condition	Max Qx	0.17	310.8	106.8	0.002	1.784	2044	1757	fc'3061	1757.1	6542	> 311	ok	1318	φVc>Vu	not required
	Min Qx	0.17	310.8	75.55			2191			1757.1	6542	> 311	ok	1318	φVc>Vu	not required

Reinforcement Along Y:

Force Values			Depth		Reinforcement Calculation							
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	$a=f_y/0.85f_c' bA_s$	A_s required	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,}$ provided
	KN-m/m		mm	mm	KN-m/m	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2
Max My	624	top face	50	1784	624.4	0.014	929	3330	32	241.61	200	4022.9
Min My	-1116	bottom face	75	1759	1116		1689		32	241.61	200	4022.9
Max Mxy	-1084	bottom face	75	1759	1084		1641		32	241.61	200	4022.9
Min Mxy	-967	bottom face	75	1759	966.8	*As	1463		32	241.61	200	4022.9

Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	$\varphi(V_c+0.66\sqrt{f_c'}bd)\geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa		KN	KN-m/m		m^2	KN	KN	KN	KN	KN			KN		
Load condition	Max Qy	0.01	1	20	17.69	0.002	1.784	1829	1757	3061	1757.1	1840	> 20	ok	1318	φVc>Vu	not required
	Min Qy	0.23	1	422	252.3			1893			1757.1	1840	>422	ok	1318	φVc>Vu	not required

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• **Area Raft Other (Part 1):**

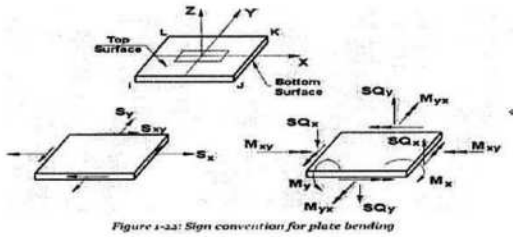
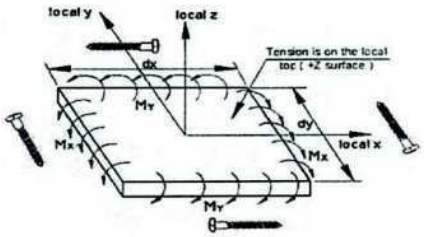


Figure 1-22: Sign convention for plate bending



Sign convention :
Local Z axis of plate is perpendicular to surface & positive towards upward direction.

Moment :
Mx : BM along X direction, +ve means tension on +Z surface of plate
My : BM along Y direction, +ve means tension on +Z surface of plate
For -ve Mx & My it is vice-virsa

Shear force :
SQx : Shear at face perpendicular to X axis
SQy : Shear at face perpendicular to Y axis

Axial force :
+ve Sx means tension along local X axis of plate & -ve Sx means compression along local X axis. for Sy this is similar.

Reinforcement in Foundation:

PLATE CENTER STRESS STAAD GROUP : RAFT_OTHER_1 At the face of the column

Load condition	Plate	Shear		Membrane			Bending Moment			M + I M_{xy} I		L/C
		SQX (local) Mpa	SQY(local)MPa	SX (local) MPa	SY (local) MPa	SXY (local) MPa	Mx (KN-m/m)	My (KN-m/m)	Mxy (KN-m/m)	Mx (KN-m/m)	My (KN-m/m)	
Max Qx	10953	0.272	0.061	0	0	0	1914	267.8	-62.35	1977	330.1	104 1.4X[D] + 1.7X[L + H1C + H3C + H5C]
Min Qx	11927	-0.272	0.061	0	0	0	1914	267.6	0	1914	267.6	104 1.4X[D] + 1.7X[L + H1C + H3C + H5C]
Max Qy	10929	-0.223	0.154	0	0	0	1750	1069	220.3	1971	1289	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Min Qy	11723	-0.229	-0.176	0	0	0	1664	1089	-180.9	1845	1270	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Max Sx	11975	0.074	-0.068	0	0	0	447.9	307.5	84.64	532.5	392.2	101 1.4X[D]
Min Sx	11975	0.074	-0.068	0	0	0	447.9	307.5	84.64	532.5	392.2	101 1.4X[D]
Max Sy	11975	0.074	-0.068	0	0	0	447.9	307.5	84.64	532.5	392.2	101 1.4X[D]
Min Sy	11975	0.074	-0.068	0	0	0	447.9	307.5	84.64	532.5	392.2	101 1.4X[D]
Max Sxy	11975	0.074	-0.068	0	0	0	447.9	307.5	84.64	532.5	392.2	101 1.4X[D]
Min Sxy	11975	0.074	-0.068	0	0	0	447.9	307.5	84.64	532.5	392.2	101 1.4X[D]
Max Mx	10953	0.092	0.079	0	0	0	2199	1205	-1.664	2200	1207	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Min Mx	11723	-0.067	-0.041	0	0	0	225.6	218	-120.1	345.7	338.1	119 1.2X[D] + 1.0X[E1A] + 1.0X[L]
Max My	10970	0.064	-0.037	0	0	0	2056	1710	-19.14	2075	1729	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Min My	11926	-0.197	0.014	0	0	0	1256	-53.25	121.3	1378	-174.5	159 0.9X[D] + 1.0X[E1B] + 1.6X 0.8X[H1C + H3C + H5C]
Max Mxy	11708	0.176	0.033	0	0	0	1697	536.5	316.5	2014	853	104 1.4X[D] + 1.7X[L + H1C + H3C + H5C]
Min Mxy	11934	-0.176	0.033	0	0	0	1697	536.5	-316.5	2014	853	104 1.4X[D] + 1.7X[L + H1C + H3C + H5C]

PLATE CENTER STRESS STAAD GROUP : RAFT_OTHER_1 At a distance "d" from the face of the column

Load condition	Plate	Shear		Membrane			Bending Moment			M + I M_{xy} I		L/C
		SQX (local) Mpa	SQY(local)MPa	SX (local) MPa	SY (local) MPa	SXY (local) MPa	Mx (KN-m/m)	My (KN-m/m)	Mxy (KN-m/m)	Mx (KN-m/m)	My (KN-m/m)	
Max Qx	10953	0.272	0.061	0	0	0	1914	267.8	-62.35	1977	330.1	104 1.4X[D] + 1.7X[L + H1C + H3C + H5C]
Min Qx	11927	-0.272	0.061	0	0	0	1914	267.6	0	1914	267.6	104 1.4X[D] + 1.7X[L + H1C + H3C + H5C]
Max Qy	10929	-0.223	0.154	0	0	0	1750	1069	220.3	1971	1289	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Min Qy	11723	-0.229	-0.176	0	0	0	1664	1089	-180.9	1845	1270	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Max Sx	11975	0.074	-0.068	0	0	0	447.9	307.5	84.64	532.5	392.2	101 1.4X[D]
Min Sx	11975	0.074	-0.068	0	0	0	447.9	307.5	84.64	532.5	392.2	101 1.4X[D]
Max Sy	11975	0.074	-0.068	0	0	0	447.9	307.5	84.64	532.5	392.2	101 1.4X[D]
Min Sy	11975	0.074	-0.068	0	0	0	447.9	307.5	84.64	532.5	392.2	101 1.4X[D]
Max Sxy	11975	0.074	-0.068	0	0	0	447.9	307.5	84.64	532.5	392.2	101 1.4X[D]
Min Sxy	11975	0.074	-0.068	0	0	0	447.9	307.5	84.64	532.5	392.2	101 1.4X[D]
Max Mx	10953	0.092	0.079	0	0	0	2199	1205	-1.664	2200	1207	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Min Mx	11723	-0.067	-0.041	0	0	0	225.6	218	-120.1	345.7	338.1	119 1.2X[D] + 1.0X[E1A] + 1.0X[L]
Max My	10970	0.064	-0.037	0	0	0	2056	1710	-19.14	2075	1729	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Min My	11926	-0.197	0.014	0	0	0	1256	-53.25	121.3	1378	-174.5	159 0.9X[D] + 1.0X[E1B] + 1.6X 0.8X[H1C + H3C + H5C]
Max Mxy	11708	0.176	0.033	0	0	0	1697	536.5	316.5	2014	853	104 1.4X[D] + 1.7X[L + H1C + H3C + H5C]
Min Mxy	11934	-0.176	0.033	0	0	0	1697	536.5	-316.5	2014	853	104 1.4X[D] + 1.7X[L + H1C + H3C + H5C]

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Material Property :

Grade of concrete =		fc'35	
Grade of reinf. =		fy420	
Clear cover =	bottom face	75	mm
	top face	50	mm
Thicknes of slab =		1850	mm

Design Constants :

φ for Moment =	0.9
φ for shear =	0.75
β_1	0.8
ε_t limit (for tension controlled section) =	0.005
Concrete compressive strain, ε_{cu} =	0.003

Reinforcement Along X:

Force Values			Depth		Reinforcement Calculation							
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	$a=f_y/0.85f_c' bA_s$	A_s required	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,}$ provided
	KN-m/m		mm	mm	KN-m/m	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2
Max Mx	2200	top face	50	1784	2200		3306		32	241.61	200	4022.9
Min Mx	45	top face	50	1784	45	0.014	66.7		32	241.61	200	4022.9
Max Mxy	2014	top face	50	1784	2014		3022	3330	32	241.61	200	4022.9
Min Mxy	2014	top face	50	1784	2014	*As	3023		32	241.61	200	4022.9

Shear Check		Qx	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{ca}	$V_{c,design}$	$\varphi(V_c+0.66\sqrt{f_c'}bd)\geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa	KN	KN-m/m		m^2	KN	KN	KN	KN	KN			KN		
Load condition	Max Qx	0.27	503.2	1977	0.002	1.784	1720	1757	fc'3061	1719.7	6514	> 503	ok	1290	φVc>Vu	not required
	Min Qx	0.27	503.2	1914			1721			1720.8	6515	> 503	ok	1291	φVc>Vu	not required

Reinforcement Along Y:

Force Values			Depth		Reinforcement Calculation							
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	$a=f_y/0.85f_c' bA_s$	A_s required	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,}$ provided
	KN-m/m		mm	mm	KN-m/m	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2
Max My	1729	top face	50	1784	1729		2590		32	241.61	200	4022.9
Min My	-175	bottom face	75	1759	174.5	0.014	263		32	241.61	200	4022.9
Max Mxy	853	top face	50	1784	853		1271	3330	32	241.61	200	4022.9
Min Mxy	853	top face	50	1784	853	*As	1271		32	241.61	200	4022.9

Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	$\varphi(V_c+0.66\sqrt{f_c'}bd)\geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa		KN	KN-m/m		m^2	KN	KN	KN	KN	KN			KN		
Load condition	Max Qy	0.15	1	285	1289	0.002	1.784	1716	1757	3061	1715.7	1809	> 285	ok	1287	φVc>Vu	not required
	Min Qy	0.18	1	326	1270			1720			1720	1812	>326	ok	1290	φVc>Vu	not required

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• **Area Raft Other (Part 2):**

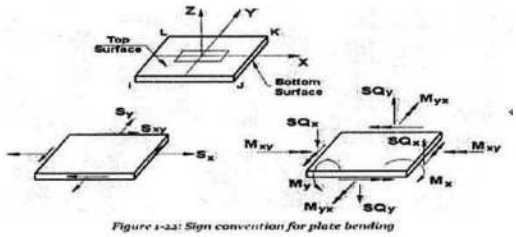
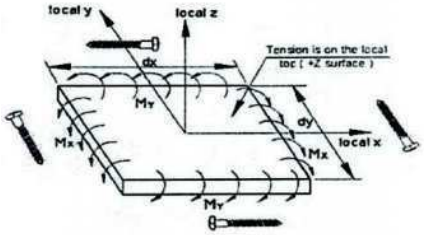


Figure 1-22: Sign convention for plate bending



Sign convention :
Local Z axis of plate is perpendicular to surface & positive towards upward direction.

Moment :
Mx : BM along X direction, +ve means tension on +Z surface of plate
My : BM along Y direction, +ve means tension on +Z surface of plate
For -ve Mx & My it is vice-virsa

Shear force :
SQx : Shear at face perpendicular to X axis
SQy : Shear at face perpendicular to Y axis

Axial force :
+ve Sx means tension along local X axis of plate & -ve Sx means compression along local X axis. for Sy this is similar.

Reinforcement in Foundation:

PLATE CENTER STRESS STAAD GROUP : RAFT_OTHER_2 At the face of the column

Load condition	Plate	Shear		Membrane			Bending Moment			M + I M_{xy} I		L/C
		SQX (local) Mpa	SQY(local)MPa	SX (local) MPa	SY (local) MPa	SXY (local) MPa	Mx (KN-m/m)	My (KN-m/m)	Mxy (KN-m/m)	Mx (KN-m/m)	My (KN-m/m)	
Max Qx	11902	0.651	0.333	0	0	0	-309.9	113.7	-670.1	-980	783.8	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Min Qx	10881	-0.651	0.333	0	0	0	-310.1	113.6	0	-310.1	113.6	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Max Qy	10917	-0.299	0.624	0	0	0	778.3	-407.9	676.8	1455	-1085	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Min Qy	11979	0.348	-0.686	0	0	0	592.6	-562.1	590.9	1184	-1153	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Max Sx	12024	0.182	-0.127	0	0	0	-146.3	-22.02	248.9	-395.2	-270.9	101 1.4X[D]
Min Sx	12024	0.182	-0.127	0	0	0	-146.3	-22.02	248.9	-395.2	-270.9	101 1.4X[D]
Max Sy	12024	0.182	-0.127	0	0	0	-146.3	-22.02	248.9	-395.2	-270.9	101 1.4X[D]
Min Sy	12024	0.182	-0.127	0	0	0	-146.3	-22.02	248.9	-395.2	-270.9	101 1.4X[D]
Max Sxy	12024	0.182	-0.127	0	0	0	-146.3	-22.02	248.9	-395.2	-270.9	101 1.4X[D]
Min Sxy	12024	0.182	-0.127	0	0	0	-146.3	-22.02	248.9	-395.2	-270.9	101 1.4X[D]
Max Mx	10933	-0.064	0.248	0	0	0	1575	1086	149	1724	1235	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Min Mx	11745	-0.627	-0.37	0	0	0	-428	141.8	-699	-1127	840.8	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Max My	11898	0.064	0.248	0	0	0	1575	1086	-149.2	1724	1235	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Min My	11775	-0.348	-0.686	0	0	0	592.7	-562.2	-590.9	1184	-1153	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Max Mxy	10905	-0.42	0.553	0	0	0	647.9	-175.9	734.6	1382	-910.5	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Min Mxy	11886	0.419	0.553	0	0	0	647.3	-175.4	-734.7	1382	-910.1	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]

PLATE CENTER STRESS STAAD GROUP : RAFT_OTHER_2 At a distance "d" from the face of the column

Load condition	Plate	Shear		Membrane			Bending Moment			M + I M_{xy} I		L/C
		SQX (local) Mpa	SQY(local)MPa	SX (local) MPa	SY (local) MPa	SXY (local) MPa	Mx (KN-m/m)	My (KN-m/m)	Mxy (KN-m/m)	Mx (KN-m/m)	My (KN-m/m)	
Max Qx	11902	0.651	0.333	0	0	0	-309.9	113.7	-670.1	-980	783.8	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Min Qx	10881	-0.651	0.333	0	0	0	-310.1	113.6	0	-310.1	113.6	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Max Qy	10917	-0.299	0.624	0	0	0	778.3	-407.9	676.8	1455	-1085	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Min Qy	11979	0.348	-0.686	0	0	0	592.6	-562.1	590.9	1184	-1153	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Max Sx	12024	0.182	-0.127	0	0	0	-146.3	-22.02	248.9	-395.2	-270.9	101 1.4X[D]
Min Sx	12024	0.182	-0.127	0	0	0	-146.3	-22.02	248.9	-395.2	-270.9	101 1.4X[D]
Max Sy	12024	0.182	-0.127	0	0	0	-146.3	-22.02	248.9	-395.2	-270.9	101 1.4X[D]
Min Sy	12024	0.182	-0.127	0	0	0	-146.3	-22.02	248.9	-395.2	-270.9	101 1.4X[D]
Max Sxy	12024	0.182	-0.127	0	0	0	-146.3	-22.02	248.9	-395.2	-270.9	101 1.4X[D]
Min Sxy	12024	0.182	-0.127	0	0	0	-146.3	-22.02	248.9	-395.2	-270.9	101 1.4X[D]
Max Mx	10933	-0.064	0.248	0	0	0	1575	1086	149	1724	1235	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Min Mx	11745	-0.627	-0.37	0	0	0	-428	141.8	-699	-1127	840.8	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Max My	11898	0.064	0.248	0	0	0	1575	1086	-149.2	1724	1235	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Min My	11775	-0.348	-0.686	0	0	0	592.7	-562.2	-590.9	1184	-1153	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Max Mxy	10905	-0.42	0.553	0	0	0	647.9	-175.9	734.6	1382	-910.5	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Min Mxy	11886	0.419	0.553	0	0	0	647.3	-175.4	-734.7	1382	-910.1	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]

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Material Property :

Grade of concrete =		fc'35	
Grade of reinf. =		fy420	
Clear cover =	bottom face	75	mm
	top face	50	mm
Thicknes of slab =		1850	mm

Design Constants :

φ for Moment =	0.9
φ for shear =	0.75
β_1	0.8
ε_t limit (for tension controlled section) =	0.005
Concrete compressive strain, ε_{cu} =	0.003

Reinforcement Along X:

Force Values			Depth		Reinforcement Calculation							
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	$a=f_y/0.85f_c' bA_s$	A_s required	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,}$ provided
	KN-m/m		mm	mm	KN-m/m	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2
Max Mx	1724	top face	50	1784	1724		2583		32	241.61	200	4022.9
Min Mx	45	top face	50	1784	45	0.014	66.7	3330	32	241.61	200	4022.9
Max Mxy	1382	top face	50	1784	1382		2067		32	241.61	200	4022.9
Min Mxy	1382	top face	50	1784	1382	*As	2066		32	241.61	200	4022.9

Shear Check		Qx	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{ca}	$V_{c,design}$	$\varphi(V_c+0.66\sqrt{f_c'}bd)\geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa	KN	KN-m/m		m^2	KN	KN	KN	KN	KN			KN		
Load condition	Max Qx	0.65	1204.4	980	0.002	1.784	1839	1757	fc'3061	1757.1	6542	> 1204	ok	1318	φVc>Vu	not required
	Min Qx	0.65	1204.4	310.1			2163			1757.1	6542	> 1204	ok	1318	φVc>Vu	not required

Reinforcement Along Y:

Force Values			Depth		Reinforcement Calculation							
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	$a=f_y/0.85f_c' bA_s$	A_s required	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,}$ provided
	KN-m/m		mm	mm	KN-m/m	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2
Max My	1235	top face	50	1784	1235		1845		32	241.61	200	4022.9
Min My	-1153	bottom face	75	1759	1153	0.014	1746	3330	32	241.61	200	4022.9
Max Mxy	-911	bottom face	75	1759	910.5		1377		32	241.61	200	4022.9
Min Mxy	-910	bottom face	75	1759	910.1	*As	1376		32	241.61	200	4022.9

Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	$\varphi(V_c+0.66\sqrt{f_c'}bd)\geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa		KN	KN-m/m		m^2	KN	KN	KN	KN	KN			KN		
Load condition	Max Qy	0.62	1	1154	1085	0.002	1.784	1819	1757	3061	1757.1	1840	> 1154	ok	1318	φVc>Vu	not required
	Min Qy	0.69	1	1269	1153			1823			1757.1	1840	>1269	ok	1318	φVc>Vu	not required

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• **Area Raft Other (Part 3):**

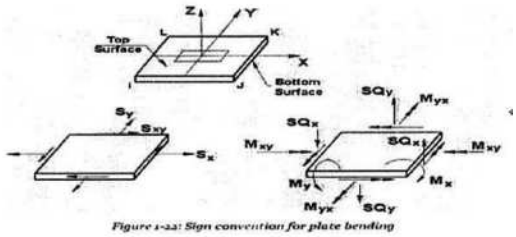
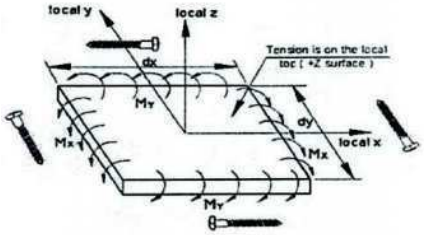


Figure 1-22: Sign convention for plate bending



Sign convention :
Local Z axis of plate is perpendicular to surface & positive towards upward direction.

Moment :
Mx : BM along X direction, +ve means tension on +Z surface of plate
My : BM along Y direction, +ve means tension on +Z surface of plate
For -ve Mx & My it is vice-virsa

Shear force :
SQx : Shear at face perpendicular to X axis
SQy : Shear at face perpendicular to Y axis

Axial force :
+ve Sx means tension along local X axis of plate & -ve Sx means compression along local X axis. for Sy this is similar.

Reinforcement in Foundation:

PLATE CENTER STRESS STAAD GROUP : RAFT_OTHER_3 At the face of the column

Load condition	Plate	Shear		Membrane			Bending Moment			M + I M_{xy} I		L/C
		SQX (local) Mpa	SQY(local)MPa	SX (local) MPa	SY (local) MPa	SXY (local) MPa	Mx (KN-m/m)	My (KN-m/m)	Mxy (KN-m/m)	Mx (KN-m/m)	My (KN-m/m)	
Max Qx	10941	0.297	0.005	0	0	0	-107.1	1335	-216.2	-323.3	1552	104 1.4X[D] + 1.7X[L + H1C + H3C + H5C]
Min Qx	11804	-0.297	0.005	0	0	0	-107.3	1336	0	-107.3	1336	104 1.4X[D] + 1.7X[L + H1C + H3C + H5C]
Max Qy	11805	-0.103	0.202	0	0	0	1343	1299	-5.301	1348	1305	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Min Qy	11937	-0.079	-0.212	0	0	0	1347	1264	-60.29	1408	1324	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Max Sx	11937	-0.002	-0.082	0	0	0	418.5	332	19.66	438.2	351.7	101 1.4X[D]
Min Sx	11937	-0.002	-0.082	0	0	0	418.5	332	19.66	438.2	351.7	101 1.4X[D]
Max Sy	11937	-0.002	-0.082	0	0	0	418.5	332	19.66	438.2	351.7	101 1.4X[D]
Min Sy	11937	-0.002	-0.082	0	0	0	418.5	332	19.66	438.2	351.7	101 1.4X[D]
Max Sxy	11937	-0.002	-0.082	0	0	0	418.5	332	19.66	438.2	351.7	101 1.4X[D]
Min Sxy	11937	-0.002	-0.082	0	0	0	418.5	332	19.66	438.2	351.7	101 1.4X[D]
Max Mx	11740	0.079	-0.212	0	0	0	1347	1264	60.25	1407	1324	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Min Mx	11681	-0.121	-0.042	0	0	0	-888.5	1256	79.12	-967.6	1335	135 0.9X[D] + 1.6X[W1B + H1C + H3C + H5C]
Max My	11681	-0.078	0.173	0	0	0	374.3	1657	15.75	390	1673	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Min My	10940	0.015	0.057	0	0	0	259.2	112.8	36.47	295.7	149.2	120 1.2X[D] + 1.0X[E1B] + 1.0X[L]
Max Mxy	11740	0.227	-0.067	0	0	0	775.5	1074	314	1090	1388	104 1.4X[D] + 1.7X[L + H1C + H3C + H5C]
Min Mxy	11937	-0.227	-0.067	0	0	0	775.5	1074	-314	1090	1388	104 1.4X[D] + 1.7X[L + H1C + H3C + H5C]

PLATE CENTER STRESS STAAD GROUP : RAFT_OTHER_3 At a distance "d" from the face of the column

Load condition	Plate	Shear		Membrane			Bending Moment			M + I M_{xy} I		L/C
		SQX (local) Mpa	SQY(local)MPa	SX (local) MPa	SY (local) MPa	SXY (local) MPa	Mx (KN-m/m)	My (KN-m/m)	Mxy (KN-m/m)	Mx (KN-m/m)	My (KN-m/m)	
Max Qx	10941	0.297	0.005	0	0	0	-107.1	1335	-216.2	-323.3	1552	104 1.4X[D] + 1.7X[L + H1C + H3C + H5C]
Min Qx	11804	-0.297	0.005	0	0	0	-107.3	1336	0	-107.3	1336	104 1.4X[D] + 1.7X[L + H1C + H3C + H5C]
Max Qy	11805	-0.103	0.202	0	0	0	1343	1299	-5.301	1348	1305	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Min Qy	11937	-0.079	-0.212	0	0	0	1347	1264	-60.29	1408	1324	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Max Sx	11937	-0.002	-0.082	0	0	0	418.5	332	19.66	438.2	351.7	101 1.4X[D]
Min Sx	11937	-0.002	-0.082	0	0	0	418.5	332	19.66	438.2	351.7	101 1.4X[D]
Max Sy	11937	-0.002	-0.082	0	0	0	418.5	332	19.66	438.2	351.7	101 1.4X[D]
Min Sy	11937	-0.002	-0.082	0	0	0	418.5	332	19.66	438.2	351.7	101 1.4X[D]
Max Sxy	11937	-0.002	-0.082	0	0	0	418.5	332	19.66	438.2	351.7	101 1.4X[D]
Min Sxy	11937	-0.002	-0.082	0	0	0	418.5	332	19.66	438.2	351.7	101 1.4X[D]
Max Mx	11740	0.079	-0.212	0	0	0	1347	1264	60.25	1407	1324	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Min Mx	11681	-0.121	-0.042	0	0	0	-888.5	1256	79.12	-967.6	1335	135 0.9X[D] + 1.6X[W1B + H1C + H3C + H5C]
Max My	11681	-0.078	0.173	0	0	0	374.3	1657	15.75	390	1673	112 1.4X[D] + 1.7X[L + H2B + H4B + H6B]
Min My	10940	0.015	0.057	0	0	0	259.2	112.8	36.47	295.7	149.2	120 1.2X[D] + 1.0X[E1B] + 1.0X[L]
Max Mxy	11740	0.227	-0.067	0	0	0	775.5	1074	314	1090	1388	104 1.4X[D] + 1.7X[L + H1C + H3C + H5C]
Min Mxy	11937	-0.227	-0.067	0	0	0	775.5	1074	-314	1090	1388	104 1.4X[D] + 1.7X[L + H1C + H3C + H5C]

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Material Property :

Grade of concrete =		fc'35	
Grade of reinf. =		fy420	
Clear cover =	bottom face	75	mm
	top face	50	mm
Thicknes of slab =		1850	mm

Design Constants :

φ for Moment =	0.9
φ for shear =	0.75
β_1	0.8
ε_t limit (for tension controlled section) =	0.005
Concrete compressive strain, ε_{cu} =	0.003

Reinforcement Along X:

Force Values			Depth		Reinforcement Calculation							
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	$a=f_y/0.85f_c' bA_s$	A_s required	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,}$ provided
	KN-m/m		mm	mm	KN-m/m	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2
Max Mx	1407	top face	50	1784	1407		2105		32	241.61	200	4022.9
Min Mx	45	top face	50	1784	45	0.014	66.7	3330	32	241.61	200	4022.9
Max Mxy	1090	top face	50	1784	1090		1626		32	241.61	200	4022.9
Min Mxy	1090	top face	50	1784	1090	*As	1626		32	241.61	200	4022.9

Shear Check		Qx	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{ca}	$V_{c,design}$	$\varphi(V_c+0.66\sqrt{f_c'}bd)\geq V_u$			Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa	KN	KN-m/m		m^2	KN	KN	KN	KN	KN				KN		
Load condition	Max Qx	0.30	549.45	323.3	0.002	1.784	1896	1757	fc'3061	1757.1	6542	> 549	ok	1318	φVc>Vu	not required
	Min Qx	0.30	549.45	107.3			2314			1757.1	6542	> 549	ok	1318	φVc>Vu	not required

Reinforcement Along Y:

Force Values			Depth		Reinforcement Calculation							
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	$a=f_y/0.85f_c' bA_s$	A_s required	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,}$ provided
	KN-m/m		mm	mm	KN-m/m	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2
Max My	1673	top face	50	1784	1673		2506		32	241.61	200	4022.9
Min My	149	top face	50	1784	149.2	0.014	221	3330	32	241.61	200	4022.9
Max Mxy	1388	top face	50	1784	1388		2076		32	241.61	200	4022.9
Min Mxy	1388	top face	50	1784	1388	*As	2076		32	241.61	200	4022.9

Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	$\varphi(V_c+0.66\sqrt{f_c'}bd)\geq V_u$			Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa		KN	KN-m/m		m^2	KN	KN	KN	KN	KN				KN		
Load condition	Max Qy	0.20	1	374	1305	0.002	1.784	1724	1757	3061	1723.6	1815	> 374	ok	1293	φVc>Vu	not required
	Min Qy	0.21	1	392	1324			1725			1724.8	1816	>392	ok	1294	φVc>Vu	not required

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12. DESIGN OF VERTICAL COMPONENTS:

12.1. Design of Concrete Column:

Design of concrete columns done directly in STAAD and design attached in **ATTACHMENT 1**
The design results summary is incorporated below.

Col. Mkd.	STAAD Member	Required Reinforement from STAAD design						Provided Reinforement						Remarks
		Main Rebar				Tie Bar		Main Rebar				Tie Bar		
		Nos.	Dia	Area	%	Dia	Spac.	Nos.	Dia	Area	%	Dia	Spac.	
-	-	-	mm	mm^2	-	mm	mm	nos.	mm	mm^2	-	mm	mm	
C1 (1400 Dia)	12070	20	32	15394	1.045	10	480	28	32	22518.9	1.46	10	300	OK
	12071	20	32	15502	1.045	10	480							
	12074	20	32	15394	1.045	10	480							
	12075	20	32	15502	1.045	10	480							
	12873	20	32	15394	1.045	10	480							
	12874	20	32	15394	1.045	10	480							
	12875	20	32	15394	1.045	10	480							
	12876	20	32	15394	1.045	10	480							
	12881	20	32	15394	1.045	10	480							
	12882	20	32	15394	1.045	10	480							
	12883	20	32	15394	1.045	10	480							
	12884	20	32	15394	1.045	10	480							
C2 (1250x2000)	12878	32	32	25000	1.029	10	480	42	32	33778.4	1.35	10	300	OK
	12879	32	32	25000	1.029	10	480							
	10802	32	32	25000	1.029	10	480							
	10803	32	32	25000	1.029	10	480							
	12870	32	32	25000	1.029	10	480							
	12871	32	32	25000	1.029	10	480							
C3 (1000x1500)	10799	24	32	16365	1.287	10	480	30	32	24127.4	1.61	10	300	OK
	10808	24	32	16365	1.287	10	480							
	12869	20	32	15420	1.072	10	480							
	12872	24	32	16365	1.287	10	480							
	12877	24	32	16260	1.287	10	480							
	12880	20	32	15420	1.072	10	480							

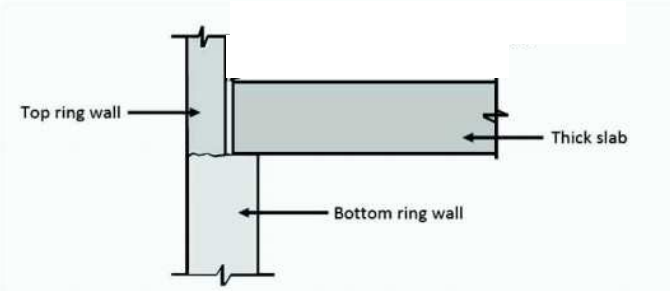
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12.2. Design of Bottom Ring Wall:

The bottom wall carries huge loads transferred from top wall by bearing and from thick slab at its continuous edge supports on bottom ring wall.

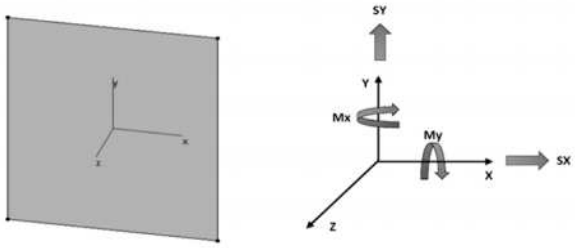


Typical section at the junction of bottom ring wall, top ring wall and thick slab

The actual stresses in the bottom ring wall due to various load combinations have been extracted from STAAD model. The bottom ring wall is designed as a structural element subjected to axial compression and uni-axial bending along the vertical axis and for axial forces with circumferential bending along the circumference.

• Nomenclature:

The bottom ring wall is modeled using four-noded plate elements with the following axis orientation.



Mx - Moment about local Y-axis

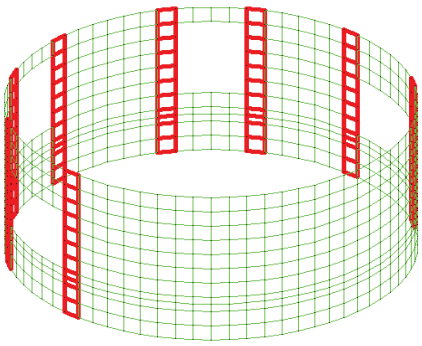
My - Moment about local X-axis

SX - Membrane stresses along local X-axis

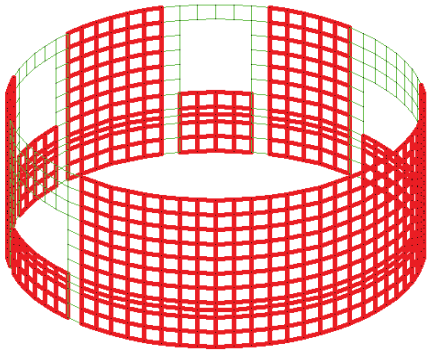
SY - Membrane stresses along local Y-axis

• STAAD Group:

The plate elements of STAAD model devided in two groups and for each group the reinforements calculated for maximum axial force and moment.



PL_NEAR OPNG



PL_OTHER

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12.2.1. Bottom Ring Wall near Opening: STAAD Plate Group Name: **PL_NEAR_OPNG**

	Plate Mkd.	Load Case No.	Shear		Membrane			Bending Moment		
			SQX	SQY	SX	SY	SXY	Mx	My	Mxy
			MPa	MPa	MPa	MPa	MPa	KN-m/m	KN-m/m	KN-m/m
Max Qx	10677	104	0.274	0.246	-0.824	-5.209	-0.724	51.145	808.562	139.948
Min Qx	10691	104	-0.277	0.247	-0.814	-5.219	0.724	50.922	811.709	-141.253
Max Qy	10413	104	-0.096	0.363	-0.957	-5.431	0.699	144.521	244.642	-167.053
Min Qy	10751	112	0.047	-0.383	-0.207	-3.302	-0.166	-59.968	-63.010	134.946
Max Sx	10739	153	0.12	0.161	0.731	-2.307	0.165	118.543	883.785	-48.092
Min Sx	10353	167	0.018	-0.031	-1.505	-5.914	-1.326	68.536	296.901	15.422
Max Sy	10739	119	0.006	-0.018	0.27	-0.145	-0.084	-15.130	-149.978	30.555
Min Sy	10353	112	0.063	0.064	-1.489	-7.584	-1.209	69.453	267.138	8.926
Max Sxy	10007	112	-0.005	0.051	-0.205	-3.594	1.506	0.875	-53.011	134.828
Min Sxy	10065	112	0.006	0.051	-0.205	-3.594	-1.507	0.917	-52.935	-134.865
Max Mx	10737	110	-0.052	0.198	-0.033	-4.067	0.268	225.507	1397.004	-299.791
Min Mx	10031	104	0.021	0.181	-0.433	-3.017	0.783	-112.827	-645.301	-20.388
Max My	10775	104	0.051	0.198	-0.031	-4.068	-0.268	224.568	1397.631	299.930
Min My	10009	104	0.002	0.272	-0.463	-3.276	0.661	-104.626	-777.494	165.873
Max Mxy	10785	110	0.185	0.16	0.216	-3.02	0.631	64.480	356.864	595.023
Min Mxy	10727	110	-0.187	0.16	0.214	-3.021	-0.632	64.114	356.561	-594.845

• Design Forces:

Thickness of wall, tw = 1425 mm Temperatue Moment = 92.82 KN-m

Forces for Vertical Reinforcement

SY = -7.455 MPa MY = 1318.302 KN-m/m

SXY = -1.478 MPa MXY = 724.436 KN-m/m

SY + SXY = -8.933 MPa Bending moment = MY + MXY = 2042.738 KN-m

Axial Force = (SY + SXY).tw Bending moment + Temperature Moment = 2135.558 KN-m
= -12729.53 KN [Compression]

Forces for Circumferential Reinforcement

SX = 1.473 MPa MX = 204.264 KN-m/m

SXY = 1.478 MPa MXY = 724.436 KN-m/m

SX + SXY = 2.951 MPa [Hoop Tension] Bending moment = MX + MXY = 928.7 KN-m

Total Hoop Force = (SX + SXY).tw = 4205.175 KN Bending moment + Temperature Moment = 1021.52 KN-m

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Design of Bottom Ring Wall: (Near opening)

Materials :

Concrete Grade, f_c' = 35 Mpa

Reinforcement Grade, f_y = 420 Mpa

Assumptions :

φ = 0.65

% of reinforcement, p_t = 0.55 % i.e. ρ_g = 0.006

β = 0.65

k = 1

Dimensions :

Depth parallel to X-X axis, h = 1425 mm

Depth parallel to Z-Z axis, b = 1000 mm

Clear Cover = 50 mm

d' = 82.5 mm

Y_h = (h-d')/h = 0.88

Y_b = (b-d')/b = 0.84

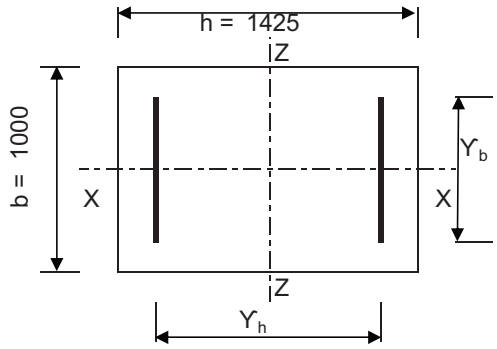
Unsupported length of column parallel to X-X axis, l_{ux} = 1.0 m kl_{ux}/r_x = 3.465 < 22 neglect slenderness effect

Unsupported length of column parallel to Z-Z axis, l_{uz} = 1.0 m kl_{uz}/r_z = 2.432

$P_{n,\,Max} = 0.8[0.85*f_c'(A_g-A_{st})+f_yA_{st}]$ = 37938.0 KN safe

Factored Force Value				Design force Value				Uniaxial moment capacity about X-X axis			Uniaxial moment capacity about ZZ axis			Check for Interaction Ratio		
L/C	P_u	M_{ux}	M_{uz}	$P_n = P_u/\varphi$	$M_{nx} = M_{ux}/\varphi$	$M_{nz} = M_{uz}/\varphi$	$K_n = P_n/f_c'bh$	$R_{nx} = M_{nx1}/f_c'b^2h$	M_{nx1}	M_{nx}/M_{nx1}	$R_{nz} = M_{nz1}/f_c'h^2b$	M_{nz1}	M_{nz}/M_{nz1}	α=log0.5/logβ	$(M_{nx}/M_{nx1})^\alpha + (M_{nz}/M_{nz1})^\alpha$	Remarks
	KN	KN-m		KN	KN-m				KN-m			KN-m				
-	12730	0.0	2136	19584	0.0	3285	0.393	-	-	0.00	0.140	9950	0.33	1.61	0.17	safe

Reinforcement Calculation								Calculation of Ties		
A_{st} (mm^2)	Combination of Reinforcement Bars,db					A_{st} provided(mm^2)	Reinforcement provided (%)	Dia, d_{tie} (cl : 25.7.2.2 SBC 304-CR-18)	Pitch (cl : 25.7.2.1 SBC 304-CR-18)	Spacing of ties provided
7837.5	16	25	+	16	20	12886	0.904	20	320	300



P_n : Nominal axial load P_u : Factored design axial load

M_n : Nominal moment M_u : Factored design moment

Use interaction diagram L35-420.9

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12.2.2. Bottom Ring Wall away from Opening: STAAD Plate Group Name: **PL_OTHER**

	Plate Mkd.	Load Case No.	Shear		Membrane			Bending Moment		
			SQX	SQY	SX	SY	SXY	Mx	My	Mxy
			MPa	MPa	MPa	MPa	MPa	KN-m/m	KN-m/m	KN-m/m
Max Qx	10283	104	0.274	0.246	-0.824	-5.209	-0.724	51.145	808.562	139.948
Min Qx	10691	104	-0.277	0.247	-0.814	-5.219	0.724	50.922	811.709	-141.253
Max Qy	10413	104	-0.096	0.363	-0.957	-5.431	0.699	144.521	244.642	-167.053
Min Qy	10751	110	0.047	-0.383	-0.207	-3.302	-0.166	-59.968	-63.010	134.946
Max Sx	10739	106	0.12	0.161	0.731	-2.307	0.165	118.543	883.785	-48.092
Min Sx	10353	165	0.018	-0.031	-1.505	-5.914	-1.326	68.536	296.901	15.422
Max Sy	10739	119	0.006	-0.018	0.27	-0.145	-0.084	-15.130	-149.978	30.555
Min Sy	10353	112	0.063	0.064	-1.489	-7.584	-1.209	69.453	267.138	8.926
Max Sxy	10007	112	-0.005	0.051	-0.205	-3.594	1.506	0.875	-53.011	134.828
Min Sxy	10065	112	0.006	0.051	-0.205	-3.594	-1.507	0.917	-52.935	-134.865
Max Mx	10737	104	-0.052	0.198	-0.033	-4.067	0.268	225.507	1397.004	-299.791
Min Mx	10031	104	0.021	0.181	-0.433	-3.017	0.783	-112.827	-645.301	-20.388
Max My	10775	104	0.051	0.198	-0.031	-4.068	-0.268	224.568	1397.631	299.930
Min My	10009	104	0.002	0.272	-0.463	-3.276	0.661	-104.626	-777.494	165.873
Max Mxy	10785	110	0.185	0.16	0.216	-3.02	0.631	64.480	356.864	595.023
Min Mxy	10727	110	-0.187	0.16	0.214	-3.021	-0.632	64.114	356.561	-594.845

• Design Forces:

Thickness of wall, tw = 1425 mm Temperatue Moment = 92.82 KN-m

Forces for Vertical Reinforcement

SY = -5.656 MPa MY = 1587.066 KN-m/m

SXY = -2.007 MPa MXY = 686.21 KN-m/m

SY + SXY = -7.663 MPa Bending moment = MY + MXY = 2273.276 KN-m

Axial Force = (SY + SXY).tw Bending moment + Temperature Moment = 2366.096 KN-m

= -10919.78 KN [Compression]

Forces for Circumferential Reinforcement

SX = 2.947 MPa MX = -331.833 KN-m/m

SXY = 2.007 MPa MXY = 686.21 KN-m/m

SX + SXY = 4.954 MPa [Hoop Tension] Bending moment = MX + MXY = 1018.043 KN-m

Total Hoop Force = (SX + SXY).tw = 7059.45 KN Bending moment + Temperature Moment = 1110.863 KN-m

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Design of Bottom Ring Wall: (away from opening)

Materials :

Concrete Grade, f_c' = 35 Mpa

Reinforcement Grade, f_y = 420 Mpa

Assumptions :

φ = 0.65

% of reinforcement, p_t = 0.55 % i.e. ρ_g = 0.006

β = 0.65

k = 1

Dimensions :

Depth parallel to X-X axis, h = 1425 mm

Depth parallel to Z-Z axis, b = 1000 mm

Clear Cover = 50 mm

d' = 82.5 mm

Y_h = (h-d')/h = 0.88

Y_b = (b-d')/b = 0.84

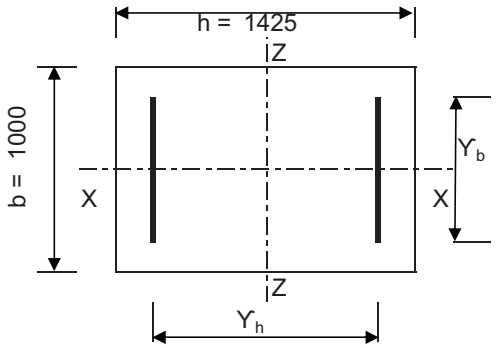
Unsupported length of column parallel to X-X axis, l_{ux} = 1.0 m kl_{ux}/r_x = 3.465 < 22 neglect slenderness effect

Unsupported length of column parallel to Z-Z axis, l_{uz} = 1.0 m kl_{uz}/r_z = 2.432

$P_{n,\,Max} = 0.8[0.85*f_c'(A_g-A_{st})+f_yA_{st}]$ = 36368.3 KN safe

Factored Force Value				Design force Value				Uniaxial moment capacity about X-X axis			Uniaxial moment capacity about ZZ axis			Check for Interaction Ratio		
L/C	P_u	M_{ux}	M_{uz}	$P_n = P_u/\varphi$	$M_{nx} = M_{ux}/\varphi$	$M_{nz} = M_{uz}/\varphi$	$K_n = P_n/f_c'bh$	$R_{nx} = M_{nx1}/f_c'b^2h$	M_{nx1}	M_{nx}/M_{nx1}	$R_{nz} = M_{nz1}/f_c'h^2b$	M_{nz1}	M_{nz}/M_{nz1}	$\alpha = \log 0.5/\log\beta$	$(M_{nx}/M_{nx1})^\alpha + (M_{nz}/M_{nz1})^\alpha$	Remarks
	KN	KN-m		KN	KN-m				KN-m			KN-m				
-	10920	0.0	2366	16800	0.0	3640	0.337	-	-	0.00	0.150	10661	0.34	1.61	0.18	safe

Reinforcement Calculation								Calculation of Ties		
A_{st} (mm^2)	Combination of Reinforcement Bars,db					A_{st} provided(mm^2)	Reinforcement provided (%)	Dia, d_{tie} (cl : 25.7.2.2 SBC 304-CR-18)	Pitch (cl : 25.7.2.1 SBC 304-CR-18)	Spacing of ties provided
7837.5	8	25	+	8	25	7858	0.551	20	400	300



P_n : Nominal axial load P_u : Factored design axial load

M_n : Nominal moment M_u : Factored design moment

Use interaction diagram L35-420.9

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REFERENCES

- **Bulk Solid Load**

[1] Eurocode 1 - Actions on structures - Part4 : Silos and Tanks; EN 1991-4 : 2007
[2] Standard Practice for Design and Construction of Concrete Silos and Stacking Tubes for storing Granular Materials; ACI 313-98

- **Wind Load**

[3] Saudi Building Code Requirements : Structural - Loading and Forces; SBC 301-2007

- **Seismic Load**

[4] Uniform Building Code; UBC 1997

- **Structural Design**

[5] Saudi concrete structures Code; SBC 304-2018
[6] Building Code Requirements for Structural Concrete; ACI 318-14
[7] Design and Construction of Silos and Bunkers by Sargis S. Safarian & Ernest C. Harris

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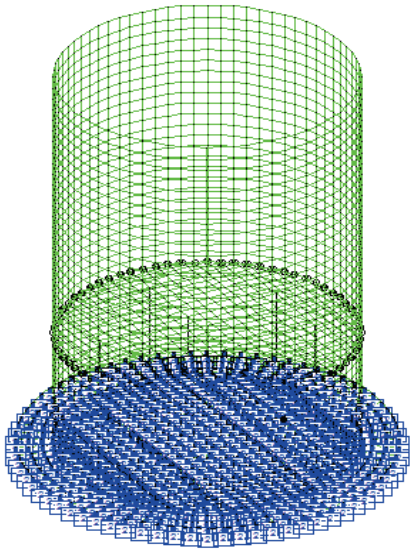
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ATTACHMENT 1 STAAD MODEL SCHEMATIC

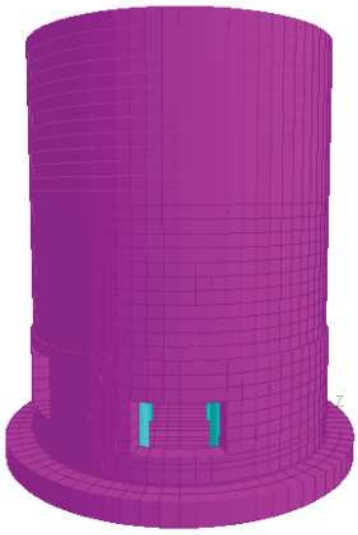
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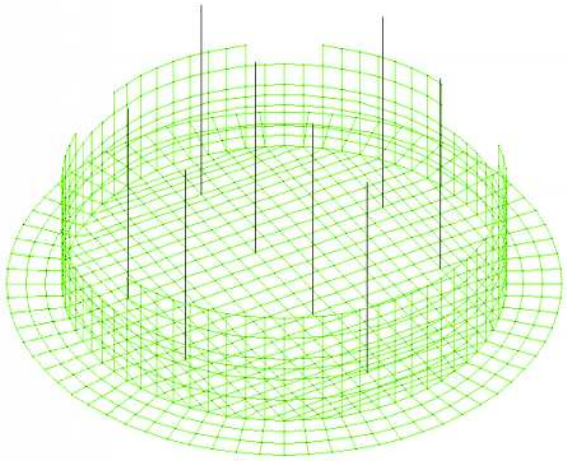
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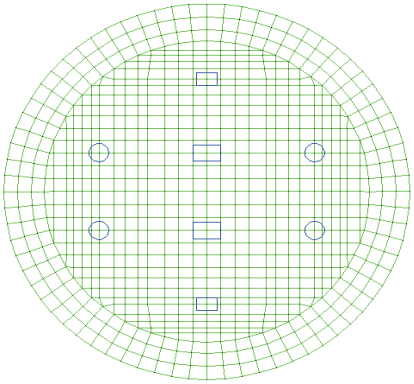
Model Mode View



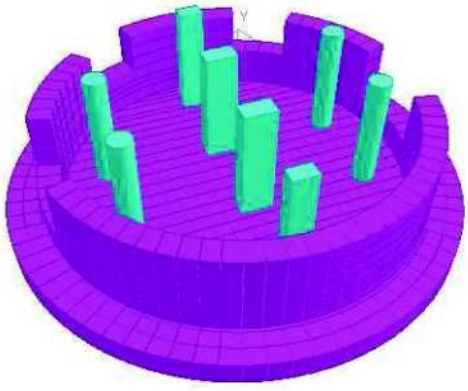
STAAD Rendered View



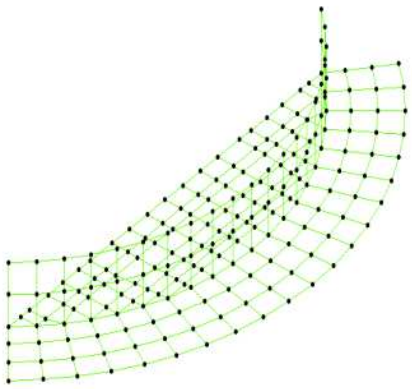
Junction of Bottom Raft & Vertical Wall



Column Position Plan



Column Rendered View



Enlarged View of portion of Wall & Vertical Raft Junction

ATTACHMENT 2 STAAD OUTPUT

PROPRIETARY & CONFIDENTIAL

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*           Version  20.07.11.70                  *
*           Proprietary Program of                *
*           Bentley Systems, Inc.                 *
*           Date=    MAR  2, 2020                 *
*           Time=    11:51:31                     *
*                                                *
*      USER ID: LNT ECC DIV                      *
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    4. ENGINEER DATE 02-03-2020
    5. JOB NAME MANSOURAH MASSARAH GOLD PROJECT
    6. JOB CLIENT MA'ADEN GOLD AND BASE METALS (MA'ADEN)
    7. ENGINEER NAME SVK
    8. CHECKER NAME UR
    9. JOB REF CRUSHED ORE STORAGE BIN
   10. JOB NO O19045
   11. JOB REV R0
   12. JOB PART 450
   13. APPROVED NAME AG
   14. CHECKER DATE 02-03-2020
   15. APPROVED DATE 02-03-2020
   16. END JOB INFORMATION
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   18. INPUT WIDTH 79
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3015. *
3016. ELEMENT PROPERTY
3017. 101 TO 172 201 TO 272 301 TO 372 401 TO 472 501 TO 572 601 TO 672 701 TO 772 -
3018. 801 TO 872 901 TO 972 1001 TO 1072 1101 TO 1172 1201 TO 1272 1301 TO 1372 -
3019. 1401 TO 1472 1501 TO 1572 1601 TO 1672 1701 TO 1772 1801 TO 1872 -
3020. 1901 TO 1972 2001 TO 2072 2101 TO 2172 2201 TO 2272 2301 TO 2372 -
3021. 2401 TO 2472 THICKNESS 1
3022. 10879 TO 10981 11198 TO 11203 11205 11206 11210 TO 11222 11225 TO 11253 11255 -
3023. 11256 TO 11259 11371 TO 11399 11487 TO 11501 11503 11511 TO 12067 -
3024. 12885 TO 12956 THICKNESS 1.85
3025. 10001 TO 10288 10295 TO 10354 10367 TO 10380 10382 10384 10386 10388 TO 10404 -
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3030. 10674 10676 TO 10692 10694 10696 10698 10700 TO 10792 THICKNESS 1.425
3031. 12172 TO 12787 THICKNESS 1.5
3032. *
3033. DEFINE MATERIAL START
3034. ISOTROPIC CONCRETE
3035. E 2.17185E+007
3036. POISSON 0.17
3037. DENSITY 24
3038. ALPHA 1E-005
3039. DAMP 0.05
3040. TYPE CONCRETE
3041. STRENGTH FCU 35000
3042. END DEFINE MATERIAL
3043. *
3044. *
3045. START GROUP DEFINITION
3046. ELEMENT
3047. *
3048. _OPNG1 134 136 138 140 234 236 238 240 334 336 338 340 434 436 438 440
3049. *
3050. *
3051. *
3052. _THICK_SLAB 12172 TO 12787
3053. *
3054. _PL_NEAR_OPNG 10007 10008 10019 10029 10043 10053 10065 10066 10079 10080 -
3055. 10091 10101 10115 10125 10137 10138 10151 10152 10163 10173 10187 10197 -
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3061. 10739 10749 10763 10773 10785 10786
3062. _PL_OTHER 10001 TO 10006 10009 TO 10018 10020 TO 10028 10030 TO 10042 10044 -
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3096. _RAFT_COL_STRIP 10883 TO 10886 10889 10890 10896 TO 10899 10902 10903 10920 -
3097. 10921 10925 10926 10947 10948 10950 10951 10955 10956 10960 10961 10977 -
3098. 10978 10980 10981 11215 TO 11220 11227 TO 11229 11234 TO 11236 -
3099. 11251 TO 11253 11257 TO 11259 11392 TO 11395 11666 TO 11670 11676 TO 11679 -
3100. 11683 TO 11686 11689 11690 11693 TO 11696 11702 11705 11706 11711 TO 11715 -
3101. 11729 TO 11732 11738 11764 11779 11780 11782 11783 11785 11786 -
3102. 11799 TO 11802 11808 11813 TO 11817 11820 11821 11829 11831 11851 11852 -
3103. 11915 11916 11922 11923 11944 TO 11948 11953 TO 11957 11962 TO 11966 12005 -
3104. 12006 TO 12008 12016 TO 12022 12027 TO 12029 12031 TO 12033 12035 TO 12037
3105. *_RAFT_CANTI 11519 TO 11662
3106. _RAFT_BELOW_WALL 11198 11202 11203 11205 11206 11213 11222 11230 11231 11242 -
3107. 11248 11371 TO 11376 11380 11396 TO 11399 11487 TO 11501 11503 -
3108. 11511 TO 11518 11663 11664 11673 11697 11698 11717 11735 11748 11757 11766 -
3109. 11769 11771 11790 11792 11794 11796 11819 11833 TO 11844 11847 11856 11860 -
3110. 11861 11864 TO 11866 11868 11869 11877 11889 11890 11905 11942 -
3111. 12045 TO 12067
3112. *_RAFT_OTHER 10879 TO 10881 10891 TO 10894 10904 TO 10907 10917 10918 10922 -
3113. *10923 10927 TO 10936 10940 TO 10942 10952 10953 10957 10958 10965 TO 10972 -
3114. *11199 TO 11201 11225 11232 11237 11243 11249 11255 11377 TO 11379 -
3115. *11381 TO 11383 11672 11681 11691 11704 11708 11709 11723 11724 11736 11739 -
3116. *11740 TO 11747 11749 TO 11754 11758 TO 11762 11765 11767 11768 11770 11772 -
3117. *11773 TO 11777 11804 11805 11810 11811 11845 11846 11848 11849 11854 11855 -
3118. *11859 11867 11870 11871 11874 TO 11876 11879 TO 11882 11884 TO 11888 11891 -
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*11892 TO 11896 11898 TO 11904 11912 11913 11920 11925 TO 11937 11940 11941 -
*11950 11951 11959 11960 11968 TO 11980 11994 TO 11998 12010 TO 12014 12024 -
*12025 TO 12026 12042 TO 12044
*
*_RAFT_BELOW_COL 10882 10883 10886 TO 10889 10895 TO 10903 10908 TO 10916 -
*10919 10920 10924 TO 10926 10937 TO 10939 10943 TO 10950 10954 10955 10959 -
*10960 TO 10964 10973 TO 10980 11210 TO 11212 11214 TO 11217 11220 11221 11226
*11227 11233 TO 11236 11238 TO 11241 11244 TO 11247 11250 TO 11253 11256 -
*11257 11384 TO 11394 11665 11666 11670 11671 11674 TO 11676 11679 11680 -
*11682 11683 11686 TO 11689 11692 11693 11696 11699 TO 11703 11705 TO 11707 -
*11710 TO 11716 11718 TO 11722 11725 TO 11729 11732 TO 11734 11737 11738 -
*11755 11756 11763 11764 11778 11779 11781 11782 11784 TO 11789 11791 11793 -
*11795 11797 TO 11803 11806 TO 11809 11812 TO 11818 11820 TO 11832 -
*11850 TO 11853 11857 11858 11862 11863 11872 11873 11878 11883 11897 11906 -
*11907 TO 11911 11914 TO 11919 11921 TO 11924 11938 11939 11943 11944 11948 -
*11949 11952 11953 11957 11958 11961 TO 11967 11981 TO 11993 11999 TO 12009 -
*12015 TO 12019 12022 12023 12029 12030 12033 TO 12041
_RAFT_CANT_MOM 11520 11522 11524 11526 11528 11530 11532 11534 11536 11538 -
11540 11542 11544 11546 11548 11550 11552 11554 11556 11558 11560 11562 -
11564 11566 11568 11570 11572 11574 11576 11578 11580 11582 11584 11586 -
11588 11590 11592 11594 11596 11598 11600 11602 11604 11606 11608 11610 -
11612 11614 11616 11618 11620 11622 11624 11626 11628 11630 11632 11634 -
11636 11638 11640 11642 11644 11646 11648 11650 11652 11654 11656 11658 -
11660 11662 12885 TO 12956
*_RAFT_COL_SHEAR 10882 10887 10888 10895 10900 10901 10908 TO 10916 10919 -
*10924 10937 TO 10939 10943 TO 10946 10949 10954 10959 10962 TO 10964 10973 -
*10974 TO 10976 10979 11210 TO 11212 11214 11221 11226 11233 11238 TO 11241 -
*11244 TO 11247 11250 11256 11384 TO 11391 11665 11671 11674 11675 11680 -
*11682 11687 11688 11692 11699 TO 11701 11703 11707 11710 11716 -
*11718 TO 11722 11725 TO 11728 11733 11734 11737 11755 11756 11763 11778 -
*11781 11784 11787 TO 11789 11791 11793 11795 11797 11798 11803 11806 11807 -
*11809 11812 11818 11822 TO 11828 11830 11832 11850 11853 11857 11858 11862 -
*11863 11872 11873 11878 11883 11897 11906 TO 11911 11914 11917 TO 11919 -
*11921 11924 11938 11939 11943 11949 11952 11958 11961 11967 11981 -
*11987 TO 11993 11999 12009 12015 12023 12030 12034 12038 TO 12041
*
_RAFT_CANT_SHEAR 12885 TO 12956
*
_RAFT_COL1_MOM 10882 10883 10886 TO 10889 10895 TO 10903 10908 TO 10916 11210 -
11211 TO 11212 11214 TO 11217 11220 11221 11226 11227 11233 TO 11236 11238 -
11239 TO 11241 11244 TO 11247 11250 TO 11253 11256 11257 11665 11666 11670 -
11671 11701 11710 TO 11716 11718 TO 11722 11812 TO 11818 11906 TO 11911 -
11943 11944 11948 11949 11952 11953 11957 11958 11961 TO 11967 -
11981 TO 11993 11999 TO 12009 12015 TO 12019 12022 12023 12029 12030 12033 -
12034 TO 12041
_RAFT_COL2_MOM 10943 TO 10950 10954 10955 10959 TO 10964 10973 TO 10980 11682 -
11683 11686 TO 11689 11692 11693 11696 11702 11703 11705 TO 11707 -
11725 TO 11727 11806 TO 11809 11820 TO 11832 11914 TO 11919
_RAFT_COL3_MOM 10919 10920 10924 TO 10926 10937 TO 10939 11384 TO 11394 11674 -
11675 TO 11676 11679 11680 11699 11700 11728 11729 11732 TO 11734 11737 11738 -
11755 11756 11763 11764 11778 11779 11781 11782 11784 TO 11789 11791 11793 -
11795 11797 TO 11803 11850 TO 11853 11857 11858 11862 11863 11872 11873 -
11878 11883 11897 11921 TO 11924 11938 11939
_RAFT_COL1_SHEAR 10882 10887 10888 10895 10900 10901 10908 TO 10916 11210 -
11211 TO 11212 11214 11221 11226 11233 11238 TO 11241 11244 TO 11247 11250 -
11256 11665 11671 11701 11710 11716 11718 TO 11722 11812 11818 -
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11906 TO 11911 11943 11949 11952 11958 11961 11967 11981 11987 TO 11993 -
11999 12009 12015 12023 12030 12034 12038 TO 12041
_RAFT_COL2_SHEAR 10943 TO 10946 10949 10954 10959 10962 TO 10964 -
10973 TO 10976 10979 11682 11687 11688 11692 11703 11707 11725 TO 11727 -
11806 11807 11809 11822 TO 11828 11830 11832 11914 11917 TO 11919
_RAFT_COL3_SHEAR 10919 10924 10937 TO 10939 11385 TO 11391 11674 11675 11680 -
11699 11728 11733 11734 11737 11755 11756 11763 11778 11781 11784 -
11787 TO 11789 11793 11795 11797 11798 11803 11850 11853 11857 11858 11863 -
11872 11873 11878 11883 11897 11921 11924 11938 11939
_RAFT_OTHR_1 10929 TO 10931 10934 TO 10936 10952 10953 10957 10958 -
10965 TO 10972 11672 11691 11708 11709 11723 11724 11810 11811 11912 11913 -
11926 TO 11935 11950 11951 11968 TO 11975
_RAFT_OTHR_2 10881 10893 10894 10905 TO 10907 10917 10918 10922 10923 10927 -
10928 10932 10933 11383 11741 TO 11745 11751 TO 11754 11760 TO 11762 11775 -
11776 TO 11777 11849 11874 11879 11880 11884 TO 11886 11893 TO 11896 11898 -
11899 TO 11902 11925 11936 11940 11941 11976 TO 11979 11994 TO 11996 12010 -
12011 12024
_RAFT_OTHER_3 10940 TO 10942 11681 11704 11736 11739 11740 11804 11805 11920 -
11937
_THICK_SLAB_EFFC 12186 TO 12189 12192 TO 12195 12199 TO 12202 12205 TO 12208 -
12220 TO 12223 12225 TO 12228 12230 12231 12233 12234 12236 12237 12239 -
12240 12250 TO 12253 12256 TO 12261 12263 12264 12266 12267 12269 12270 -
12278 12283 12284 12287 TO 12289 12292 TO 12299 12303 TO 12306 -
12308 TO 12311 12314 TO 12317 12320 TO 12323 12325 TO 12328 12336 TO 12338 -
12341 12342 12344 TO 12347 12349 12351 12353 12385 TO 12387 12416 12419 -
12421 12426 TO 12431 12434 TO 12436 12438 12441 TO 12446 12459 TO 12464 -
12467 12471 TO 12476 12488 12490 12492 TO 12494 12497 12501 12502 12504 -
12505 12508 12509 12513 12517 TO 12520 12523 TO 12533 12536 TO 12538 12544 -
12545 TO 12550 12566 12568 TO 12575 12577 12578 12583 12590 TO 12596 12613 -
12614 TO 12621 12624 12626 12627 12630 TO 12633 12636 TO 12641 12648 TO 12651 -
12654 TO 12658 12660 12663 TO 12665 12676 TO 12683 12686 TO 12691 -
12694 TO 12697 12700 TO 12703 12706 TO 12709 12712 TO 12715 12718 TO 12721 -
12724 12725 12728 TO 12731 12733 TO 12737 12748 TO 12750 12752 TO 12754 -
12756 TO 12758 12760 TO 12762
END GROUP DEFINITION
*
MEMBER PROPERTY AMERICAN
10802 10803 12870 12871 12878 12879 PRIS YD 2 ZD 1.25
10799 10808 12869 12872 12877 12880 PRIS YD 1.5 ZD 1
MEMBER PROPERTY AMERICAN
12070 12071 12074 12075 12873 TO 12876 12881 TO 12884 PRIS YD 1.4
*
*SUPPORTS
*5001 TO 5072 11962 TO 11977 12006 12008 12009 12011 12013 TO 12351 -
*12353 TO 12365 12367 TO 12391 12393 TO 12403 12405 TO 12417 12419 TO 12443 -
*12445 TO 12455 12457 TO 12470 12472 TO 12561 12563 TO 12703 -
*12704 DIRECT Y SUBGRADE 54000
*
ELEMENT RELEASE
12281 12282 12330 TO 12332 12355 TO 12359 12362 TO 12365 12369 TO 12374 12393 -
12417 12418 12539 12553 TO 12557 12559 12765 TO 12787 J1 MX MY MZ
12281 12282 12286 12301 12302 12313 12319 12330 TO 12332 12355 TO 12360 12362 -
12363 TO 12365 12377 TO 12382 12384 12393 12417 12418 12767 J2 MX MY MZ
12286 12301 12302 12313 12319 12360 12369 TO 12373 12377 TO 12382 12384 12455 -
12512 12514 12516 12584 12585 12765 12768 TO 12787 J3 MX MY MZ
12374 12455 12512 12514 12516 12539 12553 TO 12557 12559 12584 -
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*
12585 J4 MX MY MZ
*
CONSTANTS
MATERIAL CONCRETE ALL
*
SUPPORTS
10879 TO 10981 11198 TO 11203 11205 11206 11210 TO 11222 11225 TO 11253 11255 -
11256 TO 11259 11371 TO 11399 11487 TO 11501 11503 11511 TO 12067 -
12885 TO 12956 PLATE MAT DIRECT Y SUBGRADE 54000
DEFINE UBC LOAD
ZONE 0.15 I 1 RWX 3.6 RWZ 3.6 STYP 3 CT 0.0731 PX 1.11 PZ 1.11
SELFWEIGHT 1
JOINT WEIGHT
*
* FOR STORE MATERIAL
2001 TO 2072 WEIGHT 17.2
1901 TO 1972 WEIGHT 72.6
1801 TO 1872 WEIGHT 110.8
1701 TO 1772 WEIGHT 110.8
1601 TO 1672 WEIGHT 110.8
1501 TO 1572 WEIGHT 110.8
1401 TO 1472 WEIGHT 110.8
1301 TO 1372 WEIGHT 110.8
1201 TO 1272 WEIGHT 110.8
1101 TO 1172 WEIGHT 110.8
1001 TO 1072 WEIGHT 110.8
901 TO 972 WEIGHT 110.8
801 TO 872 WEIGHT 110.8
701 TO 772 WEIGHT 110.8
601 TO 672 WEIGHT 110.8
501 TO 572 WEIGHT 110.8
401 TO 472 WEIGHT 110.8
301 TO 372 WEIGHT 110.8
201 TO 272 WEIGHT 110.8
101 TO 172 WEIGHT 55.4
*
* FOR DEAD LOAD
*
******** STRUCTURAL ROOF
2501 TO 2572 WEIGHT 11.5
*
******** STRUCTURAL FLOOR
2501 TO 2572 WEIGHT 14.4
*
* FOR LIVE LOAD (CONSIDERING 50% OF LIVE LOAD AS SEISMIC WEIGHT)
*
******** STRUCTURAL ROOF
2501 TO 2572 WEIGHT 5.75
*
******** STRUCTURAL FLOOR
5436 TO 5466 5472 TO 5479 5483 TO 5491 5495 TO 5502 WEIGHT 11.55
*
******** DISCHARGE CHUTE
12798 12808 12813 12823 12965 13008 13010 13044 13087 13122 13151 13167 13169 -
13218 13230 13243 WEIGHT 30
*
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*
*
LOAD 5 SEISMIC LOAD IN X-DIRECTION (E1)
UBC LOAD X 1
*
LOAD 6 SEISMIC LOAD IN Z-DIRECTION (E2)
UBC LOAD Z 1
*
*
**********************
LOAD 1 DEAD LOAD (D)
**********************
*
SELFWEIGHT Y -1
*
ELEMENT LOAD
* SOIL FILLING
10879 TO 10981 11198 TO 11203 11205 11206 11210 TO 11222 11225 TO 11253 11255 -
11256 TO 11259 11371 TO 11399 11487 TO 11501 11503 11511 TO 12067 PR GY -20
**
** STACKED MATERIAL (VERTICAL)
*11535 TO 11578 PR GY -96
*
* LOAD FOR STRUCTURAL ROOF
JOINT LOAD
2501 TO 2572 FY -11.5
*
* LOAD FOR STRUCTURAL FLOOR
JOINT LOAD
5436 TO 5466 5472 TO 5479 5483 TO 5491 5495 TO 5502 FY -14.4
*
* STACKED MATERIAL (HORIZONTAL)
ELEMENT LOAD
10234 TRAP JT -57.6 -57.6 -50.4 -50.4
10236 TRAP JT -57.6 -57.6 -50.4 -50.4
10238 TRAP JT -57.6 -57.6 -50.4 -50.4
10240 TRAP JT -57.6 -57.6 -50.4 -50.4
10242 TRAP JT -57.6 -57.6 -50.4 -50.4
10244 TRAP JT -57.6 -57.6 -50.4 -50.4
10246 TRAP JT -57.6 -57.6 -50.4 -50.4
10248 TRAP JT -57.6 -57.6 -50.4 -50.4
10250 TRAP JT -57.6 -57.6 -50.4 -50.4
10252 TRAP JT -57.6 -57.6 -50.4 -50.4
10254 TRAP JT -57.6 -57.6 -50.4 -50.4
10256 TRAP JT -57.6 -57.6 -50.4 -50.4
10258 TRAP JT -57.6 -57.6 -50.4 -50.4
10260 TRAP JT -57.6 -57.6 -50.4 -50.4
10262 TRAP JT -57.6 -57.6 -50.4 -50.4
10264 TRAP JT -57.6 -57.6 -50.4 -50.4
10266 TRAP JT -57.6 -57.6 -50.4 -50.4
10268 TRAP JT -57.6 -57.6 -50.4 -50.4
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3338. 10272 TRAP JT -57.6 -57.6 -50.4 -50.4
3339. 10274 TRAP JT -57.6 -57.6 -50.4 -50.4
3340. 10304 TRAP JT -50.4 -50.4 -43.2 -43.2
3341. 10306 TRAP JT -50.4 -50.4 -43.2 -43.2
3342. 10308 TRAP JT -50.4 -50.4 -43.2 -43.2
3343. 10310 TRAP JT -50.4 -50.4 -43.2 -43.2
3344. 10312 TRAP JT -50.4 -50.4 -43.2 -43.2
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3468. 10700 TRAP JT -14.4 -14.4 -7.2 -7.2
3469. 10702 TRAP JT -14.4 -14.4 -7.2 -7.2
3470. 10704 TRAP JT -14.4 -14.4 -7.2 -7.2
3471. 10706 TRAP JT -14.4 -14.4 -7.2 -7.2
3472. 10736 TRAP JT -7.2 -7.2 0 0
3473. 10738 TRAP JT -7.2 -7.2 0 0
3474. 10740 TRAP JT -7.2 -7.2 0 0
3475. 10742 TRAP JT -7.2 -7.2 0 0
3476. 10744 TRAP JT -7.2 -7.2 0 0
3477. 10746 TRAP JT -7.2 -7.2 0 0
3478. 10748 TRAP JT -7.2 -7.2 0 0
3479. 10750 TRAP JT -7.2 -7.2 0 0
3480. 10752 TRAP JT -7.2 -7.2 0 0
3481. 10754 TRAP JT -7.2 -7.2 0 0
3482. 10756 TRAP JT -7.2 -7.2 0 0
3483. 10758 TRAP JT -7.2 -7.2 0 0
3484. 10760 TRAP JT -7.2 -7.2 0 0
3485. 10762 TRAP JT -7.2 -7.2 0 0
3486. 10764 TRAP JT -7.2 -7.2 0 0
3487. 10766 TRAP JT -7.2 -7.2 0 0
3488. 10768 TRAP JT -7.2 -7.2 0 0
3489. 10770 TRAP JT -7.2 -7.2 0 0
3490. 10772 TRAP JT -7.2 -7.2 0 0
3491. 10774 TRAP JT -7.2 -7.2 0 0
3492. 10776 TRAP JT -7.2 -7.2 0 0
3493. 10778 TRAP JT -7.2 -7.2 0 0
3494. 10232 TRAP JT -57.6 -57.6 -50.4 -50.4
3495. *
3496. *
3497. ************************
3498. LOAD 2 LIVE LOAD (L)
3499. ************************
3500. *
3501. * LOAD FOR STRUCTURAL ROOF
3502. JOINT LOAD
3503. 2501 TO 2572 FY -11.5
3504. *
```

```
*
* LOAD FOR STRUCTURAL FLOOR
JOINT LOAD
5436 TO 5466 5472 TO 5479 5483 TO 5491 5495 TO 5502 FY -23.1
*
* LOAD FOR MECHANICAL CHUTE
JOINT LOAD
12798 12808 12813 12823 12965 13008 13010 13044 13087 13122 13151 13167 13169 -
13218 13230 13243 FY -60
*
*
*****************************
LOAD 3 WIND LOAD ALONG X (W1)
*****************************
*
ELEMENT LOAD
* WINDWARD SIDE
10217 TO 10252 10295 TO 10324 10367 TO 10380 10382 10384 10386 10388 TO 10396 -
10439 TO 10452 10454 10456 10458 10460 TO 10468 10511 TO 10524 10526 10528 -
10530 10532 TO 10540 PR GX 0.56
10577 TO 10596 10598 10600 10602 10604 TO 10612 PR GX 0.619
10649 TO 10668 10670 10672 10674 10676 TO 10684 10721 TO 10756 PR GX 0.652
101 TO 136 201 TO 236 PR GX 0.678
301 TO 336 401 TO 436 PR GX 0.697
501 TO 536 601 TO 636 PR GX 0.717
701 TO 736 801 TO 836 PR GX 0.736
901 TO 936 1001 TO 1036 PR GX 0.756
1101 TO 1136 1201 TO 1236 PR GX 0.769
1301 TO 1336 1401 TO 1436 PR GX 0.788
1501 TO 1536 1601 TO 1636 1701 TO 1736 PR GX 0.821
1801 TO 1836 1901 TO 1936 2001 TO 2036 2101 TO 2136 PR GX 0.847
2201 TO 2236 2301 TO 2336 2401 TO 2436 PR GX 0.873
*
* LEEWARD SIDE
10253 TO 10288 10325 TO 10354 10397 TO 10404 10406 10408 10410 10412 TO 10426 -
10469 TO 10476 10478 10480 10482 10484 TO 10498 10541 TO 10548 10550 10552 -
10554 10556 TO 10570 PR GX 0.56
10613 TO 10620 10622 10624 10626 10628 TO 10648 PR GX 0.619
10685 TO 10692 10694 10696 10698 10700 TO 10720 10757 TO 10792 PR GX 0.652
137 TO 172 237 TO 272 PR GX 0.678
337 TO 372 437 TO 472 PR GX 0.697
537 TO 572 637 TO 672 PR GX 0.717
737 TO 772 837 TO 872 PR GX 0.736
937 TO 972 1037 TO 1072 PR GX 0.756
1137 TO 1172 1237 TO 1272 PR GX 0.769
1337 TO 1372 1437 TO 1472 PR GX 0.788
1537 TO 1572 1637 TO 1672 1737 TO 1772 PR GX 0.821
1837 TO 1872 1937 TO 1972 2037 TO 2072 2137 TO 2172 PR GX 0.847
2237 TO 2272 2337 TO 2372 2437 TO 2472 PR GX 0.873
*
*
*****************************
LOAD 4 WIND LOAD ALONG Z (W1B)
*****************************
*
ELEMENT LOAD
* WINDWARD SIDE
```

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*
3561. 10217 10219 10221 10223 10225 10227 10229 10231 10233 10235 10237 10239 10241 -
3562. 10243 10245 10247 10249 10251 10253 10255 10257 10259 10261 10263 10265 -
3563. 10267 10269 10271 10273 10275 10277 10279 10281 10283 10285 10287 10295 -
3564. 10297 10299 10301 10303 10305 10307 10309 10311 10313 10315 10317 10319 -
3565. 10321 10323 10325 10327 10329 10331 10333 10335 10337 10339 10341 10343 -
3566. 10345 10347 10349 10351 10353 10367 10369 10371 10373 10375 10377 10379 -
3567. 10389 10391 10393 10395 10397 10399 10401 10403 10413 10415 10417 10419 -
3568. 10421 10423 10425 10439 10441 10443 10445 10447 10449 10451 10461 10463 -
3569. 10465 10467 10469 10471 10473 10475 10485 10487 10489 10491 10493 10495 -
3570. 10497 10511 10513 10515 10517 10519 10521 10523 10533 10535 10537 10539 -
3571. 10541 10543 10545 10547 10557 10559 10561 10563 10565 10567 -
3572. 10569 PR GZ 0.56
3573. 10577 10579 10581 10583 10585 10587 10589 10591 10593 10595 10605 10607 10609 -
3574. 10611 10613 10615 10617 10619 10629 10631 10633 10635 10637 10639 10641 -
3575. 10643 10645 10647 PR GZ 0.619
3576. 10649 10651 10653 10655 10657 10659 10661 10663 10665 10667 10677 10679 10681 -
3577. 10683 10685 10687 10689 10691 10701 10703 10705 10707 10709 10711 10713 -
3578. 10715 10717 10719 PR GZ 0.652
3579. 101 103 105 107 109 111 113 115 117 119 121 123 125 127 129 131 133 135 137 -
3580. 139 141 143 145 147 149 151 153 155 157 159 161 163 165 167 169 171 10721 -
3581. 10723 10725 10727 10729 10731 10733 10735 10737 10739 10741 10743 10745 -
3582. 10747 10749 10751 10753 10755 10757 10759 10761 10763 10765 10767 10769 -
3583. 10771 10773 10775 10777 10779 10781 10783 10785 10787 10789 -
3584. 10791 PR GZ 0.678
3585. 201 203 205 207 209 211 213 215 217 219 221 223 225 227 229 231 233 235 237 -
3586. 239 241 243 245 247 249 251 253 255 257 259 261 263 265 267 269 271 301 303 -
3587. 305 307 309 311 313 315 317 319 321 323 325 327 329 331 333 335 337 339 341 -
3588. 343 345 347 349 351 353 355 357 359 361 363 365 367 369 371 PR GZ 0.697
3589. 401 403 405 407 409 411 413 415 417 419 421 423 425 427 429 431 433 435 437 -
3590. 439 441 443 445 447 449 451 453 455 457 459 461 463 465 467 469 471 501 503 -
3591. 505 507 509 511 513 515 517 519 521 523 525 527 529 531 533 535 537 539 541 -
3592. 543 545 547 549 551 553 555 557 559 561 563 565 567 569 571 PR GZ 0.717
3593. 601 603 605 607 609 611 613 615 617 619 621 623 625 627 629 631 633 635 637 -
3594. 639 641 643 645 647 649 651 653 655 657 659 661 663 665 667 669 671 701 703 -
3595. 705 707 709 711 713 715 717 719 721 723 725 727 729 731 733 735 737 739 741 -
3596. 743 745 747 749 751 753 755 757 759 761 763 765 767 769 771 PR GZ 0.736
3597. 801 803 805 807 809 811 813 815 817 819 821 823 825 827 829 831 833 835 837 -
3598. 839 841 843 845 847 849 851 853 855 857 859 861 863 865 867 869 871 901 903 -
3599. 905 907 909 911 913 915 917 919 921 923 925 927 929 931 933 935 937 939 941 -
3600. 943 945 947 949 951 953 955 957 959 961 963 965 967 969 971 PR GZ 0.756
3601. 1001 1003 1005 1007 1009 1011 1013 1015 1017 1019 1021 1023 1025 1027 1029 -
3602. 1031 1033 1035 1037 1039 1041 1043 1045 1047 1049 1051 1053 1055 1057 1059 -
3603. 1061 1063 1065 1067 1069 1071 1101 1103 1105 1107 1109 1111 1113 1115 1117 -
3604. 1119 1121 1123 1125 1127 1129 1131 1133 1135 1137 1139 1141 1143 1145 1147 -
3605. 1149 1151 1153 1155 1157 1159 1161 1163 1165 1167 1169 1171 PR GZ 0.769
3606. 1201 1203 1205 1207 1209 1211 1213 1215 1217 1219 1221 1223 1225 1227 1229 -
3607. 1231 1233 1235 1237 1239 1241 1243 1245 1247 1249 1251 1253 1255 1257 1259 -
3608. 1261 1263 1265 1267 1269 1271 1301 1303 1305 1307 1309 1311 1313 1315 1317 -
3609. 1319 1321 1323 1325 1327 1329 1331 1333 1335 1337 1339 1341 1343 1345 1347 -
3610. 1349 1351 1353 1355 1357 1359 1361 1363 1365 1367 1369 1371 PR GZ 0.788
3611. 1401 1403 1405 1407 1409 1411 1413 1415 1417 1419 1421 1423 1425 1427 1429 -
3612. 1431 1433 1435 1437 1439 1441 1443 1445 1447 1449 1451 1453 1455 1457 1459 -
3613. 1461 1463 1465 1467 1469 1471 1501 1503 1505 1507 1509 1511 1513 1515 1517 -
3614. 1519 1521 1523 1525 1527 1529 1531 1533 1535 1537 1539 1541 1543 1545 1547 -
3615. 1549 1551 1553 1555 1557 1559 1561 1563 1565 1567 1569 1571 1601 1603 1605 -
3616. 1607 1609 1611 1613 1615 1617 1619 1621 1623 1625 1627 1629 1631 1633 1635 -
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3617. 1637 1639 1641 1643 1645 1647 1649 1651 1653 1655 1657 1659 1661 1663 1665 -
3618. 1667 1669 1671 PR GZ 0.821
3619. 1701 1703 1705 1707 1709 1711 1713 1715 1717 1719 1721 1723 1725 1727 1729 -
3620. 1731 1733 1735 1737 1739 1741 1743 1745 1747 1749 1751 1753 1755 1757 1759 -
3621. 1761 1763 1765 1767 1769 1771 1801 1803 1805 1807 1809 1811 1813 1815 1817 -
3622. 1819 1821 1823 1825 1827 1829 1831 1833 1835 1837 1839 1841 1843 1845 1847 -
3623. 1849 1851 1853 1855 1857 1859 1861 1863 1865 1867 1869 1871 1901 1903 1905 -
3624. 1907 1909 1911 1913 1915 1917 1919 1921 1923 1925 1927 1929 1931 1933 1935 -
3625. 1937 1939 1941 1943 1945 1947 1949 1951 1953 1955 1957 1959 1961 1963 1965 -
3626. 1967 1969 1971 2001 2003 2005 2007 2009 2011 2013 2015 2017 2019 2021 2023 -
3627. 2025 2027 2029 2031 2033 2035 2037 2039 2041 2043 2045 2047 2049 2051 2053 -
3628. 2055 2057 2059 2061 2063 2065 2067 2069 2071 2101 2103 2105 2107 2109 2111 -
3629. 2113 2115 2117 2119 2121 2123 2125 2127 2129 2131 2133 2135 2137 2139 2141 -
3630. 2143 2145 2147 2149 2151 2153 2155 2157 2159 2161 2163 2165 2167 2169 -
3631. 2171 PR GZ 0.847
3632. 2201 2203 2205 2207 2209 2211 2213 2215 2217 2219 2221 2223 2225 2227 2229 -
3633. 2231 2233 2235 2237 2239 2241 2243 2245 2247 2249 2251 2253 2255 2257 2259 -
3634. 2261 2263 2265 2267 2269 2271 2301 2303 2305 2307 2309 2311 2313 2315 2317 -
3635. 2319 2321 2323 2325 2327 2329 2331 2333 2335 2337 2339 2341 2343 2345 2347 -
3636. 2349 2351 2353 2355 2357 2359 2361 2363 2365 2367 2369 2371 2401 2403 2405 -
3637. 2407 2409 2411 2413 2415 2417 2419 2421 2423 2425 2427 2429 2431 2433 2435 -
3638. 2437 2439 2441 2443 2445 2447 2449 2451 2453 2455 2457 2459 2461 2463 2465 -
3639. 2467 2469 2471 PR GZ 0.873
3640. *
3641. * LEEWARD SIDE
3642. 10218 10220 10222 10224 10226 10228 10230 10232 10234 10236 10238 10240 10242 -
3643. 10244 10246 10248 10250 10252 10254 10256 10258 10260 10262 10264 10266 -
3644. 10268 10270 10272 10274 10276 10278 10280 10282 10284 10286 10288 10296 -
3645. 10298 10300 10302 10304 10306 10308 10310 10312 10314 10316 10318 10320 -
3646. 10322 10324 10326 10328 10330 10332 10334 10336 10338 10340 10342 10344 -
3647. 10346 10348 10350 10352 10354 10368 10370 10372 10374 10376 10378 10380 -
3648. 10382 10384 10386 10388 10390 10392 10394 10396 10398 10400 10402 10404 -
3649. 10406 10408 10410 10412 10414 10416 10418 10420 10422 10424 10426 10440 -
3650. 10442 10444 10446 10448 10450 10452 10454 10456 10458 10460 10462 10464 -
3651. 10466 10468 10470 10472 10474 10476 10478 10480 10482 10484 10486 10488 -
3652. 10490 10492 10494 10496 10498 10512 10514 10516 10518 10520 10522 10524 -
3653. 10526 10528 10530 10532 10534 10536 10538 10540 10542 10544 10546 10548 -
3654. 10550 10552 10554 10556 10558 10560 10562 10564 10566 10568 -
3655. 10570 PR GZ 0.56
3656. 10578 10580 10582 10584 10586 10588 10590 10592 10594 10596 10598 10600 10602 -
3657. 10604 10606 10608 10610 10612 10614 10616 10618 10620 10622 10624 10626 -
3658. 10628 10630 10632 10634 10636 10638 10640 10642 10644 10646 -
3659. 10648 PR GZ 0.619
3660. 10650 10652 10654 10656 10658 10660 10662 10664 10666 10668 10670 10672 10674 -
3661. 10676 10678 10680 10682 10684 10686 10688 10690 10692 10694 10696 10698 -
3662. 10700 10702 10704 10706 10708 10710 10712 10714 10716 10718 -
3663. 10720 PR GZ 0.652
3664. 102 104 106 108 110 112 114 116 118 120 122 124 126 128 130 132 134 136 138 -
3665. 140 142 144 146 148 150 152 154 156 158 160 162 164 166 168 170 172 10722 -
3666. 10724 10726 10728 10730 10732 10734 10736 10738 10740 10742 10744 10746 -
3667. 10748 10750 10752 10754 10756 10758 10760 10762 10764 10766 10768 10770 -
3668. 10772 10774 10776 10778 10780 10782 10784 10786 10788 10790 -
3669. 10792 PR GZ 0.678
3670. 202 204 206 208 210 212 214 216 218 220 222 224 226 228 230 232 234 236 238 -
3671. 240 242 244 246 248 250 252 254 256 258 260 262 264 266 268 270 272 302 304 -
3672. 306 308 310 312 314 316 318 320 322 324 326 328 330 332 334 336 338 340 342 -
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3673. 344 346 348 350 352 354 356 358 360 362 364 366 368 370 372 PR GZ 0.697
3674. 402 404 406 408 410 412 414 416 418 420 422 424 426 428 430 432 434 436 438 -
3675. 440 442 444 446 448 450 452 454 456 458 460 462 464 466 468 470 472 502 504 -
3676. 506 508 510 512 514 516 518 520 522 524 526 528 530 532 534 536 538 540 542 -
3677. 544 546 548 550 552 554 556 558 560 562 564 566 568 570 572 PR GZ 0.717
3678. 602 604 606 608 610 612 614 616 618 620 622 624 626 628 630 632 634 636 638 -
3679. 640 642 644 646 648 650 652 654 656 658 660 662 664 666 668 670 672 702 704 -
3680. 706 708 710 712 714 716 718 720 722 724 726 728 730 732 734 736 738 740 742 -
3681. 744 746 748 750 752 754 756 758 760 762 764 766 768 770 772 PR GZ 0.736
3682. 802 804 806 808 810 812 814 816 818 820 822 824 826 828 830 832 834 836 838 -
3683. 840 842 844 846 848 850 852 854 856 858 860 862 864 866 868 870 872 902 904 -
3684. 906 908 910 912 914 916 918 920 922 924 926 928 930 932 934 936 938 940 942 -
3685. 944 946 948 950 952 954 956 958 960 962 964 966 968 970 972 PR GZ 0.756
3686. 1002 1004 1006 1008 1010 1012 1014 1016 1018 1020 1022 1024 1026 1028 1030 -
3687. 1032 1034 1036 1038 1040 1042 1044 1046 1048 1050 1052 1054 1056 1058 1060 -
3688. 1062 1064 1066 1068 1070 1072 1102 1104 1106 1108 1110 1112 1114 1116 1118 -
3689. 1120 1122 1124 1126 1128 1130 1132 1134 1136 1138 1140 1142 1144 1146 1148 -
3690. 1150 1152 1154 1156 1158 1160 1162 1164 1166 1168 1170 1172 PR GZ 0.769
3691. 1202 1204 1206 1208 1210 1212 1214 1216 1218 1220 1222 1224 1226 1228 1230 -
3692. 1232 1234 1236 1238 1240 1242 1244 1246 1248 1250 1252 1254 1256 1258 1260 -
3693. 1262 1264 1266 1268 1270 1272 1302 1304 1306 1308 1310 1312 1314 1316 1318 -
3694. 1320 1322 1324 1326 1328 1330 1332 1334 1336 1338 1340 1342 1344 1346 1348 -
3695. 1350 1352 1354 1356 1358 1360 1362 1364 1366 1368 1370 1372 PR GZ 0.788
3696. 1402 1404 1406 1408 1410 1412 1414 1416 1418 1420 1422 1424 1426 1428 1430 -
3697. 1432 1434 1436 1438 1440 1442 1444 1446 1448 1450 1452 1454 1456 1458 1460 -
3698. 1462 1464 1466 1468 1470 1472 1502 1504 1506 1508 1510 1512 1514 1516 1518 -
3699. 1520 1522 1524 1526 1528 1530 1532 1534 1536 1538 1540 1542 1544 1546 1548 -
3700. 1550 1552 1554 1556 1558 1560 1562 1564 1566 1568 1570 1572 1602 1604 1606 -
3701. 1608 1610 1612 1614 1616 1618 1620 1622 1624 1626 1628 1630 1632 1634 1636 -
3702. 1638 1640 1642 1644 1646 1648 1650 1652 1654 1656 1658 1660 1662 1664 1666 -
3703. 1668 1670 1672 PR GZ 0.821
3704. 1702 1704 1706 1708 1710 1712 1714 1716 1718 1720 1722 1724 1726 1728 1730 -
3705. 1732 1734 1736 1738 1740 1742 1744 1746 1748 1750 1752 1754 1756 1758 1760 -
3706. 1762 1764 1766 1768 1770 1772 1802 1804 1806 1808 1810 1812 1814 1816 1818 -
3707. 1820 1822 1824 1826 1828 1830 1832 1834 1836 1838 1840 1842 1844 1846 1848 -
3708. 1850 1852 1854 1856 1858 1860 1862 1864 1866 1868 1870 1872 1902 1904 1906 -
3709. 1908 1910 1912 1914 1916 1918 1920 1922 1924 1926 1928 1930 1932 1934 1936 -
3710. 1938 1940 1942 1944 1946 1948 1950 1952 1954 1956 1958 1960 1962 1964 1966 -
3711. 1968 1970 1972 2002 2004 2006 2008 2010 2012 2014 2016 2018 2020 2022 2024 -
3712. 2026 2028 2030 2032 2034 2036 2038 2040 2042 2044 2046 2048 2050 2052 2054 -
3713. 2056 2058 2060 2062 2064 2066 2068 2070 2072 2102 2104 2106 2108 2110 2112 -
3714. 2114 2116 2118 2120 2122 2124 2126 2128 2130 2132 2134 2136 2138 2140 2142 -
3715. 2144 2146 2148 2150 2152 2154 2156 2158 2160 2162 2164 2166 2168 2170 -
3716. 2172 PR GZ 0.847
3717. 2202 2204 2206 2208 2210 2212 2214 2216 2218 2220 2222 2224 2226 2228 2230 -
3718. 2232 2234 2236 2238 2240 2242 2244 2246 2248 2250 2252 2254 2256 2258 2260 -
3719. 2262 2264 2266 2268 2270 2272 2302 2304 2306 2308 2310 2312 2314 2316 2318 -
3720. 2320 2322 2324 2326 2328 2330 2332 2334 2336 2338 2340 2342 2344 2346 2348 -
3721. 2350 2352 2354 2356 2358 2360 2362 2364 2366 2368 2370 2372 2402 2404 2406 -
3722. 2408 2410 2412 2414 2416 2418 2420 2422 2424 2426 2428 2430 2432 2434 2436 -
3723. 2438 2440 2442 2444 2446 2448 2450 2452 2454 2456 2458 2460 2462 2464 2466 -
3724. 2468 2470 2472 PR GZ 0.873
3725. *
3726. *
3727. ******************************************************
3728. LOAD 11 MATERIAL FILLING WALL HORIZONTAL CASE-I (H1A)
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3729. ******************************************************
3730. ELEMENT LOAD
3731. 2001 TO 2072 PR 18.5
3732. 1901 TO 1972 PR 34.4
3733. 1801 TO 1872 PR 48.1
3734. 1701 TO 1772 PR 60.1
3735. 1601 TO 1672 PR 70.6
3736. 1501 TO 1572 PR 80
3737. 1401 TO 1472 PR 88.3
3738. 1301 TO 1372 PR 95.7
3739. 1201 TO 1272 PR 102.3
3740. 1101 TO 1172 PR 108.4
3741. 1001 TO 1072 PR 113.8
3742. 901 TO 972 PR 118.8
3743. 801 TO 872 PR 123.3
3744. 701 TO 772 PR 127.5
3745. 601 TO 672 PR 131.3
3746. 501 TO 572 PR 134.8
3747. 401 TO 472 PR 138.1
3748. 301 TO 372 PR 141.1
3749. 201 TO 272 PR 143.8
3750. 101 TO 172 PR 146.4
3751. *
3752. ******************************************************
3753. LOAD 12 MATERIAL FILLING WALL HORIZONTAL CASE-II (H1B)
3754. ******************************************************
3755. ELEMENT LOAD
3756. 2001 TO 2072 PR 22.6
3757. 1901 TO 1972 PR 41.2
3758. 1801 TO 1872 PR 56.8
3759. 1701 TO 1772 PR 70.1
3760. 1601 TO 1672 PR 81.5
3761. 1501 TO 1572 PR 91.3
3762. 1401 TO 1472 PR 99.9
3763. 1301 TO 1372 PR 107.4
3764. 1201 TO 1272 PR 114.1
3765. 1101 TO 1172 PR 120
3766. 1001 TO 1072 PR 125.3
3767. 901 TO 972 PR 130.1
3768. 801 TO 872 PR 134.4
3769. 701 TO 772 PR 138.3
3770. 601 TO 672 PR 141.8
3771. 501 TO 572 PR 145.1
3772. 401 TO 472 PR 148.1
3773. 301 TO 372 PR 150.8
3774. 201 TO 272 PR 153.3
3775. 101 TO 172 PR 155.7
3776. *
3777. ******************************************************
3778. LOAD 13 MATERIAL FILLING WALL HORIZONTAL CASE-III (H1C)
3779. ******************************************************
3780. ELEMENT LOAD
3781. 2001 TO 2072 PR 14.2
3782. 1901 TO 1972 PR 26.8
3783. 1801 TO 1872 PR 38.1
3784. 1701 TO 1772 PR 48.3
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1601 TO 1672 PR 57.5
1501 TO 1572 PR 65.8
1401 TO 1472 PR 73.4
1301 TO 1372 PR 80.4
1201 TO 1272 PR 86.8
1101 TO 1172 PR 92.6
1001 TO 1072 PR 98
901 TO 972 PR 103
801 TO 872 PR 107.6
701 TO 772 PR 111.9
601 TO 672 PR 115.9
501 TO 572 PR 119.6
401 TO 472 PR 123.1
301 TO 372 PR 126.3
201 TO 272 PR 129.4
101 TO 172 PR 132.3
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******************************************************
LOAD 14 MATERIAL EMPTYING WALL HORIZONTAL CASE-I (H2A)
******************************************************
ELEMENT LOAD
2001 TO 2072 PR 19.9
1901 TO 1972 PR 37
1801 TO 1872 PR 51.8
1701 TO 1772 PR 64.7
1601 TO 1672 PR 76.1
1501 TO 1572 PR 86.1
1401 TO 1472 PR 95.1
1301 TO 1372 PR 103.1
1201 TO 1272 PR 110.2
1101 TO 1172 PR 116.7
1001 TO 1072 PR 122.6
901 TO 972 PR 127.9
801 TO 872 PR 132.8
701 TO 772 PR 137.3
601 TO 672 PR 141.4
501 TO 572 PR 145.2
401 TO 472 PR 148.7
301 TO 372 PR 151.9
201 TO 272 PR 154.9
101 TO 172 PR 157.7
*
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LOAD 15 MATERIAL EMPTYING WALL HORIZONTAL CASE-II (H2B)
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ELEMENT LOAD
2001 TO 2072 PR 24.3
1901 TO 1972 PR 44.4
1801 TO 1872 PR 61.2
1701 TO 1772 PR 75.5
1601 TO 1672 PR 87.7
1501 TO 1572 PR 98.3
1401 TO 1472 PR 107.6
1301 TO 1372 PR 115.7
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3841. 1201 TO 1272 PR 122.9
3842. 1101 TO 1172 PR 129.2
3843. 1001 TO 1072 PR 135
3844. 901 TO 972 PR 140.1
3845. 801 TO 872 PR 144.7
3846. 701 TO 772 PR 148.9
3847. 601 TO 672 PR 152.8
3848. 501 TO 572 PR 156.3
3849. 401 TO 472 PR 159.5
3850. 301 TO 372 PR 162.4
3851. 201 TO 272 PR 165.2
3852. 101 TO 172 PR 167.7
3853. *
3854. *
3855. ******************************************************
3856. LOAD 16 MATERIAL EMPTYING WALL HORIZONTAL CASE-III (H2C)
3857. ******************************************************
3858. ELEMENT LOAD
3859. 2001 TO 2072 PR 15.2
3860. 1901 TO 1972 PR 28.8
3861. 1801 TO 1872 PR 41
3862. 1701 TO 1772 PR 52
3863. 1601 TO 1672 PR 61.9
3864. 1501 TO 1572 PR 70.9
3865. 1401 TO 1472 PR 79.1
3866. 1301 TO 1372 PR 86.6
3867. 1201 TO 1272 PR 93.4
3868. 1101 TO 1172 PR 99.8
3869. 1001 TO 1072 PR 105.6
3870. 901 TO 972 PR 110.9
3871. 801 TO 872 PR 115.9
3872. 701 TO 772 PR 120.5
3873. 601 TO 672 PR 124.8
3874. 501 TO 572 PR 128.8
3875. 401 TO 472 PR 132.6
3876. 301 TO 372 PR 136.1
3877. 201 TO 272 PR 139.4
3878. 101 TO 172 PR 142.4
3879. *
3880. *
3881. ****************************************************************
3882. LOAD 21 MATERIAL WALL VERTICAL (FILLING _EMPTYING) CASE-I (H3A)
3883. ****************************************************************
3884. ELEMENT LOAD
3885. 2001 TO 2072 PR GY -5.9
3886. 1901 TO 1972 PR GY -10.9
3887. 1801 TO 1872 PR GY -15.3
3888. 1701 TO 1772 PR GY -19.1
3889. 1601 TO 1672 PR GY -22.5
3890. 1501 TO 1572 PR GY -25.4
3891. 1401 TO 1472 PR GY -28.1
3892. 1301 TO 1372 PR GY -30.4
3893. 1201 TO 1272 PR GY -32.6
3894. 1101 TO 1172 PR GY -34.5
3895. 1001 TO 1072 PR GY -36.2
3896. 901 TO 972 PR GY -37.8
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*
801 TO 872 PR GY -39.2
701 TO 772 PR GY -40.5
601 TO 672 PR GY -41.8
501 TO 572 PR GY -42.9
401 TO 472 PR GY -43.9
301 TO 372 PR GY -44.9
201 TO 272 PR GY -45.8
101 TO 172 PR GY -46.6
*
*
*********************************************************************
LOAD 22 MATERIAL WALL VERTICAL (FILLING _EMPTYING) CASE-II (H3B)
*********************************************************************
ELEMENT LOAD
2001 TO 2072 PR GY -6.5
1901 TO 1972 PR GY -11.9
1801 TO 1872 PR GY -16.4
1701 TO 1772 PR GY -20.3
1601 TO 1672 PR GY -23.6
1501 TO 1572 PR GY -26.4
1401 TO 1472 PR GY -28.9
1301 TO 1372 PR GY -31.1
1201 TO 1272 PR GY -33
1101 TO 1172 PR GY -34.7
1001 TO 1072 PR GY -36.2
901 TO 972 PR GY -37.6
801 TO 872 PR GY -38.9
701 TO 772 PR GY -40
601 TO 672 PR GY -41
501 TO 572 PR GY -42
401 TO 472 PR GY -42.8
301 TO 372 PR GY -43.6
201 TO 272 PR GY -44.3
101 TO 172 PR GY -45
*
*
*********************************************************************
LOAD 23 MATERIAL WALL VERTICAL (FILLING _EMPTYING) CASE-III (H3C)
*********************************************************************
ELEMENT LOAD
2001 TO 2072 PR GY -5.2
1901 TO 1972 PR GY -9.8
1801 TO 1872 PR GY -14
1701 TO 1772 PR GY -17.8
1601 TO 1672 PR GY -21.1
1501 TO 1572 PR GY -24.2
1401 TO 1472 PR GY -27
1301 TO 1372 PR GY -29.6
1201 TO 1272 PR GY -31.9
1101 TO 1172 PR GY -34.1
1001 TO 1072 PR GY -36.1
901 TO 972 PR GY -37.9
801 TO 872 PR GY -39.6
701 TO 772 PR GY -41.2
601 TO 672 PR GY -42.6
501 TO 572 PR GY -44
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*
401 TO 472 PR GY -45.3
301 TO 372 PR GY -46.5
201 TO 272 PR GY -47.6
101 TO 172 PR GY -48.7
*
*
*********************************************************************
LOAD 24 MATERIAL WALL VERTICAL (FILLING _EMPTYING) CASE-I (H4A)
*********************************************************************
ELEMENT LOAD
2001 TO 2072 PR GY -9.8
1901 TO 1972 PR GY -18.2
1801 TO 1872 PR GY -25.5
1701 TO 1772 PR GY -31.8
1601 TO 1672 PR GY -37.4
1501 TO 1572 PR GY -42.4
1401 TO 1472 PR GY -46.8
1301 TO 1372 PR GY -50.7
1201 TO 1272 PR GY -54.2
1101 TO 1172 PR GY -57.4
1001 TO 1072 PR GY -60.3
901 TO 972 PR GY -63
801 TO 872 PR GY -65.4
701 TO 772 PR GY -67.6
601 TO 672 PR GY -69.6
501 TO 572 PR GY -71.4
401 TO 472 PR GY -73.2
301 TO 372 PR GY -74.8
201 TO 272 PR GY -76.2
101 TO 172 PR GY -77.6
*
*
*********************************************************************
LOAD 25 MATERIAL WALL VERTICAL (FILLING _EMPTYING) CASE-II (H4B)
*********************************************************************
ELEMENT LOAD
2001 TO 2072 PR GY -14.9
1901 TO 1972 PR GY -27.2
1801 TO 1872 PR GY -37.5
1701 TO 1772 PR GY -46.3
1601 TO 1672 PR GY -53.8
1501 TO 1572 PR GY -60.2
1401 TO 1472 PR GY -65.9
1301 TO 1372 PR GY -70.9
1201 TO 1272 PR GY -75.3
1101 TO 1172 PR GY -79.2
1001 TO 1072 PR GY -82.7
901 TO 972 PR GY -85.8
801 TO 872 PR GY -88.7
701 TO 772 PR GY -91.3
601 TO 672 PR GY -93.6
501 TO 572 PR GY -95.8
401 TO 472 PR GY -97.7
301 TO 372 PR GY -99.5
201 TO 272 PR GY -101.2
101 TO 172 PR GY -102.7
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4009. *
4010. *
4011. ************************************************************
4012. LOAD 26 MATERIAL WALL VERTICAL (FILLING _EMPTYING) CASE-III (H4C)
4013. ************************************************************
4014. ELEMENT LOAD
4015. 2001 TO 2072 PR GY -9.3
4016. 1901 TO 1972 PR GY -17.7
4017. 1801 TO 1872 PR GY -25.1
4018. 1701 TO 1772 PR GY -31.9
4019. 1601 TO 1672 PR GY -37.9
4020. 1501 TO 1572 PR GY -43.4
4021. 1401 TO 1472 PR GY -48.5
4022. 1301 TO 1372 PR GY -53.1
4023. 1201 TO 1272 PR GY -57.3
4024. 1101 TO 1172 PR GY -61.1
4025. 1001 TO 1072 PR GY -64.7
4026. 901 TO 972 PR GY -68
4027. 801 TO 872 PR GY -71
4028. 701 TO 772 PR GY -73.9
4029. 601 TO 672 PR GY -76.5
4030. 501 TO 572 PR GY -78.9
4031. 401 TO 472 PR GY -81.2
4032. 301 TO 372 PR GY -83.4
4033. 201 TO 272 PR GY -85.4
4034. 101 TO 172 PR GY -87.3
4035. *
4036. *
4037. ******************************************************
4038. LOAD 31 MATERIAL VERTICAL (ON FLAT SLAB) CASE-I (H5A)
4039. ******************************************************
4040. ELEMENT LOAD
4041. 12172 TO 12787 PR GY -320.1
4042. *
4043. ******************************************************
4044. LOAD 32 MATERIAL VERTICAL (ON FLAT SLAB) CASE-II (H5B)
4045. ******************************************************
4046. ELEMENT LOAD
4047. 12172 TO 12787 PR GY -302
4048. *
4049. *******************************************************
4050. LOAD 33 MATERIAL VERTICAL (ON FLAT SLAB) CASE-III (H5C)
4051. *******************************************************
4052. ELEMENT LOAD
4053. 12172 TO 12787 PR GY -344.5
4054. *
4055. ******************************************************
4056. LOAD 34 MATERIAL VERTICAL (ON FLAT SLAB) CASE-I (H6A)
4057. ******************************************************
4058. ELEMENT LOAD
4059. 12172 TO 12787 PR GY -247.9
4060. *
4061. *******************************************************
4062. LOAD 35 MATERIAL VERTICAL (ON FLAT SLAB) CASE-II (H6B)
4063. *******************************************************
4064. ELEMENT LOAD
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*
12172 TO 12787 PR GY -180.3
*
*******************************************************
LOAD 36 MATERIAL VERTICAL (ON FLAT SLAB) CASE-III (H6C)
*******************************************************
ELEMENT LOAD
12172 TO 12787 PR GY -233.9
*
*
********************
* LOAD COMBINATION
********************
*
* FACTORED LOAD COMBINATIONS FOR STRENGTH DESIGN
*
LOAD COMB 101 1.4X[ D ]
1 1.4
*
LOAD COMB 102 1.4X[ D ] + 1.7X[ L + H1A + H3A + H5A ]
1 1.4 2 1.7 11 1.7 21 1.7 31 1.7
*
LOAD COMB 103 1.4X[ D ] + 1.7X[ L + H1B + H3B + H5B ]
1 1.4 2 1.7 12 1.7 22 1.7 32 1.7
*
LOAD COMB 104 1.4X[ D ] + 1.7X[ L + H1C + H3C + H5C ]
1 1.4 2 1.7 13 1.7 23 1.7 33 1.7
*
LOAD COMB 105 1.4X[ D ] + 1.7X[ L + H1A + H4A + H6A ]
1 1.4 2 1.7 11 1.7 24 1.7 34 1.7
*
LOAD COMB 106 1.4X[ D ] + 1.7X[ L + H1B + H4B + H6B ]
1 1.4 2 1.7 12 1.7 25 1.7 35 1.7
*
LOAD COMB 107 1.4X[ D ] + 1.7X[ L + H1C + H4C + H6C ]
1 1.4 2 1.7 13 1.7 26 1.7 36 1.7
*
LOAD COMB 108 1.4X[ D ] + 1.7X[ L + H2A + H3A + H5A ]
1 1.4 2 1.7 14 1.7 21 1.7 31 1.7
*
LOAD COMB 109 1.4X[ D ] + 1.7X[ L + H2B + H3B + H5B ]
1 1.4 2 1.7 15 1.7 22 1.7 32 1.7
*
LOAD COMB 110 1.4X[ D ] + 1.7X[ L + H2C + H3C + H5C ]
1 1.4 2 1.7 16 1.7 23 1.7 33 1.7
*
LOAD COMB 111 1.4X[ D ] + 1.7X[ L + H2A + H4A + H6A ]
1 1.4 2 1.7 14 1.7 24 1.7 34 1.7
*
LOAD COMB 112 1.4X[ D ] + 1.7X[ L + H2B + H4B + H6B ]
1 1.4 2 1.7 15 1.7 25 1.7 35 1.7
*
LOAD COMB 113 1.4X[ D ] + 1.7X[ L + H2C + H4C + H6C ]
1 1.4 2 1.7 16 1.7 26 1.7 36 1.7
*
LOAD COMB 114 1.2X[ D ] + 1.0X[ L ]
1 1.2 2 1.0
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  *
4121. *
4122. LOAD COMB 115 1.2X[ D ] + 0.8X[ W1A ]
4123. 1 1.2 3 0.8
4124. *
4125. LOAD COMB 116 1.2X[ D ] + 0.8X[ W1B ]
4126. 1 1.2 4 0.8
4127. *
4128. LOAD COMB 117 1.2X[ D ] + 1.6X[ W1A ] + 1.0X[ L ]
4129. 1 1.2 3 1.6 2 1.0
4130. *
4131. LOAD COMB 118 1.2X[ D ] + 1.6X[ W1B ] + 1.0X[ L ]
4132. 1 1.2 4 1.6 2 1.0
4133. *
4134. LOAD COMB 119 1.2X[ D ] + 1.0X[ E1A ] + 1.0X[ L ]
4135. 1 1.2 5 1.0 2 1.0
4136. *
4137. LOAD COMB 120 1.2X[ D ] + 1.0X[ E1B ] + 1.0X[ L ]
4138. 1 1.2 6 1.0 2 1.0
4139. *
4140. LOAD COMB 121 0.9X[ D ] + 1.6X[ W1A + H1A + H3A + H5A ]
4141. 1 0.9 3 1.6 11 1.6 21 1.6 31 1.6
4142. *
4143. LOAD COMB 122 0.9X[ D ] + 1.6X[ W1A + H1B + H3B + H5B ]
4144. 1 0.9 3 1.6 12 1.6 22 1.6 32 1.6
4145. *
4146. LOAD COMB 123 0.9X[ D ] + 1.6X[ W1A + H1C + H3C + H5C ]
4147. 1 0.9 3 1.6 13 1.6 23 1.6 33 1.6
4148. *
4149. LOAD COMB 124 0.9X[ D ] + 1.6X[ W1A + H1A + H4A + H6A ]
4150. 1 0.9 3 1.6 11 1.6 24 1.6 34 1.6
4151. *
4152. LOAD COMB 125 0.9X[ D ] + 1.6X[ W1A + H1B + H4B + H6B ]
4153. 1 0.9 3 1.6 12 1.6 25 1.6 35 1.6
4154. *
4155. LOAD COMB 126 0.9X[ D ] + 1.6X[ W1A + H1C + H4C + H6C ]
4156. 1 0.9 3 1.6 13 1.6 26 1.6 36 1.6
4157. *
4158. LOAD COMB 127 0.9X[ D ] + 1.6X[ W1A + H2A + H3A + H5A ]
4159. 1 0.9 3 1.6 14 1.6 21 1.6 31 1.6
4160. *
4161. LOAD COMB 128 0.9X[ D ] + 1.6X[ W1A + H2B + H3B + H5B ]
4162. 1 0.9 3 1.6 15 1.6 22 1.6 32 1.6
4163. *
4164. LOAD COMB 129 0.9X[ D ] + 1.6X[ W1A + H2C + H3C + H5C ]
4165. 1 0.9 3 1.6 16 1.6 23 1.6 33 1.6
4166. *
4167. LOAD COMB 130 0.9X[ D ] + 1.6X[ W1A + H2A + H4A + H6A ]
4168. 1 0.9 3 1.6 14 1.6 24 1.6 34 1.6
4169. *
4170. LOAD COMB 131 0.9X[ D ] + 1.6X[ W1A + H2B + H4B + H6B ]
4171. 1 0.9 3 1.6 15 1.6 25 1.6 35 1.6
4172. *
4173. LOAD COMB 132 0.9X[ D ] + 1.6X[ W1A + H2C + H4C + H6C ]
4174. 1 0.9 3 1.6 16 1.6 26 1.6 36 1.6
4175. *
4176. LOAD COMB 133 0.9X[ D ] + 1.6X[ W1B + H1A + H3A + H5A ]
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*
4177. 1 0.9 4 1.6 11 1.6 21 1.6 31 1.6
4178. *
4179. LOAD COMB 134 0.9X[ D ] + 1.6X[ W1B + H1B + H3B + H5B ]
4180. 1 0.9 4 1.6 12 1.6 22 1.6 32 1.6
4181. *
4182. LOAD COMB 135 0.9X[ D ] + 1.6X[ W1B + H1C + H3C + H5C ]
4183. 1 0.9 4 1.6 13 1.6 23 1.6 33 1.6
4184. *
4185. LOAD COMB 136 0.9X[ D ] + 1.6X[ W1B + H1A + H4A + H6A ]
4186. 1 0.9 4 1.6 11 1.6 24 1.6 34 1.6
4187. *
4188. LOAD COMB 137 0.9X[ D ] + 1.6X[ W1B + H1B + H4B + H6B ]
4189. 1 0.9 4 1.6 12 1.6 25 1.6 35 1.6
4190. *
4191. LOAD COMB 138 0.9X[ D ] + 1.6X[ W1B + H1C + H4C + H6C ]
4192. 1 0.9 4 1.6 13 1.6 26 1.6 36 1.6
4193. *
4194. LOAD COMB 139 0.9X[ D ] + 1.6X[ W1B + H2A + H3A + H5A ]
4195. 1 0.9 4 1.6 14 1.6 21 1.6 31 1.6
4196. *
4197. LOAD COMB 140 0.9X[ D ] + 1.6X[ W1B + H2B + H3B + H5B ]
4198. 1 0.9 4 1.6 15 1.6 22 1.6 32 1.6
4199. *
4200. LOAD COMB 141 0.9X[ D ] + 1.6X[ W1B + H2C + H3C + H5C ]
4201. 1 0.9 4 1.6 16 1.6 23 1.6 33 1.6
4202. *
4203. LOAD COMB 142 0.9X[ D ] + 1.6X[ W1B + H2A + H4A + H6A ]
4204. 1 0.9 4 1.6 14 1.6 24 1.6 34 1.6
4205. *
4206. LOAD COMB 143 0.9X[ D ] + 1.6X[ W1B + H2B + H4B + H6B ]
4207. 1 0.9 4 1.6 15 1.6 25 1.6 35 1.6
4208. *
4209. LOAD COMB 144 0.9X[ D ] + 1.6X[ W1B + H2C + H4C + H6C ]
4210. 1 0.9 4 1.6 16 1.6 26 1.6 36 1.6
4211. *
4212. LOAD COMB 145 0.9X[ D ] + 1.0X[ E1A ] + 1.6X 0.8X [ H1A + H3A + H5A ]
4213. 1 0.9 5 1.0 11 1.28 21 1.28 31 1.28
4214. *
4215. LOAD COMB 146 0.9X[ D ] + 1.0X[ E1A ] + 1.6X 0.8X[ H1B + H3B + H5B ]
4216. 1 0.9 5 1.0 12 1.28 22 1.28 32 1.28
4217. *
4218. LOAD COMB 147 0.9X[ D ] + 1.0X[ E1A ] + 1.6X 0.8X[ H1C + H3C + H5C ]
4219. 1 0.9 5 1.0 13 1.28 23 1.28 33 1.28
4220. *
4221. LOAD COMB 148 0.9X[ D ] + 1.0X[ E1A ] + 1.6X 0.8X[ H1A + H4A + H6A ]
4222. 1 0.9 5 1.0 11 1.28 24 1.28 34 1.28
4223. *
4224. LOAD COMB 149 0.9X[ D ] + 1.0X[ E1A ] + 1.6X 0.8X[ H1B + H4B + H6B ]
4225. 1 0.9 5 1.0 12 1.28 25 1.28 35 1.28
4226. *
4227. LOAD COMB 150 0.9X[ D ] + 1.0X[ E1A ] + 1.6X 0.8X[ H1C + H4C + H6C ]
4228. 1 0.9 5 1.0 13 1.28 26 1.28 36 1.28
4229. *
4230. LOAD COMB 151 0.9X[ D ] + 1.0X[ E1A ] + 1.6X 0.8X[ H2A + H3A + H5A ]
4231. 1 0.9 5 1.0 14 1.28 21 1.28 31 1.28
4232. *
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  *
4233. LOAD COMB 152 0.9X[ D ] + 1.0X[ E1A ] + 1.6X 0.8X[ H2B + H3B + H5B ]
4234. 1 0.9 5 1.0 15 1.28 22 1.28 32 1.28
4235. *
4236. LOAD COMB 153 0.9X[ D ] + 1.0X[ E1A ] + 1.6X 0.8X[ H2C + H3C + H5C ]
4237. 1 0.9 5 1.0 16 1.28 23 1.28 33 1.28
4238. *
4239. LOAD COMB 154 0.9X[ D ] + 1.0X[ E1A ] + 1.6X 0.8X[ H2A + H4A + H6A ]
4240. 1 0.9 5 1.0 14 1.28 24 1.28 34 1.28
4241. *
4242. LOAD COMB 155 0.9X[ D ] + 1.0X[ E1A ] + 1.6X 0.8X[ H2B + H4B + H6B ]
4243. 1 0.9 5 1.0 15 1.28 25 1.28 35 1.28
4244. *
4245. LOAD COMB 156 0.9X[ D ] + 1.0X[ E1A ] + 1.6X 0.8X[ H2C + H4C + H6C ]
4246. 1 0.9 5 1.0 16 1.28 26 1.28 36 1.28
4247. *
4248. LOAD COMB 157 0.9X[ D ] + 1.0X[ E1B ] + 1.6X 0.8X[ H1A + H3A + H5A ]
4249. 1 0.9 6 1.0 11 1.28 21 1.28 31 1.28
4250. *
4251. LOAD COMB 158 0.9X[ D ] + 1.0X[ E1B ] + 1.6X 0.8X[ H1B + H3B + H5B ]
4252. 1 0.9 6 1.0 12 1.28 22 1.28 32 1.28
4253. *
4254. LOAD COMB 159 0.9X[ D ] + 1.0X[ E1B ] + 1.6X 0.8X[ H1C + H3C + H5C ]
4255. 1 0.9 6 1.0 13 1.28 23 1.28 33 1.28
4256. *
4257. LOAD COMB 160 0.9X[ D ] + 1.0X[ E1B ] + 1.6X 0.8X[ H1A + H4A + H6A ]
4258. 1 0.9 6 1.0 11 1.28 24 1.28 34 1.28
4259. *
4260. LOAD COMB 161 0.9X[ D ] + 1.0X[ E1B ] + 1.6X 0.8X[ H1B + H4B + H6B ]
4261. 1 0.9 6 1.0 12 1.28 25 1.28 35 1.28
4262. *
4263. LOAD COMB 162 0.9X[ D ] + 1.0X[ E1B ] + 1.6X 0.8X[ H1C + H4C + H6C ]
4264. 1 0.9 6 1.0 13 1.28 26 1.28 36 1.28
4265. *
4266. LOAD COMB 163 0.9X[ D ] + 1.0X[ E1B ] + 1.6X 0.8X[ H2A + H3A + H5A ]
4267. 1 0.9 6 1.0 14 1.28 21 1.28 31 1.28
4268. *
4269. LOAD COMB 164 0.9X[ D ] + 1.0X[ E1B ] + 1.6X 0.8X[ H2B + H3B + H5B ]
4270. 1 0.9 6 1.0 15 1.28 22 1.28 32 1.28
4271. *
4272. LOAD COMB 165 0.9X[ D ] + 1.0X[ E1B ] + 1.6X 0.8X[ H2C + H3C + H5C ]
4273. 1 0.9 6 1.0 16 1.28 23 1.28 33 1.28
4274. *
4275. LOAD COMB 166 0.9X[ D ] + 1.0X[ E1B ] + 1.6X 0.8X[ H2A + H4A + H6A ]
4276. 1 0.9 6 1.0 14 1.28 24 1.28 34 1.28
4277. *
4278. LOAD COMB 167 0.9X[ D ] + 1.0X[ E1B ] + 1.6X 0.8X[ H2B + H4B + H6B ]
4279. 1 0.9 6 1.0 15 1.28 25 1.28 35 1.28
4280. *
4281. LOAD COMB 168 0.9X[ D ] + 1.0X[ E1B ] + 1.6X 0.8X[ H2C + H4C + H6C ]
4282. 1 0.9 6 1.0 16 1.28 26 1.28 36 1.28
4283. *
4284. *
4285. * LOAD COMBINATIONS FOR SERVICEABILITY AND ALLOWABLE STRESS DESIGN
4286. *
4287. LOAD COMB 501 1X[ D + H1A + H3A + H5A + L ]
4288. 1 1.0 11 1.0 21 1.0 31 1.0 2 1.0
```

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  *
4289. *
4290. LOAD COMB 502 1X[ D + H1B + H3B + H5B + L ]
4291. 1 1.0 12 1.0 22 1.0 32 1.0 2 1.0
4292. *
4293. LOAD COMB 503 1X[ D + H1C + H3C + H5C + L ]
4294. 1 1.0 13 1.0 23 1.0 33 1.0 2 1.0
4295. *
4296. LOAD COMB 504 1X[ D + H1A + H4A + H6A + L ]
4297. 1 1.0 11 1.0 24 1.0 34 1.0 2 1.0
4298. *
4299. LOAD COMB 505 1X[ D + H1B + H4B + H6B + L ]
4300. 1 1.0 12 1.0 25 1.0 35 1.0 2 1.0
4301. *
4302. LOAD COMB 506 1X[ D + H1C + H4C + H6C + L ]
4303. 1 1.0 13 1.0 26 1.0 36 1.0 2 1.0
4304. *
4305. LOAD COMB 507 1X[ D + H2A + H3A + H5A + L ]
4306. 1 1.0 14 1.0 21 1.0 31 1.0 2 1.0
4307. *
4308. LOAD COMB 508 1X[ D + H2B + H3B + H5B + L ]
4309. 1 1.0 15 1.0 22 1.0 32 1.0 2 1.0
4310. *
4311. LOAD COMB 509 1X[ D + H2C + H3C + H5C + L ]
4312. 1 1.0 16 1.0 23 1.0 33 1.0 2 1.0
4313. *
4314. LOAD COMB 510 1X[ D + H2A + H4A + H6A + L ]
4315. 1 1.0 14 1.0 24 1.0 34 1.0 2 1.0
4316. *
4317. LOAD COMB 511 1X[ D + H2B + H4B + H6B + L ]
4318. 1 1.0 15 1.0 25 1.0 35 1.0 2 1.0
4319. *
4320. LOAD COMB 512 1X[ D + H2C + H4C + H6C + L ]
4321. 1 1.0 16 1.0 26 1.0 36 1.0 2 1.0
4322. *
4323. LOAD COMB 513 1X[ D + H1A + H3A + H5A ]
4324. 1 1.0 11 1.0 21 1.0 31 1.0
4325. *
4326. LOAD COMB 514 1X[ D + H1B + H3B + H5B ]
4327. 1 1.0 12 1.0 22 1.0 32 1.0
4328. *
4329. LOAD COMB 515 1X[ D + H1C + H3C + H5C ]
4330. 1 1.0 13 1.0 23 1.0 33 1.0
4331. *
4332. LOAD COMB 516 1X[ D + H1A + H4A + H6A ]
4333. 1 1.0 11 1.0 24 1.0 34 1.0
4334. *
4335. LOAD COMB 517 1X[ D + H1B + H4B + H6B ]
4336. 1 1.0 12 1.0 25 1.0 35 1.0
4337. *
4338. LOAD COMB 518 1X[ D + H1C + H4C + H6C ]
4339. 1 1.0 13 1.0 26 1.0 36 1.0
4340. *
4341. LOAD COMB 519 1X[ D + H2A + H3A + H5A ]
4342. 1 1.0 14 1.0 21 1.0 31 1.0
4343. *
4344. LOAD COMB 520 1X[ D + H2B + H3B + H5B ]
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  *
4345. 1 1.0 15 1.0 22 1.0 32 1.0
4346. *
4347. LOAD COMB 521 1X[ D + H2C + H3C + H5C ]
4348. 1 1.0 16 1.0 23 1.0 33 1.0
4349. *
4350. LOAD COMB 522 1X[ D + H2A + H4A + H6A ]
4351. 1 1.0 14 1.0 24 1.0 34 1.0
4352. *
4353. LOAD COMB 523 1X[ D + H2B + H4B + H6B ]
4354. 1 1.0 15 1.0 25 1.0 35 1.0
4355. *
4356. LOAD COMB 524 1X[ D + H2C + H4C + H6C ]
4357. 1 1.0 16 1.0 26 1.0 36 1.0
4358. *
4359. LOAD COMB 525 1X[ D + H1A + H3A + H5A ] + 0.75X[ L ]
4360. 1 1.0 11 1.0 21 1.0 31 1.0 2 0.75
4361. *
4362. LOAD COMB 526 1X[ D + H1B + H3B + H5B ] + 0.75X[ L ]
4363. 1 1.0 12 1.0 22 1.0 32 1.0 2 0.75
4364. *
4365. LOAD COMB 527 1X[ D + H1C + H3C + H5C ] + 0.75X[ L ]
4366. 1 1.0 13 1.0 23 1.0 33 1.0 2 0.75
4367. *
4368. LOAD COMB 528 1X[ D + H1A + H4A + H6A ] + 0.75X[ L ]
4369. 1 1.0 11 1.0 24 1.0 34 1.0 2 0.75
4370. *
4371. LOAD COMB 529 1X[ D + H1B + H4B + H6B ] + 0.75X[ L ]
4372. 1 1.0 12 1.0 25 1.0 35 1.0 2 0.75
4373. *
4374. LOAD COMB 530 1X[ D + H1C + H4C + H6C ] + 0.75X[ L ]
4375. 1 1.0 13 1.0 26 1.0 36 1.0 2 0.75
4376. *
4377. LOAD COMB 531 1X[ D + H2A + H3A + H5A ] + 0.75X[ L ]
4378. 1 1.0 14 1.0 21 1.0 31 1.0 2 0.75
4379. *
4380. LOAD COMB 532 1X[ D + H2B + H3B + H5B ] + 0.75X[ L ]
4381. 1 1.0 15 1.0 22 1.0 32 1.0 2 0.75
4382. *
4383. LOAD COMB 533 1X[ D + H2C + H3C + H5C ] + 0.75X[ L ]
4384. 1 1.0 16 1.0 23 1.0 33 1.0 2 0.75
4385. *
4386. LOAD COMB 534 1X[ D + H2A + H4A + H6A ] + 0.75X[ L ]
4387. 1 1.0 14 1.0 24 1.0 34 1.0 2 0.75
4388. *
4389. LOAD COMB 535 1X[ D + H2B + H4B + H6B ] + 0.75X[ L ]
4390. 1 1.0 15 1.0 25 1.0 35 1.0 2 0.75
4391. *
4392. LOAD COMB 536 1X[ D + H2C + H4C + H6C ] + 0.75X[ L ]
4393. 1 1.0 16 1.0 26 1.0 36 1.0 2 0.75
4394. *
4395. LOAD COMB 537 1X[ D + H1A + H3A + H5A + W1A ]
4396. 1 1.0 11 1.0 21 1.0 31 1.0 3 1.0
4397. *
4398. LOAD COMB 538 1X[ D + H1B + H3B + H5B + W1A ]
4399. 1 1.0 12 1.0 22 1.0 32 1.0 3 1.0
4400. *
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  *
4401. LOAD COMB 539 1X[ D + H1C + H3C + H5C + W1A ]
4402. 1 1.0 13 1.0 23 1.0 33 1.0 3 1.0
4403. *
4404. LOAD COMB 540 1X[ D + H1A + H4A + H6A + W1A ]
4405. 1 1.0 11 1.0 24 1.0 34 1.0 3 1.0
4406. *
4407. LOAD COMB 541 1X[ D + H1B + H4B + H6B + W1A ]
4408. 1 1.0 12 1.0 25 1.0 35 1.0 3 1.0
4409. *
4410. LOAD COMB 542 1X[ D + H1C + H4C + H6C + W1A ]
4411. 1 1.0 13 1.0 26 1.0 36 1.0 3 1.0
4412. *
4413. LOAD COMB 543 1X[ D + H2A + H3A + H5A + W1A ]
4414. 1 1.0 14 1.0 21 1.0 31 1.0 3 1.0
4415. *
4416. LOAD COMB 544 1X[ D + H2B + H3B + H5B + W1A ]
4417. 1 1.0 15 1.0 22 1.0 32 1.0 3 1.0
4418. *
4419. LOAD COMB 545 1X[ D + H2C + H3C + H5C + W1A ]
4420. 1 1.0 16 1.0 23 1.0 33 1.0 3 1.0
4421. *
4422. LOAD COMB 546 1X[ D + H2A + H4A + H6A + W1A ]
4423. 1 1.0 14 1.0 24 1.0 34 1.0 3 1.0
4424. *
4425. LOAD COMB 547 1X[ D + H2B + H4B + H6B + W1A ]
4426. 1 1.0 15 1.0 25 1.0 35 1.0 3 1.0
4427. *
4428. LOAD COMB 548 1X[ D + H2C + H4C + H6C + W1A ]
4429. 1 1.0 16 1.0 26 1.0 36 1.0 3 1.0
4430. *
4431. LOAD COMB 549 1X[ D + H1A + H3A + H5A + W1B ]
4432. 1 1.0 11 1.0 21 1.0 31 1.0 4 1.0
4433. *
4434. LOAD COMB 550 1X[ D + H1B + H3B + H5B + W1B ]
4435. 1 1.0 12 1.0 22 1.0 32 1.0 4 1.0
4436. *
4437. LOAD COMB 551 1X[ D + H1C + H3C + H5C + W1B ]
4438. 1 1.0 13 1.0 23 1.0 33 1.0 4 1.0
4439. *
4440. LOAD COMB 552 1X[ D + H1A + H4A + H6A + W1B ]
4441. 1 1.0 11 1.0 24 1.0 34 1.0 4 1.0
4442. *
4443. LOAD COMB 553 1X[ D + H1B + H4B + H6B + W1B ]
4444. 1 1.0 12 1.0 25 1.0 35 1.0 4 1.0
4445. *
4446. LOAD COMB 554 1X[ D + H1C + H4C + H6C + W1B ]
4447. 1 1.0 13 1.0 26 1.0 36 1.0 4 1.0
4448. *
4449. LOAD COMB 555 1X[ D + H2A + H3A + H5A + W1B ]
4450. 1 1.0 14 1.0 21 1.0 31 1.0 4 1.0
4451. *
4452. LOAD COMB 556 1X[ D + H2B + H3B + H5B + W1B ]
4453. 1 1.0 15 1.0 22 1.0 32 1.0 4 1.0
4454. *
4455. LOAD COMB 557 1X[ D + H2C + H3C + H5C + W1B ]
4456. 1 1.0 16 1.0 23 1.0 33 1.0 4 1.0
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4457. *
4458. LOAD COMB 558 1X[ D + H2A + H4A + H6A + W1B ]
4459. 1 1.0 14 1.0 24 1.0 34 1.0 4 1.0
4460. *
4461. LOAD COMB 559 1X[ D + H2B + H4B + H6B + W1B ]
4462. 1 1.0 15 1.0 25 1.0 35 1.0 4 1.0
4463. *
4464. LOAD COMB 560 1X[ D + H2C + H4C + H6C + W1B ]
4465. 1 1.0 16 1.0 26 1.0 36 1.0 4 1.0
4466. *
4467. LOAD COMB 561 1X[ D ] + 0.8X[ H1A + H3A + H5A ] + 0.7X[ E1A ]
4468. 1 1.0 11 0.8 21 0.8 31 0.8 5 0.7
4469. *
4470. LOAD COMB 562 1X[ D ] + 0.8X[ H1B + H3B + H5B ] + 0.7X[ E1A ]
4471. 1 1.0 12 0.8 22 0.8 32 0.8 5 0.7
4472. *
4473. LOAD COMB 563 1X[ D ] + 0.8X[ H1C + H3C + H5C ] + 0.7X[ E1A ]
4474. 1 1.0 13 0.8 23 0.8 33 0.8 5 0.7
4475. *
4476. LOAD COMB 564 1X[ D ] + 0.8X[ H1A + H4A + H6A ] + 0.7X[ E1A ]
4477. 1 1.0 11 0.8 24 0.8 34 0.8 5 0.7
4478. *
4479. LOAD COMB 565 1X[ D ] + 0.8X[ H1B + H4B + H6B ] + 0.7X[ E1A ]
4480. 1 1.0 12 0.8 25 0.8 35 0.8 5 0.7
4481. *
4482. LOAD COMB 566 1X[ D ] + 0.8X[ H1C + H4C + H6C ] + 0.7X[ E1A ]
4483. 1 1.0 13 0.8 26 0.8 36 0.8 5 0.7
4484. *
4485. LOAD COMB 567 1X[ D ] + 0.8X[ H2A + H3A + H5A ] + 0.7X[ E1A ]
4486. 1 1.0 14 0.8 21 0.8 31 0.8 5 0.7
4487. *
4488. LOAD COMB 568 1X[ D ] + 0.8X[ H2B + H3B + H5B ] + 0.7X[ E1A ]
4489. 1 1.0 15 0.8 22 0.8 32 0.8 5 0.7
4490. *
4491. LOAD COMB 569 1X[ D ] + 0.8X[ H2C + H3C + H5C ] + 0.7X[ E1A ]
4492. 1 1.0 16 0.8 23 0.8 33 0.8 5 0.7
4493. *
4494. LOAD COMB 570 1X[ D ] + 0.8X[ H2A + H4A + H6A ] + 0.7X[ E1A ]
4495. 1 1.0 14 0.8 24 0.8 34 0.8 5 0.7
4496. *
4497. LOAD COMB 571 1X[ D ] + 0.8X[ H2B + H4B + H6B ] + 0.7X[ E1A ]
4498. 1 1.0 15 0.8 25 0.8 35 0.8 5 0.7
4499. *
4500. LOAD COMB 572 1X[ D ] + 0.8X[ H2C + H4C + H6C ] + 0.7X[ E1A ]
4501. 1 1.0 16 0.8 26 0.8 36 0.8 5 0.7
4502. *
4503. LOAD COMB 573 1X[ D ] + 0.8X[ H1A + H3A + H5A ] + 0.7X[ E1B ]
4504. 1 1.0 11 0.8 21 0.8 31 0.8 6 0.7
4505. *
4506. LOAD COMB 574 1X[ D ] + 0.8X[ H1B + H3B + H5B ] + 0.7X[ E1B ]
4507. 1 1.0 12 0.8 22 0.8 32 0.8 6 0.7
4508. *
4509. LOAD COMB 575 1X[ D ] + 0.8X[ H1C + H3C + H5C ] + 0.7X[ E1B ]
4510. 1 1.0 13 0.8 23 0.8 33 0.8 6 0.7
4511. *
4512. LOAD COMB 576 1X[ D ] + 0.8X[ H1A + H4A + H6A ] + 0.7X[ E1B ]
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  *
4513. 1 1.0 11 0.8 24 0.8 34 0.8 6 0.7
4514. *
4515. LOAD COMB 577 1X[ D ] + 0.8X[ H1B + H4B + H6B ] + 0.7X[ E1B ]
4516. 1 1.0 12 0.8 25 0.8 35 0.8 6 0.7
4517. *
4518. LOAD COMB 578 1X[ D ] + 0.8X[ H1C + H4C + H6C ] + 0.7X[ E1B ]
4519. 1 1.0 13 0.8 26 0.8 36 0.8 6 0.7
4520. *
4521. LOAD COMB 579 1X[ D ] + 0.8X[ H2A + H3A + H5A ] + 0.7X[ E1B ]
4522. 1 1.0 14 0.8 21 0.8 31 0.8 6 0.7
4523. *
4524. LOAD COMB 580 1X[ D ] + 0.8X[ H2B + H3B + H5B ] + 0.7X[ E1B ]
4525. 1 1.0 15 0.8 22 0.8 32 0.8 6 0.7
4526. *
4527. LOAD COMB 581 1X[ D ] + 0.8X[ H2C + H3C + H5C ] + 0.7X[ E1B ]
4528. 1 1.0 16 0.8 23 0.8 33 0.8 6 0.7
4529. *
4530. LOAD COMB 582 1X[ D ] + 0.8X[ H2A + H4A + H6A ] + 0.7X[ E1B ]
4531. 1 1.0 14 0.8 24 0.8 34 0.8 6 0.7
4532. *
4533. LOAD COMB 583 1X[ D ] + 0.8X[ H2B + H4B + H6B ] + 0.7X[ E1B ]
4534. 1 1.0 15 0.8 25 0.8 35 0.8 6 0.7
4535. *
4536. LOAD COMB 584 1X[ D ] + 0.8X[ H2C + H4C + H6C ] + 0.7X[ E1B ]
4537. 1 1.0 16 0.8 26 0.8 36 0.8 6 0.7
4538. *
4539. LOAD COMB 585 1X[ D + H1A + H3A + H5A ] + 0.75X[ W1A + L ]
4540. 1 1.0 11 1.0 21 1.0 31 1.0 3 0.75 2 0.75
4541. *
4542. LOAD COMB 586 1X[ D + H1B + H3B + H5B ] + 0.75X[ W1A + L ]
4543. 1 1.0 12 1.0 22 1.0 32 1.0 3 0.75 2 0.75
4544. *
4545. LOAD COMB 587 1X[ D + H1C + H3C + H5C ] + 0.75X[ W1A + L ]
4546. 1 1.0 13 1.0 23 1.0 33 1.0 3 0.75 2 0.75
4547. *
4548. LOAD COMB 588 1X[ D + H1A + H4A + H6A ] + 0.75X[ W1A + L ]
4549. 1 1.0 11 1.0 24 1.0 34 1.0 3 0.75 2 0.75
4550. *
4551. LOAD COMB 589 1X[ D + H1B + H4B + H6B ] + 0.75X[ W1A + L ]
4552. 1 1.0 12 1.0 25 1.0 35 1.0 3 0.75 2 0.75
4553. *
4554. LOAD COMB 590 1X[ D + H1C + H4C + H6C ] + 0.75X[ W1A + L ]
4555. 1 1.0 13 1.0 26 1.0 36 1.0 3 0.75 2 0.75
4556. *
4557. LOAD COMB 591 1X[ D + H2A + H3A + H5A ] + 0.75X[ W1A + L ]
4558. 1 1.0 14 1.0 21 1.0 31 1.0 3 0.75 2 0.75
4559. *
4560. LOAD COMB 592 1X[ D + H2B + H3B + H5B ] + 0.75X[ W1A + L ]
4561. 1 1.0 15 1.0 22 1.0 32 1.0 3 0.75 2 0.75
4562. *
4563. LOAD COMB 593 1X[ D + H2C + H3C + H5C ] + 0.75X[ W1A + L ]
4564. 1 1.0 16 1.0 23 1.0 33 1.0 3 0.75 2 0.75
4565. *
4566. LOAD COMB 594 1X[ D + H2A + H4A + H6A ] + 0.75X[ W1A + L ]
4567. 1 1.0 14 1.0 24 1.0 34 1.0 3 0.75 2 0.75
4568. *
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*
LOAD COMB 595 1X[ D + H2B + H4B + H6B ] + 0.75X[ W1A + L ]
1 1.0 15 1.0 25 1.0 35 1.0 3 0.75 2 0.75
*
LOAD COMB 596 1X[ D + H2C + H4C + H6C ] + 0.75X[ W1A + L ]
1 1.0 16 1.0 26 1.0 36 1.0 3 0.75 2 0.75
*
LOAD COMB 597 1X[ D + H1A + H3A + H5A ] + 0.75X[ W1B + L ]
1 1.0 11 1.0 21 1.0 31 1.0 4 0.75 2 0.75
*
LOAD COMB 598 1X[ D + H1B + H3B + H5B ] + 0.75X[ W1B + L ]
1 1.0 12 1.0 22 1.0 32 1.0 4 0.75 2 0.75
*
LOAD COMB 599 1X[ D + H1C + H3C + H5C ] + 0.75X[ W1B + L ]
1 1.0 13 1.0 23 1.0 33 1.0 4 0.75 2 0.75
*
LOAD COMB 600 1X[ D + H1A + H4A + H6A ] + 0.75X[ W1B + L ]
1 1.0 11 1.0 24 1.0 34 1.0 4 0.75 2 0.75
*
LOAD COMB 601 1X[ D + H1B + H4B + H6B ] + 0.75X[ W1B + L ]
1 1.0 12 1.0 25 1.0 35 1.0 4 0.75 2 0.75
*
LOAD COMB 602 1X[ D + H1C + H4C + H6C ] + 0.75X[ W1B + L ]
1 1.0 13 1.0 26 1.0 36 1.0 4 0.75 2 0.75
*
LOAD COMB 603 1X[ D + H2A + H3A + H5A ] + 0.75X[ W1B + L ]
1 1.0 14 1.0 21 1.0 31 1.0 4 0.75 2 0.75
*
LOAD COMB 604 1X[ D + H2B + H3B + H5B ] + 0.75X[ W1B + L ]
1 1.0 15 1.0 22 1.0 32 1.0 4 0.75 2 0.75
*
LOAD COMB 605 1X[ D + H2C + H3C + H5C ] + 0.75X[ W1B + L ]
1 1.0 16 1.0 23 1.0 33 1.0 4 0.75 2 0.75
*
LOAD COMB 606 1X[ D + H2A + H4A + H6A ] + 0.75X[ W1B + L ]
1 1.0 14 1.0 24 1.0 34 1.0 4 0.75 2 0.75
*
LOAD COMB 607 1X[ D + H2B + H4B + H6B ] + 0.75X[ W1B + L ]
1 1.0 15 1.0 25 1.0 35 1.0 4 0.75 2 0.75
*
LOAD COMB 608 1X[ D + H2C + H4C + H6C ] + 0.75X[ W1B + L ]
1 1.0 16 1.0 26 1.0 36 1.0 4 0.75 2 0.75
*
LOAD COMB 609 1X[ D ] + 0.8X[ H1A + H3A + H5A ] + 0.75X0.7X[ E1A ] + 0.75X[ L ]
1 1.0 11 0.8 21 0.8 31 0.8 5 0.525 2 0.75
*
LOAD COMB 610 1X[ D ] + 0.8X[ H1B + H3B + H5B ] + 0.75X0.7X[ E1A ] + 0.75X[ L ]
1 1.0 12 0.8 22 0.8 32 0.8 5 0.525 2 0.75
*
LOAD COMB 611 1X[ D ] + 0.8X[ H1C + H3C + H5C ] + 0.75X0.7X[ E1A ] + 0.75X[ L ]
1 1.0 13 0.8 23 0.8 33 0.8 5 0.525 2 0.75
*
LOAD COMB 612 1X[ D ] + 0.8X[ H1A + H4A + H6A ] + 0.75X0.7X[ E1A ] + 0.75X[ L ]
1 1.0 11 0.8 24 0.8 34 0.8 5 0.525 2 0.75
*
LOAD COMB 613 1X[ D ] + 0.8X[ H1B + H4B + H6B ] + 0.75X0.7X[ E1A ] + 0.75X[ L ]
1 1.0 12 0.8 25 0.8 35 0.8 5 0.525 2 0.75
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*
4625. *
4626. LOAD COMB 614 1X[ D ] + 0.8X[ H1C + H4C + H6C ] + 0.75X0.7X[ E1A ] + 0.75X[ L ]
4627. 1 1.0 13 0.8 26 0.8 36 0.8 5 0.525 2 0.75
4628. *
4629. LOAD COMB 615 1X[ D ] + 0.8X[ H2A + H3A + H5A ] + 0.75X0.7X[ E1A ] + 0.75X[ L ]
4630. 1 1.0 14 0.8 21 0.8 31 0.8 5 0.525 2 0.75
4631. *
4632. LOAD COMB 616 1X[ D ] + 0.8X[ H2B + H3B + H5B ] + 0.75X0.7X[ E1A ] + 0.75X[ L ]
4633. 1 1.0 15 0.8 22 0.8 32 0.8 5 0.525 2 0.75
4634. *
4635. LOAD COMB 617 1X[ D ] + 0.8X[ H2C + H3C + H5C ] + 0.75X0.7X[ E1A ] + 0.75X[ L ]
4636. 1 1.0 16 0.8 23 0.8 33 0.8 5 0.525 2 0.75
4637. *
4638. LOAD COMB 618 1X[ D ] + 0.8X[ H2A + H4A + H6A ] + 0.75X0.7X[ E1A ] + 0.75X[ L ]
4639. 1 1.0 14 0.8 24 0.8 34 0.8 5 0.525 2 0.75
4640. *
4641. LOAD COMB 619 1X[ D ] + 0.8X[ H2B + H4B + H6B ] + 0.75X0.7X[ E1A ] + 0.75X[ L ]
4642. 1 1.0 15 0.8 25 0.8 35 0.8 5 0.525 2 0.75
4643. *
4644. LOAD COMB 620 1X[ D ] + 0.8X[ H2C + H4C + H6C ] + 0.75X0.7X[ E1A ] + 0.75X[ L ]
4645. 1 1.0 16 0.8 26 0.8 36 0.8 5 0.525 2 0.75
4646. *
4647. LOAD COMB 621 1X[ D ] + 0.8X[ H1A + H3A + H5A ] + 0.75X0.7X[ E1B ] + 0.75X[ L ]
4648. 1 1.0 11 0.8 21 0.8 31 0.8 6 0.525 2 0.75
4649. *
4650. LOAD COMB 622 1X[ D ] + 0.8X[ H1B + H3B + H5B ] + 0.75X0.7X[ E1B ] + 0.75X[ L ]
4651. 1 1.0 12 0.8 22 0.8 32 0.8 6 0.525 2 0.75
4652. *
4653. LOAD COMB 623 1X[ D ] + 0.8X[ H1C + H3C + H5C ] + 0.75X0.7X[ E1B ] + 0.75X[ L ]
4654. 1 1.0 13 0.8 23 0.8 33 0.8 6 0.525 2 0.75
4655. *
4656. LOAD COMB 624 1X[ D ] + 0.8X[ H1A + H4A + H6A ] + 0.75X0.7X[ E1B ] + 0.75X[ L ]
4657. 1 1.0 11 0.8 24 0.8 34 0.8 6 0.525 2 0.75
4658. *
4659. LOAD COMB 625 1X[ D ] + 0.8X[ H1B + H4B + H6B ] + 0.75X0.7X[ E1B ] + 0.75X[ L ]
4660. 1 1.0 12 0.8 25 0.8 35 0.8 6 0.525 2 0.75
4661. *
4662. LOAD COMB 626 1X[ D ] + 0.8X[ H1C + H4C + H6C ] + 0.75X0.7X[ E1B ] + 0.75X[ L ]
4663. 1 1.0 13 0.8 26 0.8 36 0.8 6 0.525 2 0.75
4664. *
4665. LOAD COMB 627 1X[ D ] + 0.8X[ H2A + H3A + H5A ] + 0.75X0.7X[ E1B ] + 0.75X[ L ]
4666. 1 1.0 14 0.8 21 0.8 31 0.8 6 0.525 2 0.75
4667. *
4668. LOAD COMB 628 1X[ D ] + 0.8X[ H2B + H3B + H5B ] + 0.75X0.7X[ E1B ] + 0.75X[ L ]
4669. 1 1.0 15 0.8 22 0.8 32 0.8 6 0.525 2 0.75
4670. *
4671. LOAD COMB 629 1X[ D ] + 0.8X[ H2C + H3C + H5C ] + 0.75X0.7X[ E1B ] + 0.75X[ L ]
4672. 1 1.0 16 0.8 23 0.8 33 0.8 6 0.525 2 0.75
4673. *
4674. LOAD COMB 630 1X[ D ] + 0.8X[ H2A + H4A + H6A ] + 0.75X0.7X[ E1B ] + 0.75X[ L ]
4675. 1 1.0 14 0.8 24 0.8 34 0.8 6 0.525 2 0.75
4676. *
4677. LOAD COMB 631 1X[ D ] + 0.8X[ H2B + H4B + H6B ] + 0.75X0.7X[ E1B ] + 0.75X[ L ]
4678. 1 1.0 15 0.8 25 0.8 35 0.8 6 0.525 2 0.75
4679. *
4680. LOAD COMB 632 1X[ D ] + 0.8X[ H2C + H4C + H6C ] + 0.75X0.7X[ E1B ] + 0.75X[ L ]
```

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  *
4681. 1 1.0 16 0.8 26 0.8 36 0.8 6 0.525 2 0.75
4682. *
4683. LOAD COMB 633 0.6X[ D ] + [ W1A + H1A + H3A + H5A ]
4684. 1 0.6 3 1.0 11 1.0 21 1.0 31 1.0
4685. *
4686. LOAD COMB 634 0.6X[ D ] + [ W1A + H1B + H3B + H5B ]
4687. 1 0.6 3 1.0 12 1.0 22 1.0 32 1.0
4688. *
4689. LOAD COMB 635 0.6X[ D ] + [ W1A + H1C + H3C + H5C ]
4690. 1 0.6 3 1.0 13 1.0 23 1.0 33 1.0
4691. *
4692. LOAD COMB 636 0.6X[ D ] + [ W1A + H1A + H4A + H6A ]
4693. 1 0.6 3 1.0 11 1.0 24 1.0 34 1.0
4694. *
4695. LOAD COMB 637 0.6X[ D ] + [ W1A + H1B + H4B + H6B ]
4696. 1 0.6 3 1.0 12 1.0 25 1.0 35 1.0
4697. *
4698. LOAD COMB 638 0.6X[ D ] + [ W1A + H1C + H4C + H6C ]
4699. 1 0.6 3 1.0 13 1.0 26 1.0 36 1.0
4700. *
4701. LOAD COMB 639 0.6X[ D ] + [ W1A + H2A + H3A + H5A ]
4702. 1 0.6 3 1.0 14 1.0 21 1.0 31 1.0
4703. *
4704. LOAD COMB 640 0.6X[ D ] + [ W1A + H2B + H3B + H5B ]
4705. 1 0.6 3 1.0 15 1.0 22 1.0 32 1.0
4706. *
4707. LOAD COMB 641 0.6X[ D ] + [ W1A + H2C + H3C + H5C ]
4708. 1 0.6 3 1.0 16 1.0 23 1.0 33 1.0
4709. *
4710. LOAD COMB 642 0.6X[ D ] + [ W1A + H2A + H4A + H6A ]
4711. 1 0.6 3 1.0 14 1.0 24 1.0 34 1.0
4712. *
4713. LOAD COMB 643 0.6X[ D ] + [ W1A + H2B + H4B + H6B ]
4714. 1 0.6 3 1.0 15 1.0 25 1.0 35 1.0
4715. *
4716. LOAD COMB 644 0.6X[ D ] + [ W1A + H2C + H4C + H6C ]
4717. 1 0.6 3 1.0 16 1.0 26 1.0 36 1.0
4718. *
4719. LOAD COMB 645 0.6X[ D ] + [ W1B + H1A + H3A + H5A ]
4720. 1 0.6 4 1.0 11 1.0 21 1.0 31 1.0
4721. *
4722. LOAD COMB 646 0.6X[ D ] + [ W1B + H1B + H3B + H5B ]
4723. 1 0.6 4 1.0 12 1.0 22 1.0 32 1.0
4724. *
4725. LOAD COMB 647 0.6X[ D ] + [ W1B + H1C + H3C + H5C ]
4726. 1 0.6 4 1.0 13 1.0 23 1.0 33 1.0
4727. *
4728. LOAD COMB 648 0.6X[ D ] + [ W1B + H1A + H4A + H6A ]
4729. 1 0.6 4 1.0 11 1.0 24 1.0 34 1.0
4730. *
4731. LOAD COMB 649 0.6X[ D ] + [ W1B + H1B + H4B + H6B ]
4732. 1 0.6 4 1.0 12 1.0 25 1.0 35 1.0
4733. *
4734. LOAD COMB 650 0.6X[ D ] + [ W1B + H1C + H4C + H6C ]
4735. 1 0.6 4 1.0 13 1.0 26 1.0 36 1.0
4736. *
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  *
4737. LOAD COMB 651 0.6X[ D ] + [ W1B + H2A + H3A + H5A ]
4738. 1 0.6 4 1.0 14 1.0 21 1.0 31 1.0
4739. *
4740. LOAD COMB 652 0.6X[ D ] + [ W1B + H2B + H3B + H5B ]
4741. 1 0.6 4 1.0 15 1.0 22 1.0 32 1.0
4742. *
4743. LOAD COMB 653 0.6X[ D ] + [ W1B + H2C + H3C + H5C ]
4744. 1 0.6 4 1.0 16 1.0 23 1.0 33 1.0
4745. *
4746. LOAD COMB 654 0.6X[ D ] + [ W1B + H2A + H4A + H6A ]
4747. 1 0.6 4 1.0 14 1.0 24 1.0 34 1.0
4748. *
4749. LOAD COMB 655 0.6X[ D ] + [ W1B + H2B + H4B + H6B ]
4750. 1 0.6 4 1.0 15 1.0 25 1.0 35 1.0
4751. *
4752. LOAD COMB 656 0.6X[ D ] + [ W1B + H2C + H4C + H6C ]
4753. 1 0.6 4 1.0 16 1.0 26 1.0 36 1.0
4754. *
4755. LOAD COMB 657 0.6X[ D ] + 0.7X[ E1A ] + 0.8X[ H1A + H3A + H5A ]
4756. 1 0.6 5 0.7 11 0.8 21 0.8 31 0.8
4757. *
4758. LOAD COMB 658 0.6X[ D ] + 0.7X[ E1A ] + 0.8X[ H1B + H3B + H5B ]
4759. 1 0.6 5 0.7 12 0.8 22 0.8 32 0.8
4760. *
4761. LOAD COMB 659 0.6X[ D ] + 0.7X[ E1A ] + 0.8X[ H1C + H3C + H5C ]
4762. 1 0.6 5 0.7 13 0.8 23 0.8 33 0.8
4763. *
4764. LOAD COMB 660 0.6X[ D ] + 0.7X[ E1A ] + 0.8X[ H1A + H4A + H6A ]
4765. 1 0.6 5 0.7 11 0.8 24 0.8 34 0.8
4766. *
4767. LOAD COMB 661 0.6X[ D ] + 0.7X[ E1A ] + 0.8X[ H1B + H4B + H6B ]
4768. 1 0.6 5 0.7 12 0.8 25 0.8 35 0.8
4769. *
4770. LOAD COMB 662 0.6X[ D ] + 0.7X[ E1A ] + 0.8X[ H1C + H4C + H6C ]
4771. 1 0.6 5 0.7 13 0.8 26 0.8 36 0.8
4772. *
4773. LOAD COMB 663 0.6X[ D ] + 0.7X[ E1A ] + 0.8X[ H2A + H3A + H5A ]
4774. 1 0.6 5 0.7 14 0.8 21 0.8 31 0.8
4775. *
4776. LOAD COMB 664 0.6X[ D ] + 0.7X[ E1A ] + 0.8X[ H2B + H3B + H5B ]
4777. 1 0.6 5 0.7 15 0.8 22 0.8 32 0.8
4778. *
4779. LOAD COMB 665 0.6X[ D ] + 0.7X[ E1A ] + 0.8X[ H2C + H3C + H5C ]
4780. 1 0.6 5 0.7 16 0.8 23 0.8 33 0.8
4781. *
4782. LOAD COMB 666 0.6X[ D ] + 0.7X[ E1A ] + 0.8X[ H2A + H4A + H6A ]
4783. 1 0.6 5 0.7 14 0.8 24 0.8 34 0.8
4784. *
4785. LOAD COMB 667 0.6X[ D ] + 0.7X[ E1A ] + 0.8X[ H2B + H4B + H6B ]
4786. 1 0.6 5 0.7 15 0.8 25 0.8 35 0.8
4787. *
4788. LOAD COMB 668 0.6X[ D ] + 0.7X[ E1A ] + 0.8X[ H2C + H4C + H6C ]
4789. 1 0.6 5 0.7 16 0.8 26 0.8 36 0.8
4790. *
4791. LOAD COMB 669 0.6X[ D ] + 0.7X[ E1B ] + 0.8X[ H1A + H3A + H5A ]
4792. 1 0.6 6 0.7 11 0.8 21 0.8 31 0.8
```

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  *
4793. *
4794. LOAD COMB 670 0.6X[ D ] + 0.7X[ E1B ] + 0.8X[ H1B + H3B + H5B ]
4795. 1 0.6 6 0.7 12 0.8 22 0.8 32 0.8
4796. *
4797. LOAD COMB 671 0.6X[ D ] + 0.7X[ E1B ] + 0.8X[ H1C + H3C + H5C ]
4798. 1 0.6 6 0.7 13 0.8 23 0.8 33 0.8
4799. *
4800. LOAD COMB 672 0.6X[ D ] + 0.7X[ E1B ] + 0.8X[ H1A + H4A + H6A ]
4801. 1 0.6 6 0.7 11 0.8 24 0.8 34 0.8
4802. *
4803. LOAD COMB 673 0.6X[ D ] + 0.7X[ E1B ] + 0.8X[ H1B + H4B + H6B ]
4804. 1 0.6 6 0.7 12 0.8 25 0.8 35 0.8
4805. *
4806. LOAD COMB 674 0.6X[ D ] + 0.7X[ E1B ] + 0.8X[ H1C + H4C + H6C ]
4807. 1 0.6 6 0.7 13 0.8 26 0.8 36 0.8
4808. *
4809. LOAD COMB 675 0.6X[ D ] + 0.7X[ E1B ] + 0.8X[ H2A + H3A + H5A ]
4810. 1 0.6 6 0.7 14 0.8 21 0.8 31 0.8
4811. *
4812. LOAD COMB 676 0.6X[ D ] + 0.7X[ E1B ] + 0.8X[ H2B + H3B + H5B ]
4813. 1 0.6 6 0.7 15 0.8 22 0.8 32 0.8
4814. *
4815. LOAD COMB 677 0.6X[ D ] + 0.7X[ E1B ] + 0.8X[ H2C + H3C + H5C ]
4816. 1 0.6 6 0.7 16 0.8 23 0.8 33 0.8
4817. *
4818. LOAD COMB 678 0.6X[ D ] + 0.7X[ E1B ] + 0.8X[ H2A + H4A + H6A ]
4819. 1 0.6 6 0.7 14 0.8 24 0.8 34 0.8
4820. *
4821. LOAD COMB 679 0.6X[ D ] + 0.7X[ E1B ] + 0.8X[ H2B + H4B + H6B ]
4822. 1 0.6 6 0.7 15 0.8 25 0.8 35 0.8
4823. *
4824. LOAD COMB 680 0.6X[ D ] + 0.7X[ E1B ] + 0.8X[ H2C + H4C + H6C ]
4825. 1 0.6 6 0.7 16 0.8 26 0.8 36 0.8
4826. *
4827. *
4828. PERFORM ANALYSIS PRINT STATICS CHECK
```

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS	**3888**	**NUMBER OF MEMBERS**	**24**
NUMBER OF PLATES	**3880**	**NUMBER OF SOLIDS**	**0**
NUMBER OF SURFACES	**0**	**NUMBER OF SUPPORTS**	**847**

SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

ORIGINAL/FINAL BAND-WIDTH= 3672/ 152/ 885 DOF
TOTAL PRIMARY LOAD CASES = 24, TOTAL DEGREES OF FREEDOM = 20787
TOTAL LOAD COMBINATION CASES = 248 SO FAR.
SIZE OF STIFFNESS MATRIX = 18397 DOUBLE KILO-WORDS

*

REQRD/AVAIL. DISK SPACE = 235.2/ 87876.5 MB

*

**WARNING: IF THIS UBC/IBC ANALYSIS HAS TENSION/COMPRESSION
OR REPEAT LOAD OR RE-ANALYSIS OR SELECT OPTIMIZE, THEN EACH
UBC/IBC CASE SHOULD BE FOLLOWED BY PERFORM ANALYSIS _CHANGE.

```
************************************************************
*                                                          *
*  X  DIRECTION : Ta =  1.034 Tb =  0.509 Tuser =  1.110  *
*  T =  1.110, LOAD FACTOR =  1.000                        *
* UBC TYPE = 97                                            *
* UBC FACTOR V = 0.0626 x     261659.19 =   16370.07 KN    *
*                                                          *
************************************************************

************************************************************
*                                                          *
*  Z  DIRECTION : Ta =  1.034 Tb =  0.511 Tuser =  1.110  *
*  T =  1.110, LOAD FACTOR =  1.000                        *
* UBC TYPE = 97                                            *
* UBC FACTOR V = 0.0626 x     261659.19 =   16370.07 KN    *
*                                                          *
************************************************************

*** LOADING     5 : ADDITIONAL APPLIED FORCES AND MOMENTS ABOUT ORIGIN
    DUE TO SEISMIC LOADS(KN  ,METE)

    FORCE-X =     16370.03  FORCE-Y =         0.00  FORCE-Z =         0.00
    MOM-X   =         0.00  MOM-Y   =       344.68  MOM-Z   =   -326313.50


*** LOADING     6 : ADDITIONAL APPLIED FORCES AND MOMENTS ABOUT ORIGIN
    DUE TO SEISMIC LOADS(KN  ,METE)

    FORCE-X =         0.00  FORCE-Y =         0.00  FORCE-Z =     16370.03
    MOM-X   =    326313.50  MOM-Y   =        -0.00  MOM-Z   =         0.00



       STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO.     5
       SEISMIC LOAD IN X-DIRECTION (E1)


        CENTER OF FORCE BASED ON X FORCES ONLY (METE).
      (FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

                    X = -0.381148102E-08
                    Y =  0.199334014E+02
                    Z =  0.210556182E-01
```

*

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 5)
SUMMATION FORCE-X = 16370.07
SUMMATION FORCE-Y = 0.00
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 0.00 MY= 344.68 MZ= -326311.15

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 5)
SUMMATION FORCE-X = -16370.07
SUMMATION FORCE-Y = 0.00
SUMMATION FORCE-Z = -0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -0.00 MY= -344.68 MZ= 326311.15

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 5)

	MAXIMUMS	AT NODE
X =	9.22899E-01	2502
Y =	-2.85749E-01	2572
Z =	-2.08376E-02	2512
RX=	7.98116E-05	5322
RY=	2.57327E-05	5219
RZ=	-3.04147E-04	13104

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 6
SEISMIC LOAD IN Z-DIRECTION (E2)

CENTER OF FORCE BASED ON Z FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = -0.381148102E-08
Y = 0.199334014E+02
Z = 0.210556182E-01

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 6)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = 0.00
SUMMATION FORCE-Z = 16370.07

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 326311.15 MY= 0.00 MZ= 0.00

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 6)
SUMMATION FORCE-X = -0.00
SUMMATION FORCE-Y = -0.00
SUMMATION FORCE-Z = -16370.07

*

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -326311.15 MY= -0.00 MZ= 0.00

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 6)

	MAXIMUMS	AT NODE
X =	-1.76420E-02	2570
Y =	-2.89668E-01	13294
Z =	9.35402E-01	2536
RX=	3.11286E-04	5436
RY=	3.95599E-05	2550
RZ=	7.00714E-05	5407

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 1
DEAD LOAD (D)

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.384029210E-10
Y = 0.794245261E+01
Z = 0.117560242E+00

CENTER OF FORCE BASED ON Z FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.581540163E-09
Y = 0.298967551E+01
Z = 0.101258054E+02

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 1)
SUMMATION FORCE-X = -0.00
SUMMATION FORCE-Y = -125250.91
SUMMATION FORCE-Z = -3869.64

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 3155.55 MY= 0.00 MZ= 0.00

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 1)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = 125250.91
SUMMATION FORCE-Z = 3869.64

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -3155.55 MY= -0.00 MZ= -0.00

*

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 1)

	MAXIMUMS	AT NODE
X =	2.66530E-02	2572
Y =	-4.61170E-01	2553
Z =	-5.70824E-02	2536
RX=	-1.67336E-04	12073
RY=	-3.41019E-05	2554
RZ=	1.49764E-04	12026

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 2
LIVE LOAD (L)

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.000000000E+00
Y = 0.124034267E+02
Z = 0.451169626E+00

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 2)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = -3081.60
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 1390.32 MY= 0.00 MZ= 0.00

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 2)
SUMMATION FORCE-X = -0.00
SUMMATION FORCE-Y = 3081.60
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -1390.32 MY= 0.00 MZ= -0.00

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 2)

	MAXIMUMS	AT NODE
X =	5.34901E-04	5551
Y =	-1.30840E-02	2536
Z =	2.53105E-03	2536
RX=	7.24163E-06	12269
RY=	8.07455E-07	5646
RZ=	6.13296E-06	12845

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 3
WIND LOAD ALONG X (W1)

*

CENTER OF FORCE BASED ON X FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = -0.372892098E-15
Y = 0.179970026E+02
Z = 0.144637195E+00

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 3)
SUMMATION FORCE-X = 1653.44
SUMMATION FORCE-Y = 0.00
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 0.00 MY= 239.15 MZ= -29756.96

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 3)
SUMMATION FORCE-X = -1653.44
SUMMATION FORCE-Y = 0.00
SUMMATION FORCE-Z = -0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -0.00 MY= -239.15 MZ= 29756.96

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 3)

	MAXIMUMS	AT NODE
X =	8.43303E-02	2504
Y =	-2.62129E-02	2572
Z =	-1.80633E-03	2510
RX=	7.62591E-06	5322
RY=	2.48933E-06	5219
RZ=	-2.79825E-05	5567

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 4
WIND LOAD ALONG Z (W1B)

CENTER OF FORCE BASED ON Z FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = -0.154046555E-15
Y = 0.179798173E+02
Z = 0.143403596E+00

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 4)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = 0.00
SUMMATION FORCE-Z = 1667.66

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 29984.28 MY= 0.00 MZ= 0.00

*

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 4)
SUMMATION FORCE-X = -0.00
SUMMATION FORCE-Y = -0.00
SUMMATION FORCE-Z = -1667.66

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -29984.28 MY= 0.00 MZ= 0.00

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 4)

	MAXIMUMS	AT NODE
X =	-1.80787E-03	2568
Y =	-2.70086E-02	13294
Z =	8.63971E-02	2536
RX=	2.93521E-05	5466
RY=	3.71952E-06	2550
RZ=	6.85304E-06	5407

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 11
MATERIAL FILLING WALL HORIZONTAL CASE-I (H1A)

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 11)
SUMMATION FORCE-X = 2.4744445E-12
SUMMATION FORCE-Y = 0.0000000E+00
SUMMATION FORCE-Z = -3.4372086E-12

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -1.5280279E-10 MY= 1.8364427E-12 MZ= -5.0936403E-11

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 11)
SUMMATION FORCE-X = -1.3108605E-10
SUMMATION FORCE-Y = -1.8765850E-10
SUMMATION FORCE-Z = -1.4739600E-10

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -7.5264467E-07 MY= 7.8878690E-10 MZ= 2.5205026E-09

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 11)

	MAXIMUMS	AT NODE
X =	8.40485E-02	772
Y =	-1.83278E-02	2313
Z =	8.29012E-02	736
RX=	-1.88745E-04	237
RY=	7.45675E-05	12927
RZ=	-1.94753E-04	272

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 12
MATERIAL FILLING WALL HORIZONTAL CASE-II (H1B)

*

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 12)
SUMMATION FORCE-X = 3.0068168E-12
SUMMATION FORCE-Y = 0.0000000E+00
SUMMATION FORCE-Z = -2.1115128E-11

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 6.3494850E-10 MY= 1.9274356E-12 MZ= 3.1707606E-11

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 12)
SUMMATION FORCE-X = -1.3088653E-10
SUMMATION FORCE-Y = -1.9595361E-10
SUMMATION FORCE-Z = -1.6455142E-10

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -8.1087616E-07 MY= 8.7474489E-10 MZ= 2.7665576E-09

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 12)

	MAXIMUMS	AT NODE
X =	9.09028E-02	772
Y =	-2.01745E-02	2313
Z =	8.96782E-02	736
RX=	-2.02276E-04	237
RY=	7.94677E-05	12927
RZ=	-2.08683E-04	272

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 13
MATERIAL FILLING WALL HORIZONTAL CASE-III (H1C)

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 13)
SUMMATION FORCE-X = 2.4420972E-12
SUMMATION FORCE-Y = 0.0000000E+00
SUMMATION FORCE-Z = 1.7055666E-11

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -2.1340497E-10 MY= 2.2937407E-12 MZ= 3.5156458E-11

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 13)
SUMMATION FORCE-X = -1.1543885E-10
SUMMATION FORCE-Y = -1.6273769E-10
SUMMATION FORCE-Z = -1.2793824E-10

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -6.6612220E-07 MY= 8.3235195E-10 MZ= 1.9992763E-09

*

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 13)

	MAXIMUMS	AT NODE
X =	7.40815E-02	772
Y =	-1.58281E-02	2313
Z =	7.30513E-02	736
RX=	-1.68428E-04	237
RY=	6.71109E-05	12927
RZ=	-1.73830E-04	272

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 14
MATERIAL EMPTYING WALL HORIZONTAL CASE-I (H2A)

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 14)
SUMMATION FORCE-X = -1.9867658E-12
SUMMATION FORCE-Y = 0.0000000E+00
SUMMATION FORCE-Z = -7.2971544E-12

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 5.1127792E-10 MY= -2.3021451E-12 MZ= -3.4552748E-12

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 14)
SUMMATION FORCE-X = -1.5202339E-10
SUMMATION FORCE-Y = -1.9897660E-10
SUMMATION FORCE-Z = -1.5211525E-10

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -8.1037984E-07 MY= 1.0199853E-09 MZ= 2.8828231E-09

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 14)

	MAXIMUMS	AT NODE
X =	9.05134E-02	772
Y =	-1.97376E-02	2313
Z =	8.92778E-02	736
RX=	-2.03256E-04	237
RY=	8.03101E-05	12927
RZ=	-2.09726E-04	272

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 15
MATERIAL EMPTYING WALL HORIZONTAL CASE-II (H2B)

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 15)
SUMMATION FORCE-X = 2.0159031E-12
SUMMATION FORCE-Y = 0.0000000E+00
SUMMATION FORCE-Z = -2.9008856E-11

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 1.7587220E-10 MY= -1.4203134E-12 MZ= 1.8907675E-11

*

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 15)
SUMMATION FORCE-X = -1.4507983E-10
SUMMATION FORCE-Y = -2.1724838E-10
SUMMATION FORCE-Z = -1.6320420E-10

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -8.7325613E-07 MY= 1.0325492E-09 MZ= 3.0944892E-09

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 15)

	MAXIMUMS	AT NODE
X =	9.79057E-02	772
Y =	-2.17284E-02	2313
Z =	9.65867E-02	736
RX=	-2.17881E-04	237
RY=	8.56038E-05	12927
RZ=	-2.24782E-04	272

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 16
MATERIAL EMPTYING WALL HORIZONTAL CASE-III (H2C)

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 16)
SUMMATION FORCE-X = -3.4233808E-12
SUMMATION FORCE-Y = 0.0000000E+00
SUMMATION FORCE-Z = -6.8033026E-12

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -1.7854394E-11 MY= 6.1170266E-12 MZ= 2.4398865E-11

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 16)
SUMMATION FORCE-X = -1.1193546E-10
SUMMATION FORCE-Y = -1.7218020E-10
SUMMATION FORCE-Z = -1.4370100E-10

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -7.1778855E-07 MY= 6.7449149E-10 MZ= 1.9685529E-09

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 16)

	MAXIMUMS	AT NODE
X =	7.97867E-02	772
Y =	-1.70462E-02	2313
Z =	7.86771E-02	736
RX=	-1.81424E-04	237
RY=	7.22762E-05	12927
RZ=	-1.87241E-04	272

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 21
MATERIAL WALL VERTICAL (FILLING _EMPTYING) CASE-I (H3A)

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = -0.699704083E-16
Y = 0.159410976E+02
Z = 0.419822450E-15

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 21)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = -46995.46
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -1.6576328E-11 MY= 0.0000000E+00 MZ= 2.9133811E-12

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 21)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = 46995.46
SUMMATION FORCE-Z = -0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -3.6161891E-06 MY= 7.1984550E-09 MZ= 6.4924229E-09

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 21)

	MAXIMUMS	AT NODE
X =	2.01033E-02	2572
Y =	-2.14867E-01	2161
Z =	-3.12071E-02	2536
RX=	-1.41328E-04	12941
RY=	2.51237E-05	2555
RZ=	-1.63917E-04	13105

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 22
MATERIAL WALL VERTICAL (FILLING _EMPTYING) CASE-II (H3B)

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = -0.804784600E-15
Y = 0.160995021E+02
Z = 0.167080281E-14

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 22)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = -46988.16
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 2.3833689E-11 MY= 0.0000000E+00 MZ= 2.0829591E-11

*

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 22)
SUMMATION FORCE-X = -0.00
SUMMATION FORCE-Y = 46988.16
SUMMATION FORCE-Z = -0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -3.6161051E-06 MY= 7.5240023E-09 MZ= 6.5446245E-09

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 22)

	MAXIMUMS	AT NODE
X =	2.00960E-02	2572
Y =	-2.15304E-01	2161
Z =	-3.12072E-02	2536
RX=	-1.41318E-04	12941
RY=	2.51204E-05	2555
RZ=	-1.63901E-04	13105

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 23
MATERIAL WALL VERTICAL (FILLING _EMPTYING) CASE-III (H3C)

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.699812690E-16
Y = 0.157483689E+02
Z = -0.944747131E-15

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 23)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = -46988.16
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -1.4379852E-11 MY= 0.0000000E+00 MZ= 4.3046431E-12

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 23)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = 46988.16
SUMMATION FORCE-Z = -0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -3.6148812E-06 MY= 7.5432856E-09 MZ= 6.4932450E-09

*

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 23)

	MAXIMUMS	AT NODE
X =	2.01048E-02	2572
Y =	-2.14262E-01	2161
Z =	-3.11966E-02	2536
RX=	-1.41290E-04	12941
RY=	2.51192E-05	2555
RZ=	-1.63880E-04	13105

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 24
MATERIAL WALL VERTICAL (FILLING _EMPTYING) CASE-I (H4A)

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.126009319E-15
Y = 0.159413480E+02
Z = 0.114458464E-14

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 24)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = -78286.86
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -1.9831222E-10 MY= 0.0000000E+00 MZ= -9.3558952E-12

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 24)
SUMMATION FORCE-X = -0.00
SUMMATION FORCE-Y = 78286.86
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -6.0241588E-06 MY= 1.2655304E-08 MZ= 1.0510818E-08

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 24)

	MAXIMUMS	AT NODE
X =	3.34888E-02	2572
Y =	-3.57936E-01	2161
Z =	-5.19860E-02	2536
RX=	-2.35429E-04	12941
RY=	4.18521E-05	2555
RZ=	-2.73060E-04	13105

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 25
MATERIAL WALL VERTICAL (FILLING _EMPTYING) CASE-II (H4B)

*

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.214646677E-15
Y = 0.160993050E+02
Z = -0.245310488E-15

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 25)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = -107236.88
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -1.8682889E-10 MY= 0.0000000E+00 MZ= 3.5994587E-11

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 25)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = 107236.88
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -8.2528096E-06 MY= 1.5330849E-08 MZ= 1.5672407E-08

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 25)

	MAXIMUMS	AT NODE
X =	4.58632E-02	2572
Y =	-4.91368E-01	2161
Z =	-7.12216E-02	2536
RX=	-3.22517E-04	12941
RY=	5.73300E-05	2555
RZ=	-3.74058E-04	13105

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 26
MATERIAL WALL VERTICAL (FILLING _EMPTYING) CASE-III (H4C)

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = -0.273081973E-15
Y = 0.157499988E+02
Z = -0.212613821E-14

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 26)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = -84289.85
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 4.2553187E-10 MY= 0.0000000E+00 MZ= 2.9414793E-11

*

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 26)
SUMMATION FORCE-X = -0.00
SUMMATION FORCE-Y = 84289.85
SUMMATION FORCE-Z = -0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -6.4842385E-06 MY= 1.2532494E-08 MZ= 1.1725841E-08

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 26)

	MAXIMUMS	AT NODE
X =	3.60649E-02	2572
Y =	-3.84363E-01	2161
Z =	-5.59622E-02	2536
RX=	-2.53454E-04	12941
RY=	4.50601E-05	2555
RZ=	-2.93976E-04	13105

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 31
MATERIAL VERTICAL (ON FLAT SLAB) CASE-I (H5A)

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.215768958E-09
Y = 0.799999996E+01
Z = -0.100449909E-09

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 31)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = -135436.45
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -2.9512414E-10 MY= 0.0000000E+00 MZ= 1.6338697E-11

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 31)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = 135436.45
SUMMATION FORCE-Z = -0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 5.6693442E-06 MY= 1.2663968E-08 MZ= 2.7138638E-07

*

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 31)

	MAXIMUMS	AT NODE
X =	5.02151E-02	5623
Y =	-9.87601E-01	13097
Z =	-5.58978E-02	5769
RX=	9.03017E-04	12927
RY=	-8.32904E-05	5695
RZ=	8.95038E-04	13105

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 32
MATERIAL VERTICAL (ON FLAT SLAB) CASE-II (H5B)

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.215768765E-09
Y = 0.799999996E+01
Z = -0.100449597E-09

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 32)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = -127778.24
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -2.7580543E-10 MY= 0.0000000E+00 MZ= -1.3975238E-11

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 32)
SUMMATION FORCE-X = -0.00
SUMMATION FORCE-Y = 127778.24
SUMMATION FORCE-Z = -0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 5.3499868E-06 MY= 1.2384681E-08 MZ= 2.5574850E-07

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 32)

	MAXIMUMS	AT NODE
X =	4.73757E-02	5623
Y =	-9.31758E-01	13097
Z =	-5.27370E-02	5769
RX=	8.51956E-04	12927
RY=	-7.85807E-05	5695
RZ=	8.44428E-04	13105

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 33
MATERIAL VERTICAL (ON FLAT SLAB) CASE-III (H5C)

*

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.215768766E-09
Y = 0.799999996E+01
Z = -0.100450751E-09

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 33)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = -145760.27
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -4.6447115E-11 MY= 0.0000000E+00 MZ= 2.6820126E-11

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 33)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = 145760.27
SUMMATION FORCE-Z = -0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 6.1012709E-06 MY= 1.3062467E-08 MZ= 2.9270355E-07

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 33)

	MAXIMUMS	AT NODE
X =	5.40428E-02	5623
Y =	-1.06288E+00	13097
Z =	-6.01587E-02	5769
RX=	9.71851E-04	12927
RY=	-8.96393E-05	5695
RZ=	9.63263E-04	13105

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 34
MATERIAL VERTICAL (ON FLAT SLAB) CASE-I (H6A)

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.215768800E-09
Y = 0.799999996E+01
Z = -0.100449777E-09

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 34)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = -104888.15
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -2.1538308E-10 MY= 0.0000000E+00 MZ= -7.5784838E-12

*

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 34)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = 104888.15
SUMMATION FORCE-Z = -0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 4.3912296E-06 MY= 9.6124783E-09 MZ= 2.0994301E-07

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 34)

	MAXIMUMS	AT NODE
X =	3.88889E-02	5623
Y =	-7.64843E-01	13097
Z =	-4.32898E-02	5769
RX=	6.99338E-04	12927
RY=	-6.45038E-05	5695
RZ=	6.93158E-04	13105

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 35
MATERIAL VERTICAL (ON FLAT SLAB) CASE-II (H6B)

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.215768713E-09
Y = 0.799999996E+01
Z = -0.100450962E-09

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 35)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = -76286.15
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -4.8502297E-11 MY= 0.0000000E+00 MZ= 1.8419569E-11

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 35)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = 76286.15
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 3.1930990E-06 MY= 7.7425966E-09 MZ= 1.5295723E-07

*

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 35)

	MAXIMUMS	AT NODE
X =	2.82842E-02	5623
Y =	-5.56278E-01	13097
Z =	-3.14851E-02	5769
RX=	5.08635E-04	12927
RY=	-4.69143E-05	5695
RZ=	5.04140E-04	13105

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 36
MATERIAL VERTICAL (ON FLAT SLAB) CASE-III (H6C)

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.215768517E-09
Y = 0.799999996E+01
Z = -0.100449885E-09

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 36)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = -98964.65
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 6.7821008E-12 MY= 0.0000000E+00 MZ= 1.3358683E-11

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 36)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = 98964.65
SUMMATION FORCE-Z = -0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 4.1426723E-06 MY= 9.5819267E-09 MZ= 1.9867959E-07

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 36)

	MAXIMUMS	AT NODE
X =	3.66926E-02	5623
Y =	-7.21649E-01	13097
Z =	-4.08450E-02	5769
RX=	6.59843E-04	12927
RY=	-6.08610E-05	5695
RZ=	6.54012E-04	13105

********** END OF DATA FROM INTERNAL STORAGE **********

4829. *
4830. LOAD LIST 101 TO 168
4831. *
4832. START CONCRETE DESIGN

```
*
4833. CODE ACI
4834. FYMAIN 420000 MEMB 101 TO 172 201 TO 272 301 TO 372 401 TO 472 501 TO 572 -
4835. 601 TO 672 701 TO 772 801 TO 872 901 TO 972 1001 TO 1072 1101 TO 1172 1201 -
4836. 1202 TO 1272 1301 TO 1372 1401 TO 1472 1501 TO 1572 1601 TO 1672 1701 TO 1772 -
4837. 1801 TO 1872 1901 TO 1972 2001 TO 2072 2101 TO 2172 2201 TO 2272 -
4838. 2301 TO 2372 2401 TO 2472 10001 TO 10288 10295 TO 10354 10367 TO 10380 10382 -
4839. 10384 10386 10388 TO 10404 10406 10408 10410 10412 TO 10426 10439 TO 10452 -
4840. 10454 10456 10458 10460 TO 10476 10478 10480 10482 10484 TO 10498 -
4841. 10511 TO 10524 10526 10528 10530 10532 TO 10548 10550 10552 10554 -
4842. 10556 TO 10570 10577 TO 10596 10598 10600 10602 10604 TO 10620 10622 10624 -
4843. 10626 10628 TO 10668 10670 10672 10674 10676 TO 10692 10694 10696 10698 -
4844. 10700 TO 10792 10799 10802 10803 10808 10879 TO 10981 11198 TO 11203 11205 -
4845. 11206 11210 TO 11222 11225 TO 11253 11255 TO 11259 11371 TO 11399 -
4846. 11487 TO 11501 11503 11511 TO 12067 12070 12071 12074 12075 12869 TO 12956
4847. MINSEC 10 ALL
4848. FYSEC 420000 ALL
4849. CLS 0.06 ALL
4850. CLB 0.06 MEMB 101 TO 172 201 TO 272 301 TO 372 401 TO 472 501 TO 572 -
4851. 601 TO 672 701 TO 772 801 TO 872 901 TO 972 1001 TO 1072 1101 TO 1172 1201 -
4852. 1202 TO 1272 1301 TO 1372 1401 TO 1472 1501 TO 1572 1601 TO 1672 1701 TO 1772 -
4853. 1801 TO 1872 1901 TO 1972 2001 TO 2072 2101 TO 2172 2201 TO 2272 -
4854. 2301 TO 2372 2401 TO 2472 10001 TO 10288 10295 TO 10354 10367 TO 10380 10382 -
4855. 10384 10386 10388 TO 10404 10406 10408 10410 10412 TO 10426 10439 TO 10452 -
4856. 10454 10456 10458 10460 TO 10476 10478 10480 10482 10484 TO 10498 -
4857. 10511 TO 10524 10526 10528 10530 10532 TO 10548 10550 10552 10554 -
4858. 10556 TO 10570 10577 TO 10596 10598 10600 10602 10604 TO 10620 10622 10624 -
4859. 10626 10628 TO 10668 10670 10672 10674 10676 TO 10692 10694 10696 10698 -
4860. 10700 TO 10792 10799 10802 10803 10808 10879 TO 10981 11198 TO 11203 11205 -
4861. 11206 11210 TO 11222 11225 TO 11253 11255 TO 11259 11371 TO 11399 -
4862. 11487 TO 11501 11503 11511 TO 12067 12070 12071 12074 12075 12869 TO 12956
4863. CLT 0.06 MEMB 101 TO 172 201 TO 272 301 TO 372 401 TO 472 501 TO 572 -
4864. 601 TO 672 701 TO 772 801 TO 872 901 TO 972 1001 TO 1072 1101 TO 1172 1201 -
4865. 1202 TO 1272 1301 TO 1372 1401 TO 1472 1501 TO 1572 1601 TO 1672 1701 TO 1772 -
4866. 1801 TO 1872 1901 TO 1972 2001 TO 2072 2101 TO 2172 2201 TO 2272 -
4867. 2301 TO 2372 2401 TO 2472 10001 TO 10288 10295 TO 10354 10367 TO 10380 10382 -
4868. 10384 10386 10388 TO 10404 10406 10408 10410 10412 TO 10426 10439 TO 10452 -
4869. 10454 10456 10458 10460 TO 10476 10478 10480 10482 10484 TO 10498 -
4870. 10511 TO 10524 10526 10528 10530 10532 TO 10548 10550 10552 10554 -
4871. 10556 TO 10570 10577 TO 10596 10598 10600 10602 10604 TO 10620 10622 10624 -
4872. 10626 10628 TO 10668 10670 10672 10674 10676 TO 10692 10694 10696 10698 -
4873. 10700 TO 10792 10799 10802 10803 10808 10879 TO 10981 11198 TO 11203 11205 -
4874. 11206 11210 TO 11222 11225 TO 11253 11255 TO 11259 11371 TO 11399 -
4875. 11487 TO 11501 11503 11511 TO 12067 12070 12071 12074 12075 12869 TO 12956
4876. MAXMAIN 32 ALL
4877. MINMAIN 32 ALL
4878. FC 35000 MEMB 101 TO 172 201 TO 272 301 TO 372 401 TO 472 501 TO 572 -
4879. 601 TO 672 701 TO 772 801 TO 872 901 TO 972 1001 TO 1072 1101 TO 1172 1201 -
4880. 1202 TO 1272 1301 TO 1372 1401 TO 1472 1501 TO 1572 1601 TO 1672 1701 TO 1772 -
4881. 1801 TO 1872 1901 TO 1972 2001 TO 2072 2101 TO 2172 2201 TO 2272 -
4882. 2301 TO 2372 2401 TO 2472 10001 TO 10288 10295 TO 10354 10367 TO 10380 10382 -
4883. 10384 10386 10388 TO 10404 10406 10408 10410 10412 TO 10426 10439 TO 10452 -
4884. 10454 10456 10458 10460 TO 10476 10478 10480 10482 10484 TO 10498 -
4885. 10511 TO 10524 10526 10528 10530 10532 TO 10548 10550 10552 10554 -
4886. 10556 TO 10570 10577 TO 10596 10598 10600 10602 10604 TO 10620 10622 10624 -
4887. 10626 10628 TO 10668 10670 10672 10674 10676 TO 10692 10694 10696 10698 -
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4888. 10700 TO 10792 10799 10802 10803 10808 10879 TO 10981 11198 TO 11203 11205 -
4889. 11206 11210 TO 11222 11225 TO 11253 11255 TO 11259 11371 TO 11399 -
4890. 11487 TO 11501 11503 11511 TO 12067 12070 12071 12074 12075 12869 TO 12956
4891. TRACK 0 ALL
4892. DESIGN COLUMN 10799 10802 10803 10808 12070 12071 12074 12075 12869 TO 12884
```

ACI 318-11 COLUMN NO. 10799 DESIGN RESULTS
==========

FY - 420.0 FC - 35.0 MPA, RECT SIZE -1000.0 X1500.0 MMS, TIED
AREA OF STEEL REQUIRED = 16365.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
24 - 32 MM	1.287	158	STA	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 480.00 MM

ACI 318-11 COLUMN NO. 10802 DESIGN RESULTS
==========

FY - 420.0 FC - 35.0 MPA, RECT SIZE -1250.0 X2000.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 25000.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
32 - 32 MM	1.029	101	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 480.00 MM

ACI 318-11 COLUMN NO. 10803 DESIGN RESULTS
==========

FY - 420.0 FC - 35.0 MPA, RECT SIZE -1250.0 X2000.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 25000.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI

*

32 - 32 MM 1.029 101 END 0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 480.00 MM

ACI 318-11 COLUMN NO. 10808 DESIGN RESULTS

FY - 420.0 FC - 35.0 MPA, RECT SIZE -1000.0 X1500.0 MMS, TIED
AREA OF STEEL REQUIRED = 16365.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
24 - 32 MM	1.287	106	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 480.00 MM

ACI 318-11 COLUMN NO. 12070 DESIGN RESULTS

FY - 420.0 FC - 35.0 MPA, CIRC SIZE 1400.0 MMS DIAMETER TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 15393.8 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
20 - 32 MM	1.045	101	END	0.650

(EQUALLY SPACED)
TIE BAR NUMBER 10 SPACING 480.00 MM

ACI 318-11 COLUMN NO. 12071 DESIGN RESULTS

FY - 420.0 FC - 35.0 MPA, CIRC SIZE 1400.0 MMS DIAMETER TIED
AREA OF STEEL REQUIRED = 15501.6 SQ. MM

*

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
20 - 32 MM	1.045	108	STA	0.650

(EQUALLY SPACED)
TIE BAR NUMBER 10 SPACING 480.00 MM

ACI 318-11 COLUMN NO. 12074 DESIGN RESULTS

FY - 420.0 FC - 35.0 MPA, CIRC SIZE 1400.0 MMS DIAMETER TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 15393.8 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
20 - 32 MM	1.045	101	END	0.650

(EQUALLY SPACED)
TIE BAR NUMBER 10 SPACING 480.00 MM

ACI 318-11 COLUMN NO. 12075 DESIGN RESULTS

FY - 420.0 FC - 35.0 MPA, CIRC SIZE 1400.0 MMS DIAMETER TIED
AREA OF STEEL REQUIRED = 15501.6 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
20 - 32 MM	1.045	108	STA	0.650

(EQUALLY SPACED)
TIE BAR NUMBER 10 SPACING 480.00 MM

ACI 318-11 COLUMN NO. 12869 DESIGN RESULTS

*

FY - 420.0 FC - 35.0 MPA, RECT SIZE -1000.0 X1500.0 MMS, TIED
AREA OF STEEL REQUIRED = 15420.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
20 - 32 MM	1.072	163	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 480.00 MM

ACI 318-11 COLUMN NO. 12870 DESIGN RESULTS

FY - 420.0 FC - 35.0 MPA, RECT SIZE -1250.0 X2000.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 25000.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
32 - 32 MM	1.029	101	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 480.00 MM

ACI 318-11 COLUMN NO. 12871 DESIGN RESULTS

FY - 420.0 FC - 35.0 MPA, RECT SIZE -1250.0 X2000.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 25000.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
32 - 32 MM	1.029	101	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 480.00 MM

ACI 318-11 COLUMN NO. 12872 DESIGN RESULTS
==========

FY - 420.0 FC - 35.0 MPA, RECT SIZE -1000.0 X1500.0 MMS, TIED
AREA OF STEEL REQUIRED = 16365.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
24 - 32 MM	1.287	106	STA	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 480.00 MM

ACI 318-11 COLUMN NO. 12873 DESIGN RESULTS
==========

FY - 420.0 FC - 35.0 MPA, CIRC SIZE 1400.0 MMS DIAMETER TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 15393.8 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
20 - 32 MM	1.045	101	END	0.650

(EQUALLY SPACED)
TIE BAR NUMBER 10 SPACING 480.00 MM

ACI 318-11 COLUMN NO. 12874 DESIGN RESULTS
==========

FY - 420.0 FC - 35.0 MPA, CIRC SIZE 1400.0 MMS DIAMETER TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 15393.8 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI

*

20 - 32 MM 1.045 101 END 0.650

(EQUALLY SPACED)

TIE BAR NUMBER 10 SPACING 480.00 MM

ACI 318-11 COLUMN NO. 12875 DESIGN RESULTS

==========

FY - 420.0 FC - 35.0 MPA, CIRC SIZE 1400.0 MMS DIAMETER TIED

ONLY MINIMUM STEEL IS REQUIRED.

AREA OF STEEL REQUIRED = 15393.8 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
20 - 32 MM	1.045	101	END	0.650

(EQUALLY SPACED)

TIE BAR NUMBER 10 SPACING 480.00 MM

ACI 318-11 COLUMN NO. 12876 DESIGN RESULTS

==========

FY - 420.0 FC - 35.0 MPA, CIRC SIZE 1400.0 MMS DIAMETER TIED

ONLY MINIMUM STEEL IS REQUIRED.

AREA OF STEEL REQUIRED = 15393.8 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
20 - 32 MM	1.045	101	END	0.650

(EQUALLY SPACED)

TIE BAR NUMBER 10 SPACING 480.00 MM

ACI 318-11 COLUMN NO. 12877 DESIGN RESULTS

==========

FY - 420.0 FC - 35.0 MPA, RECT SIZE -1000.0 X1500.0 MMS, TIED

AREA OF STEEL REQUIRED = 16260.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
24 - 32 MM	1.287	152	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 480.00 MM

ACI 318-11 COLUMN NO. 12878 DESIGN RESULTS

FY - 420.0 FC - 35.0 MPA, RECT SIZE -1250.0 X2000.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 25000.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
32 - 32 MM	1.029	101	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 480.00 MM

ACI 318-11 COLUMN NO. 12879 DESIGN RESULTS

FY - 420.0 FC - 35.0 MPA, RECT SIZE -1250.0 X2000.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 25000.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
32 - 32 MM	1.029	101	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 480.00 MM

ACI 318-11 COLUMN NO. 12880 DESIGN RESULTS

*

FY - 420.0 FC - 35.0 MPA, RECT SIZE -1000.0 X1500.0 MMS, TIED
AREA OF STEEL REQUIRED = 15420.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
20 - 32 MM	1.072	130	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 480.00 MM

ACI 318-11 COLUMN NO. 12881 DESIGN RESULTS

FY - 420.0 FC - 35.0 MPA, CIRC SIZE 1400.0 MMS DIAMETER TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 15393.8 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
20 - 32 MM	1.045	101	END	0.650

(EQUALLY SPACED)
TIE BAR NUMBER 10 SPACING 480.00 MM

ACI 318-11 COLUMN NO. 12882 DESIGN RESULTS

FY - 420.0 FC - 35.0 MPA, CIRC SIZE 1400.0 MMS DIAMETER TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 15393.8 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
20 - 32 MM	1.045	101	END	0.650

(EQUALLY SPACED)
TIE BAR NUMBER 10 SPACING 480.00 MM

ACI 318-11 COLUMN NO. 12883 DESIGN RESULTS
==========

FY - 420.0 FC - 35.0 MPA, CIRC SIZE 1400.0 MMS DIAMETER TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 15393.8 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
20 - 32 MM	1.045	101	END	0.650

(EQUALLY SPACED)
TIE BAR NUMBER 10 SPACING 480.00 MM

ACI 318-11 COLUMN NO. 12884 DESIGN RESULTS
==========

FY - 420.0 FC - 35.0 MPA, CIRC SIZE 1400.0 MMS DIAMETER TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 15393.8 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
20 - 32 MM	1.045	101	END	0.650

(EQUALLY SPACED)
TIE BAR NUMBER 10 SPACING 480.00 MM

**********END OF COLUMN DESIGN RESULTS**********

```
4893. *DESIGN ELEMENT 101 TO 172 201 TO 272 301 TO 372 401 TO 472 501 TO 572 601 -
4894. *602 TO 672 701 TO 772 801 TO 872 901 TO 972 1001 TO 1072 1101 TO 1172 1201 -
4895. *1202 TO 1272 1301 TO 1372 1401 TO 1472 1501 TO 1572 1601 TO 1672 1701 TO 1772
4896. *1801 TO 1872 1901 TO 1972 2001 TO 2072 2101 TO 2172 2201 TO 2272 -
4897. *2301 TO 2372 2401 TO 2472 10001 TO 10288 10295 TO 10354 10367 TO 10380 10382 -
4898. *10384 10386 10388 TO 10404 10406 10408 10410 10412 TO 10426 10439 TO 10452 -
4899. *10454 10456 10458 10460 TO 10476 10478 10480 10482 10484 TO 10498 -
4900. *10511 TO 10524 10526 10528 10530 10532 TO 10548 10550 10552 10554 -
4901. *10556 TO 10570 10577 TO 10596 10598 10600 10602 10604 TO 10620 10622 10624 -
4902. *10626 10628 TO 10668 10670 10672 10674 10676 TO 10692 10694 10696 10698 -
4903. *10700 TO 10792 10879 TO 10981 11198 TO 11203 11205 11206 11210 TO 11222 -
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  *
4904. *11225 TO 11253 11255 TO 11259 11371 TO 11399 11487 TO 11501 11503 -
4905. *11511 TO 12067 12885 TO 12956
4906. END CONCRETE DESIGN
4907. FINISH


     ********** END OF THE STAAD.Pro RUN **********

       **** DATE= MAR  2,2020   TIME= 12: 1:56 ****

    ************************************************************
    *   For technical assistance on STAAD.Pro, please visit    *
    *   http://selectservices.bentley.com/en-US/               *
    *                                                          *
    *   Details about additional assistance from               *
    *   Bentley and Partners can be found at program menu      *
    *   Help->Technical Support                                *
    *                                                          *
    *      Copyright (c) 1997-2016 Bentley Systems, Inc.       *
    *                http://www.bentley.com                    *
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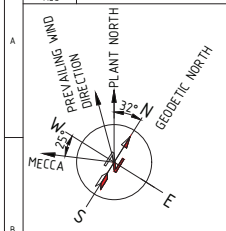
ATTACHMENT 3 MECHANICAL INPUT DRAWINGS

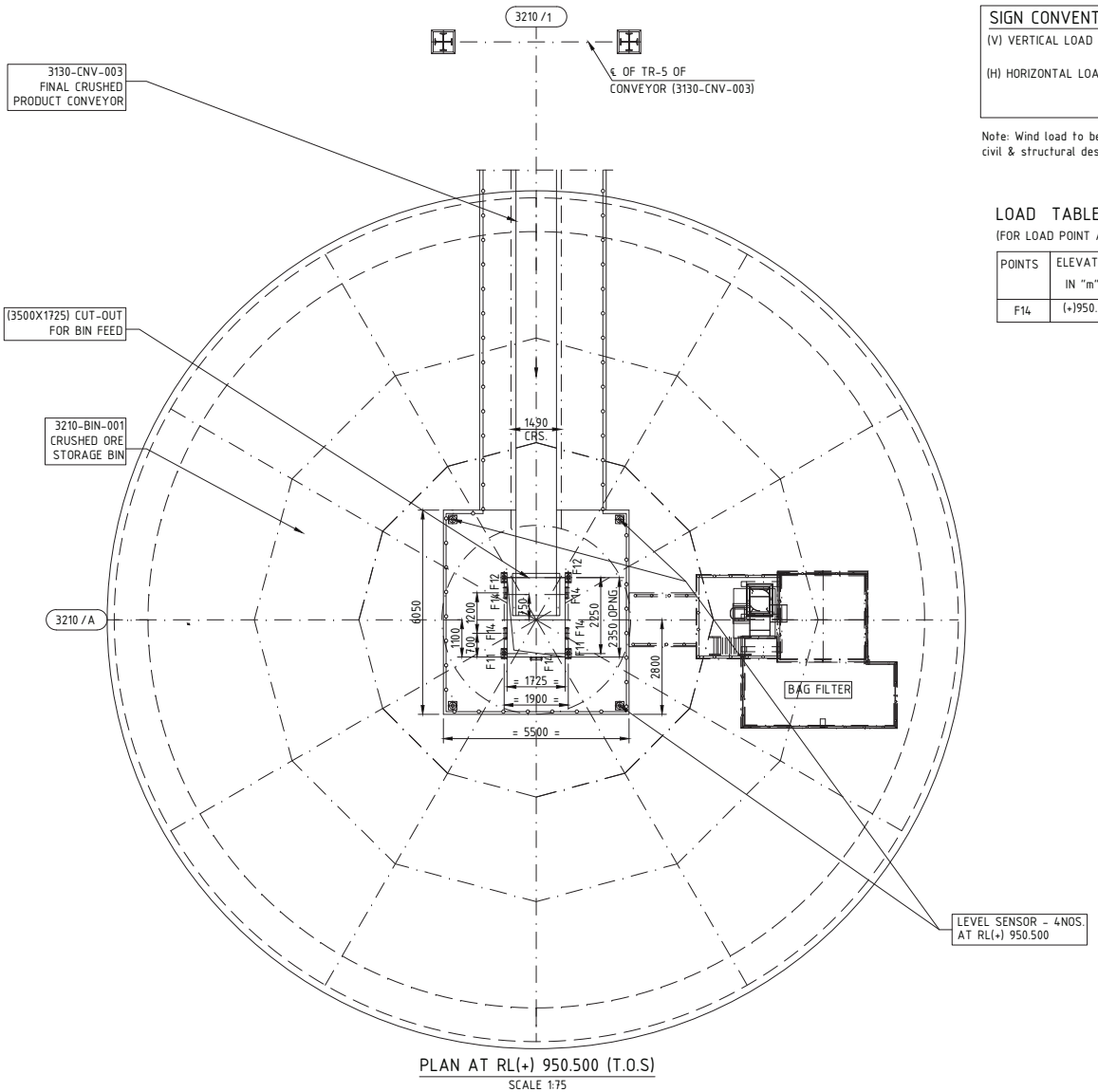
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DRAWING NUMBER MD-216-3210-EG-ME-LPX-1006 REV A01





PLAN AT RL(+) 950.500 (T.O.S)

SCALE 1:75

SIGN CONVENTION

(V) VERTICAL LOAD : UPWARD (-)
: DOWNWARD (+)

(H) HORIZONTAL LOAD : TOWARDS HEAD PULLEY=(+)
: TOWARDS TAIL PULLEY=(-)
: REVERSIBLE LOAD=(±)

Note: Wind load to be considered during civil & structural design

NOTE:
1. ALL DIMENSIONS ARE IN MILLIMETER, CO-ORDINATES AND ELEVATION ARE IN METER, UNLESS OTHERWISE STATED.
2. FINISHED FLOOR LEVEL (F.F.L) FOR CRUSHING AREA = RL(+)913.000 m
3. FINISHED GROUND LEVEL (F.G.L) FOR CRUSHING AREA = RL(+)912.500 m
4. F.F.L, F.G.L, T.O.C, = FINISH FLOOR LEVEL, FINISH GROUND LEVEL, TOP OF CONCRETE, T.O.S, B.O.S, B.O.B= TOP OF STEEL, BOTTOM OF STEEL, BOTTOM OF BASE PLATE
5. ALL MAINTENANCE FACILITIES, HANDLING PROVISIONS, ACCESS WAYS, PLATFORMS FOR MAINTENANCE ARE SHOWN TENTATIVE.

LOAD TABLE -5

(FOR LOAD POINT AT CHUTE SADDLE)

POINTS	ELEVATIONS IN "m"	V (IN "t")
F14	(+)950.250	(+)1.5

LOAD TABLE-6 : LOAD TABLE FOR HEAD FRAME

FOR CONV. 3130-CNV-003

POINT	LOAD CASE I			LOAD CASE II		
	V (IN "t")	H (IN "t")	ACROSS	V (IN "t")	H (IN "t")	ACROSS
F11	(-)4.2	(-)7.1	(±)0.5	(-)9.2	(-)10.4	(±)0.9
F12	(+)13.8	(-)7.8	(±)1.4	(+)17.1	(-)10.0	(±)1.7

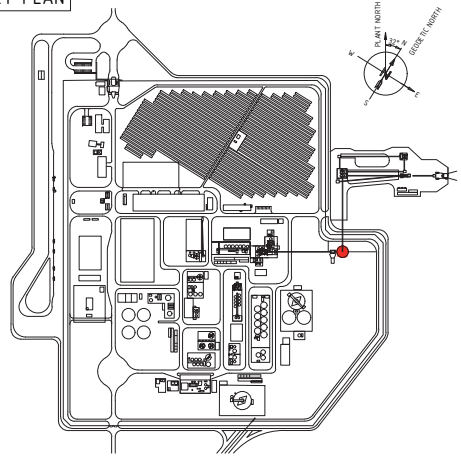
NOTE:CASE I :CONV. NORMAL RUNNING+CHUTE BLOCKED+DEAD
CASE II :CONV. STARTING +CHUTE EMPTY+DEAD

ABBREVIATION

W.P.:	WORKING POINT
℄:	CENTER LINE
DN:	DOWNWARD
UP:	UPWARD
CRS.:	CENTERS
thk.:	THICKNESS
RCC:	REINFORCED CEMENT CONCRETE
deg.:	DEGREE
B.O.M.R:	BOTTOM OF MONORAIL
OPNG.:	OPENING
TR:	TRESTLE
TYP.:	TYPICAL
PROJ:	PROJECTED
W:	WIDTH
H:	HEIGHT

CONCRETE FLOOR
CHECKERED PLATE
GRATING
BRICK WALL
CONCRETE WALL

KEY PLAN



DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-3210-EG-ME-LPX-1005	CRUSHED ORE STORAGE BIN & TRANSFER - PLAN								
MD-216-3210-EG-ME-LPX-1004	CRUSHED ORE STORAGE BIN & TRANSFER - PLAN								
MD-216-3210-EG-ME-ELV-1010	CRUSHED ORE STORAGE BIN & TRANSFER - SECTION								
MD-216-3210-EG-ME-ELV-1009	CRUSHED ORE STORAGE BIN & TRANSFER - ELEVATION								
MD-216-3210-EG-ME-LPX-1003	CRUSHED ORE STORAGE BIN & TRANSFER - PLAN	A01	21/02/2020	ISSUED FOR APPROVAL	HKD	BRD	BD	BGR	

RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

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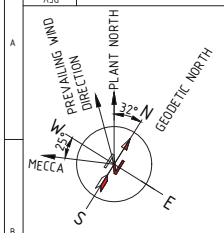
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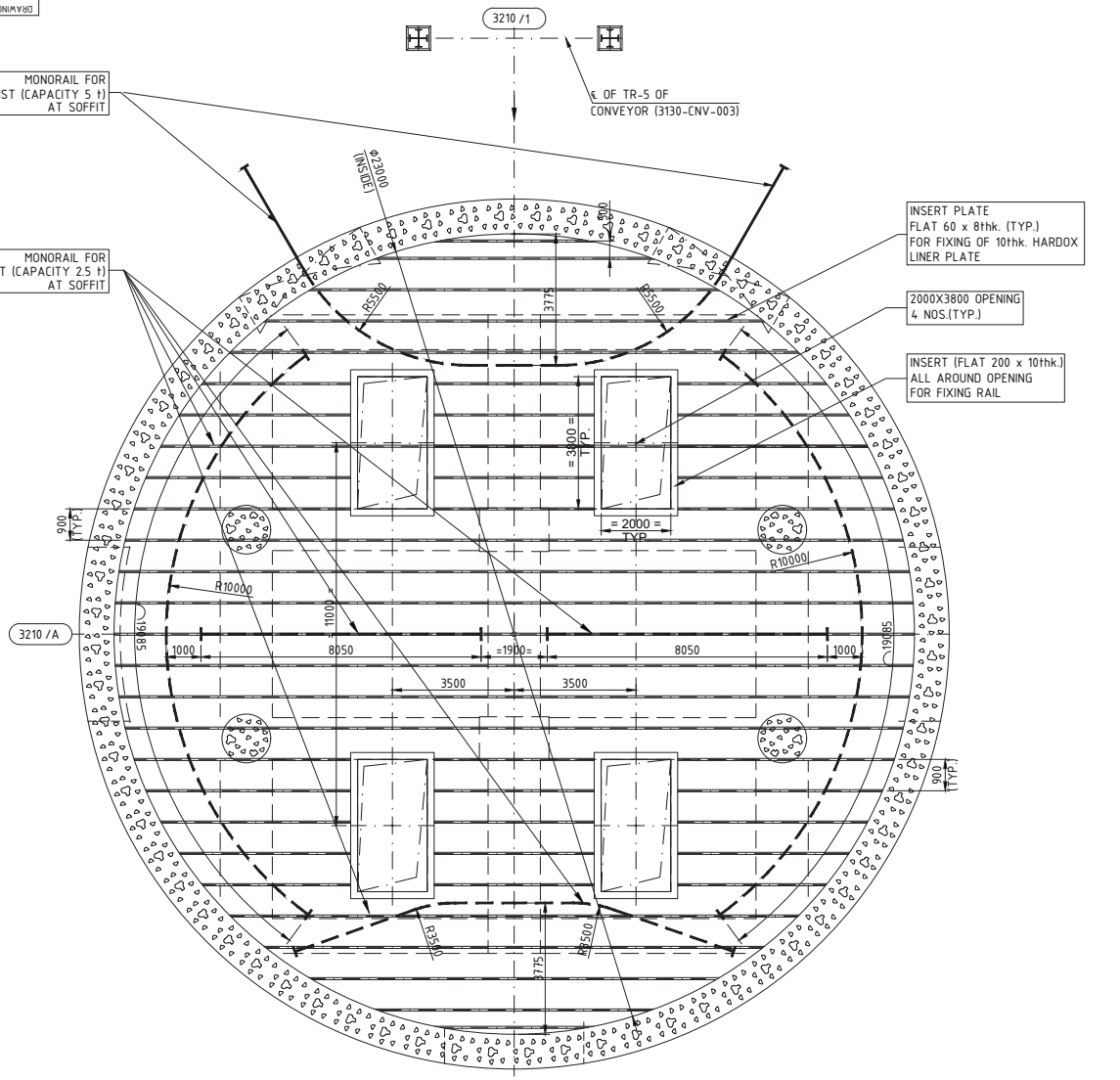
DRAWN	HKD	18/02/2020
CHECKED	BRD	20/02/2020
ENGINEER	BD	20/02/2020
APPROVED	BGR	21/02/2020
EMPLOYER		
SCALE	1:75	A1

MANSOURAH - MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT DRAWING
GRINDING & CLASSIFICATION
CRUSHED ORE STORAGE BIN & TRANSFER - PLAN

DRAWING NUMBER: MD-216-3210-EG-ME-LPX-1006 — REV: A01

DRAWING NUMBER: MD-216-3210-EG-ME-LPX-1005 REV: A01





PLAN AT RL(+) 920.800 (SOFFIT)
SCALE 1:75

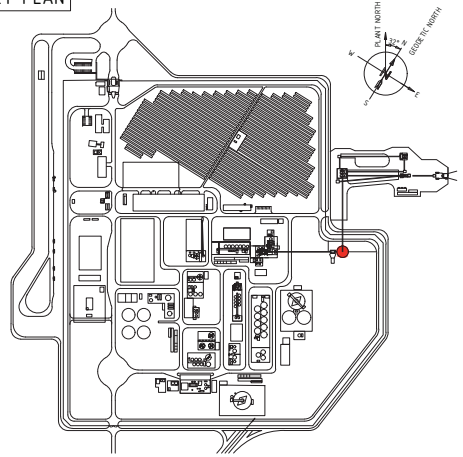
NOTE:
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2. FINISHED FLOOR LEVEL (F.F.L) FOR CRUSHING AREA = RL(+)913.000 m
3. FINISHED GROUND LEVEL (F.G.L) FOR CRUSHING AREA = RL(+)912.500 m
4. F.F.L, F.G.L, T.O.C, = FINISH FLOOR LEVEL, FINISH GROUND LEVEL, TOP OF CONCRETE, T.O.S, B.O.S, B.O.B= TOP OF STEEL, BOTTOM OF STEEL, BOTTOM OF BASE PLATE
5. ALL MAINTENANCE FACILITIES, HANDLING PROVISIONS, ACCESS WAYS, PLATFORMS FOR MAINTENANCE ARE SHOWN TENTATIVE.

ABBREVIATION

W.P.:	WORKING POINT
℄:	CENTER LINE
DN:	DOWNWARD
UP:	UPWARD
CRS.:	CENTERS
thk.:	THICKNESS
RCC:	REINFORCED CEMENT CONCRETE
deg.:	DEGREE
B.O.M.R:	BOTTOM OF MONORAIL
OPNG.:	OPENING
TR:	TRESTLE
TYP.:	TYPICAL
PROJ:	PROJECTED
W:	WIDTH
H:	HEIGHT

CONCRETE FLOOR
CHECKERED PLATE
GRATING
BRICK WALL
CONCRETE WALL

KEY PLAN



DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-3210-EG-ME-LPX-1006	CRUSHED ORE STORAGE BIN & TRANSFER - PLAN								
MD-216-3210-EG-ME-LPX-1004	CRUSHED ORE STORAGE BIN & TRANSFER - PLAN								
MD-216-3210-EG-ME-LPX-1010	CRUSHED ORE STORAGE BIN & TRANSFER - SECTION								
MD-216-3210-EG-ME-ELV-1009	CRUSHED ORE STORAGE BIN & TRANSFER - ELEVATION								
MD-216-3210-EG-ME-LPX-1003	CRUSHED ORE STORAGE BIN & TRANSFER - PLAN	A01	21/02/2020	ISSUED FOR APPROVAL	HKD	BRD	BD	BGR	

RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

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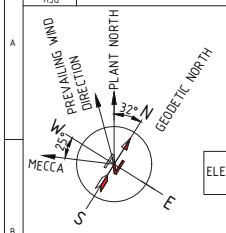
DRAWN	HKD	18/02/2020
CHECKED	BRD	20/02/2020
ENGINEER	BD	20/02/2020
APPROVED	BGR	21/02/2020
EMPLOYER		
SCALE	1:75	A1

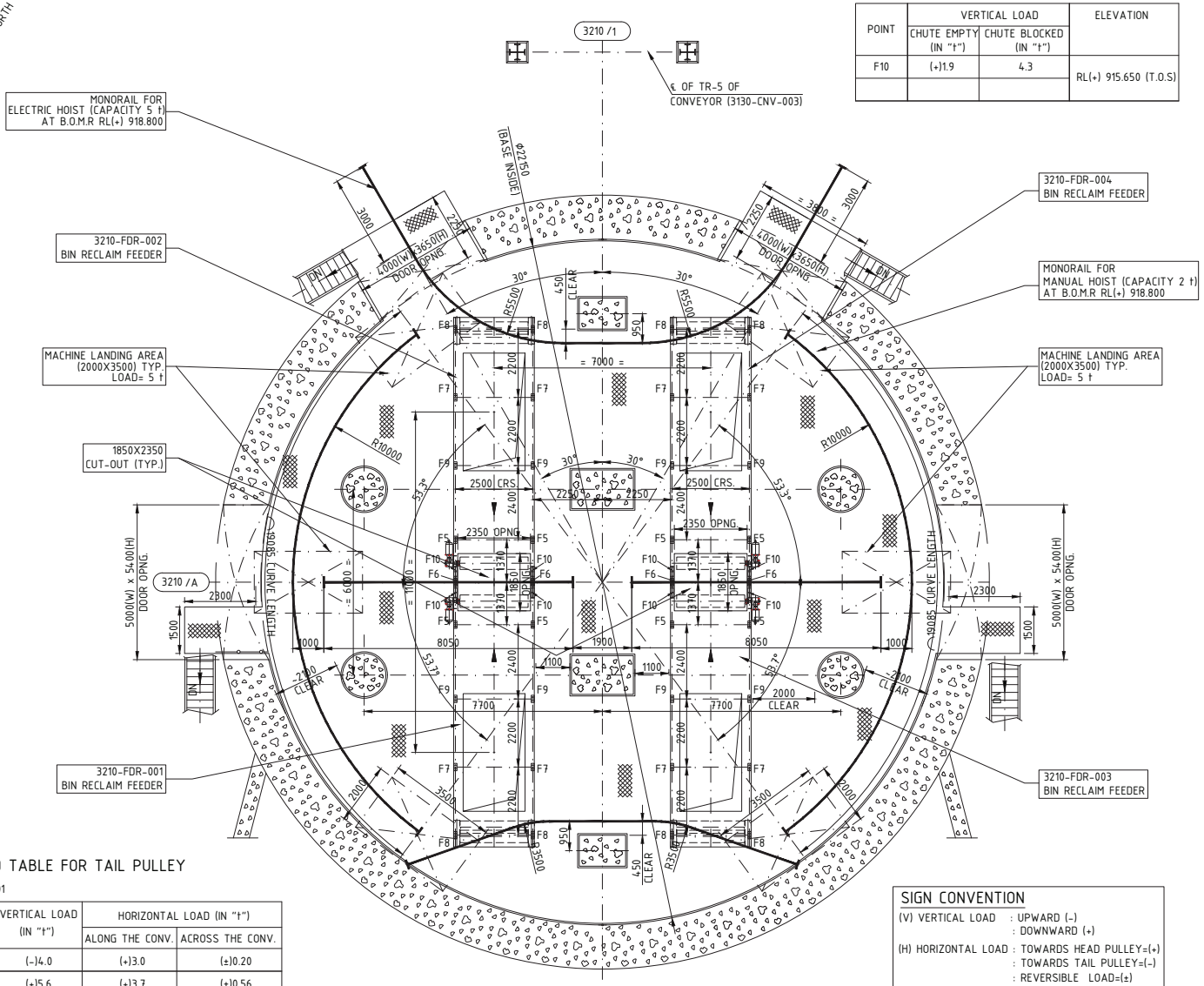
LARSEN & TOUBRO DRAWING NUMBER: 019045_100_H_G_16126_005 — REVISION: 0
OUTOTEC DRAWING NUMBER: 900707__GAU01__US01__MTL01__32105 — DOCUMENT ID: OU100669119 — REVISION: 0

MANSOURAH - MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT DRAWING
GRINDING & CLASSIFICATION
CRUSHED ORE STORAGE BIN & TRANSFER - PLAN

DRAWING NUMBER: MD-216-3210-EG-ME-LPX-1005 — REV: A01

DRAWING NUMBER: MD-216-3210-EG-ME-LPX-1004 | REV: A01





PLAN AT RL(+)915.650 (T.O.S)

SCALE 1:75

LOAD TABLE-4 (LOAD TABLE FOR CHUTE SADDLE)

FOR CONV. 3210-FDR-001, 3210-FDR-002, 3210-FDR-003,3210-FDR-004

POINT	VERTICAL LOAD		ELEVATION
	CHUTE EMPTY (IN "t")	CHUTE BLOCKED (IN "t")	
F10	(+)1.9	4.3	RL(+) 915.650 (T.O.S)

NOTE:

1. ALL DIMENSIONS ARE IN MILLIMETER, CO-ORDINATES AND ELEVATION ARE IN METER, UNLESS OTHERWISE STATED.
2. FINISHED FLOOR LEVEL (F.F.L) FOR CRUSHING AREA = RL(+)913.000 m
3. FINISHED GROUND LEVEL (F.G.L) FOR CRUSHING AREA = RL(+)912.500 m
4. F.F.L, F.G.L, T.O.C, = FINISH FLOOR LEVEL, FINISH GROUND LEVEL, TOP OF CONCRETE, T.O.S, B.O.S, B.O.B= TOP OF STEEL, BOTTOM OF STEEL, BOTTOM OF BASE PLATE
5. ALL MAINTENANCE FACILITIES, HANDLING PROVISIONS, ACCESS WAYS, PLATFORMS FOR MAINTENANCE ARE SHOWN TENTATIVE.

LOAD TABLE-3 : LOAD TABLE FOR BIN RECLAIM FEEDER

3210-520 HOLD

FOR CONV. 3210-FDR-001, 3210-FDR-002, 3210-FDR-003,3210-FDR-004

POINT	ELEVATION (IN "m")	DEAD LOAD		DEAD LOAD + MATERIAL LOAD	
		V (IN "t")	H (IN "t")	V (IN "t")	H (IN "t")
F5	RL(+) 915.650	1.9	0.19	3.8	0.38
F6		1.9	0.19	3.8	0.38
F7		1.9	0.19	4.5	0.45
F8		1.9	0.19	4.5	0.45
F9		1.9	0.19	6.5	0.65

TABLE-1 : LOAD TABLE FOR TAIL PULLEY

FOR CONV. 3210-CNV-001

POINT	ELEVATION (IN "m")	VERTICAL LOAD (IN "t")	HORIZONTAL LOAD (IN "t")	
			ALONG THE CONV.	ACROSS THE CONV.
F1	RL(+) 912.750	(-)4.0	(+)3.0	(±)0.20
F2		(+)5.6	(+)3.7	(±)0.56

SIGN CONVENTION

(V) VERTICAL LOAD : UPWARD (-)
: DOWNWARD (+)

(H) HORIZONTAL LOAD : TOWARDS HEAD PULLEY=(+)
: TOWARDS TAIL PULLEY=(-)
: REVERSIBLE LOAD=(±)

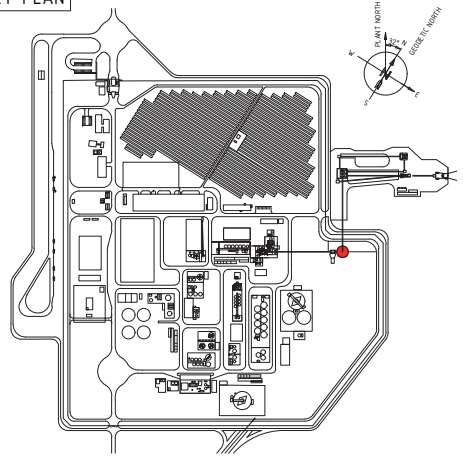
Note: Wind load to be considered during civil & structural design

ABBREVIATION

W.P.:	WORKING POINT
℄:	CENTER LINE
DN:	DOWNWARD
UP:	UPWARD
CRS.:	CENTERS
thk.:	THICKNESS
RCC:	REINFORCED CEMENT CONCRETE
deg.:	DEGREE
B.O.M.R:	BOTTOM OF MONORAIL
OPNG.:	OPENING
TR:	TRESTLE
TYP.:	TYPICAL
PROJ:	PROJECTED
W:	WIDTH
H:	HEIGHT

CONCRETE FLOOR
CHECKERED PLATE
GRATING
BRICK WALL
CONCRETE WALL

KEY PLAN



DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-3210-EG-ME-LPX-1006	CRUSHED ORE STORAGE BIN & TRANSFER - PLAN								
MD-216-3210-EG-ME-LPX-1005	CRUSHED ORE STORAGE BIN & TRANSFER - PLAN								
MD-216-3210-EG-ME-LPX-1010	CRUSHED ORE STORAGE BIN & TRANSFER - SECTION								
MD-216-3210-EG-ME-ELV-1009	CRUSHED ORE STORAGE BIN & TRANSFER - ELEVATION								
MD-216-3210-EG-ME-LPX-1003	CRUSHED ORE STORAGE BIN & TRANSFER - PLAN	A01	21/02/2020	ISSUED FOR APPROVAL	HKD	BRD	BD	BGR	

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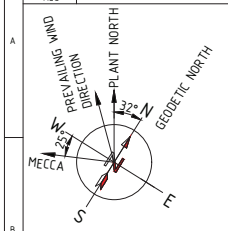
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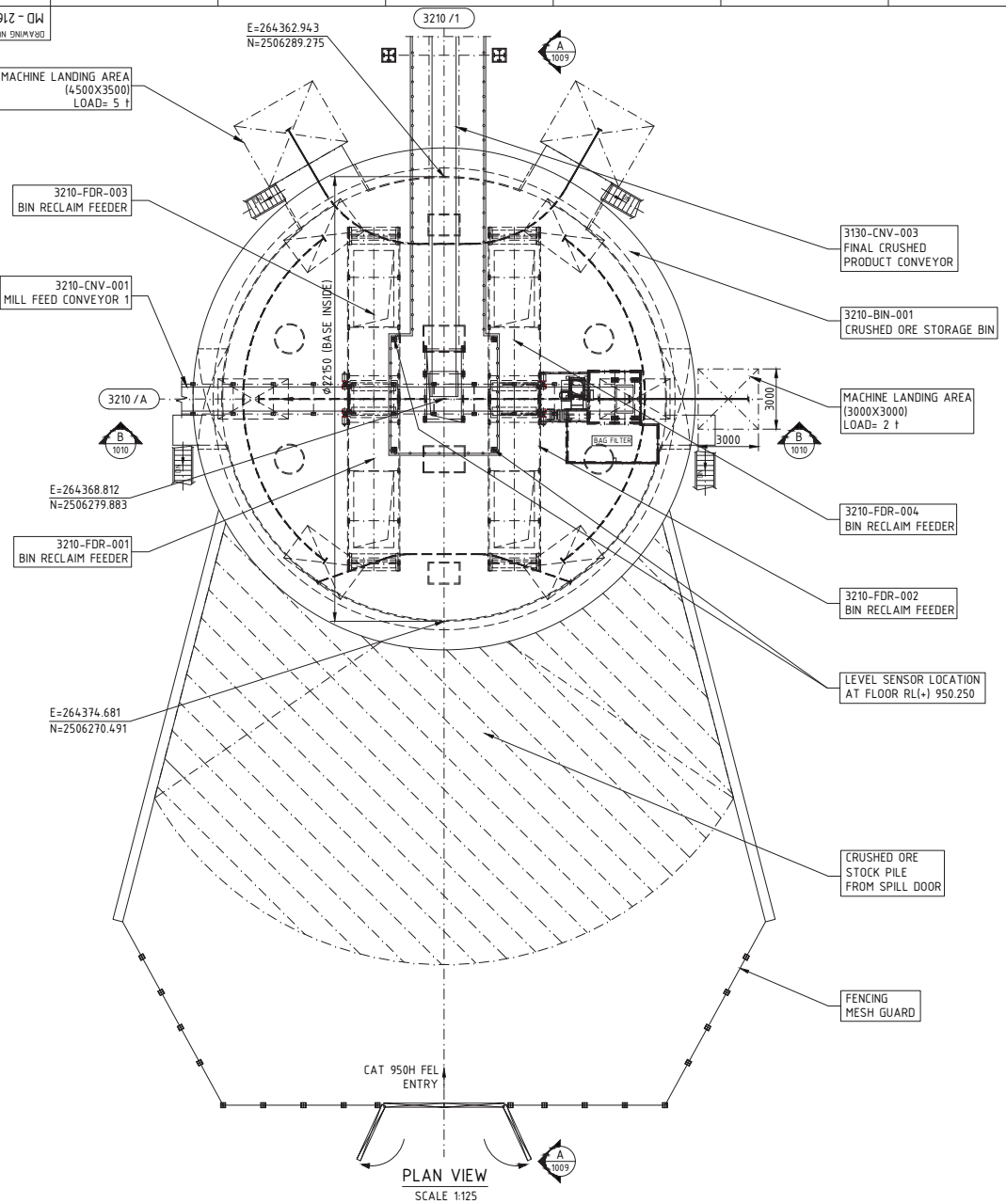
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CHECKED	BRD	20/02/2020	GENERAL ARRANGEMENT DRAWING
ENGINEER	BD	20/02/2020	GRINDING & CLASSIFICATION
APPROVED	BGR	21/02/2020	CRUSHED ORE STORAGE BIN & TRANSFER - PLAN
EMPLOYER			
SCALE	1:75	A1	

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OUTOTEC DRAWING NUMBER: 900707__GAU01__US01__MTL01__32104 | DOCUMENT ID: OU100669118 | REVISION: 0
DRAWING NUMBER: MD-216-3210-EG-ME-LPX-1004 | REV: A01





PLAN VIEW
SCALE 1:125

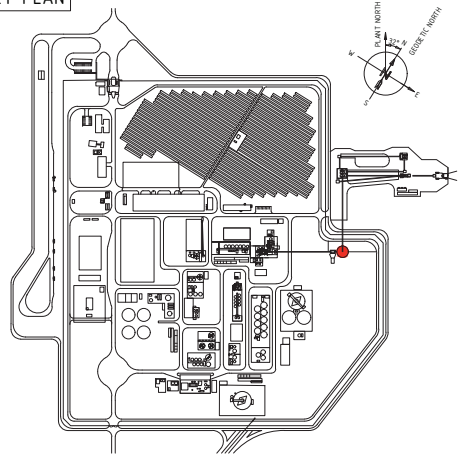
NOTE:
1. ALL DIMENSIONS ARE IN MILLIMETER, CO-ORDINATES AND ELEVATION ARE IN METER, UNLESS OTHERWISE STATED.
2. FINISHED FLOOR LEVEL (F.F.L) FOR CRUSHING AREA = RL(+)913.000 m
3. FINISHED GROUND LEVEL (F.G.L) FOR CRUSHING AREA = RL(+)912.500 m
4. F.F.L, F.G.L, T.O.C, = FINISH FLOOR LEVEL, FINISH GROUND LEVEL, TOP OF CONCRETE, T.O.S, B.O.S, B.O.B= TOP OF STEEL, BOTTOM OF STEEL, BOTTOM OF BASE PLATE
5. ALL MAINTENANCE FACILITIES, HANDLING PROVISIONS, ACCESS WAYS, PLATFORMS FOR MAINTENANCE ARE SHOWN TENTATIVE.

ABBREVIATION

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UP:	UPWARD
CRS.:	CENTERS
thk.:	THICKNESS
RCC:	REINFORCED CEMENT CONCRETE
deg.:	DEGREE
B.O.M.R:	BOTTOM OF MONORAIL
OPNG.:	OPENING
TR:	TRESTLE
TYP.:	TYPICAL
PROJ:	PROJECTED
W:	WIDTH
H:	HEIGHT

CONCRETE FLOOR
CHECKERED PLATE
GRATING
BRICK WALL
CONCRETE WALL

KEY PLAN



DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-3210-EG-ME-LPX-1006	CRUSHED ORE STORAGE BIN & TRANSFER - PLAN								
MD-216-3210-EG-ME-LPX-1005	CRUSHED ORE STORAGE BIN & TRANSFER - PLAN								
MD-216-3210-EG-ME-LPX-1004	CRUSHED ORE STORAGE BIN & TRANSFER - PLAN								
MD-216-3210-EG-ME-ELV-1010	CRUSHED ORE STORAGE BIN & TRANSFER - SECTION								
MD-216-3210-EG-ME-ELV-1009	CRUSHED ORE STORAGE BIN & TRANSFER - ELEVATION	A01	21/02/2020	ISSUED FOR APPROVAL	HKD	BRD	BD	BGR	

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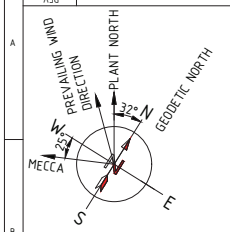
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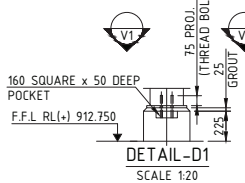
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CHECKED	BRD	20/02/2020	GENERAL ARRANGEMENT DRAWING
ENGINEER	BD	20/02/2020	GRINDING & CLASSIFICATION
APPROVED	BGR	21/02/2020	CRUSHED ORE STORAGE BIN & TRANSFER - PLAN
EMPLOYER			
SCALE	1:125	A1	

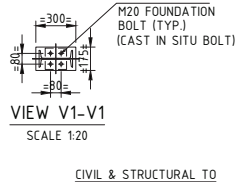
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OUTOTEC DRAWING NUMBER: 900707__GAU01__US01__MTL01__32103 — DOCUMENT ID: OU100669117 — REVISION: 0
DRAWING NUMBER: MD-216-3210-EG-ME-LPX-1003 — REV: A01

DRAWING NUMBER: MD-216-3120-EG-ME-ELV-1010 REV A01

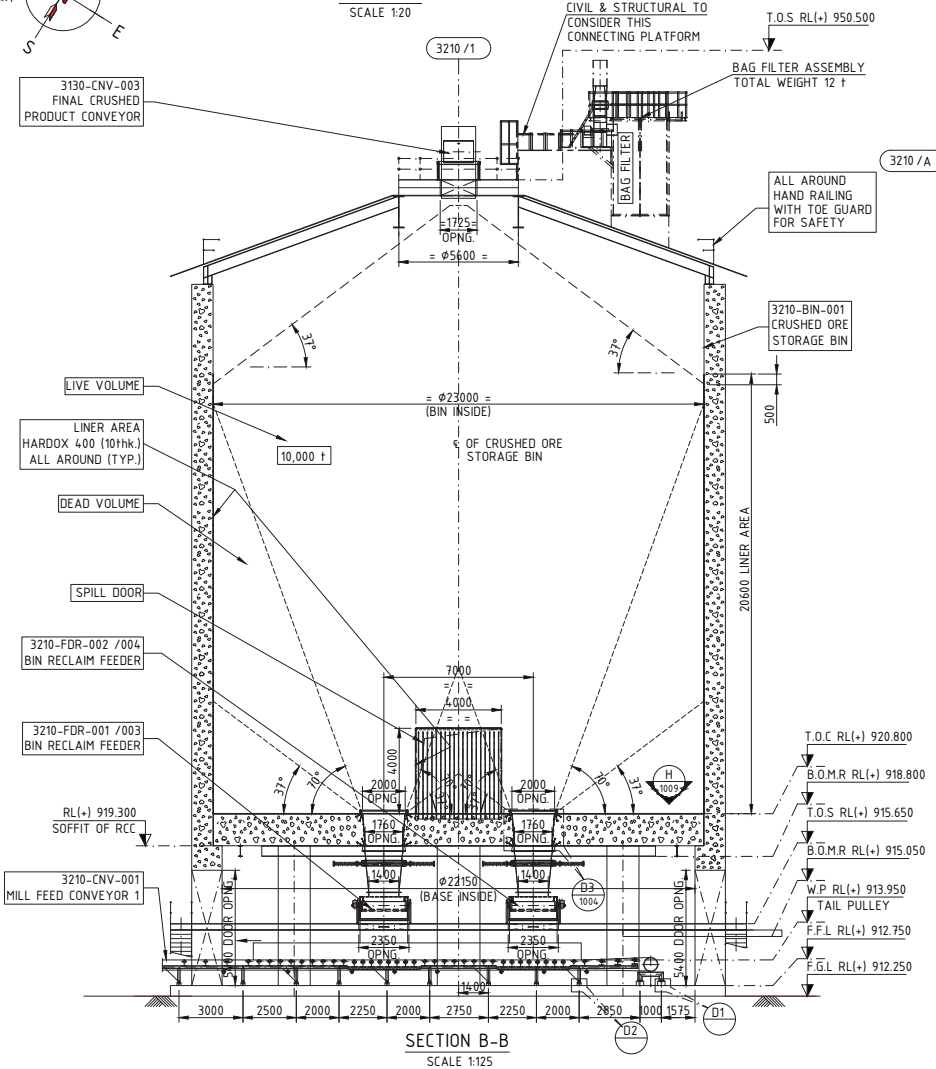




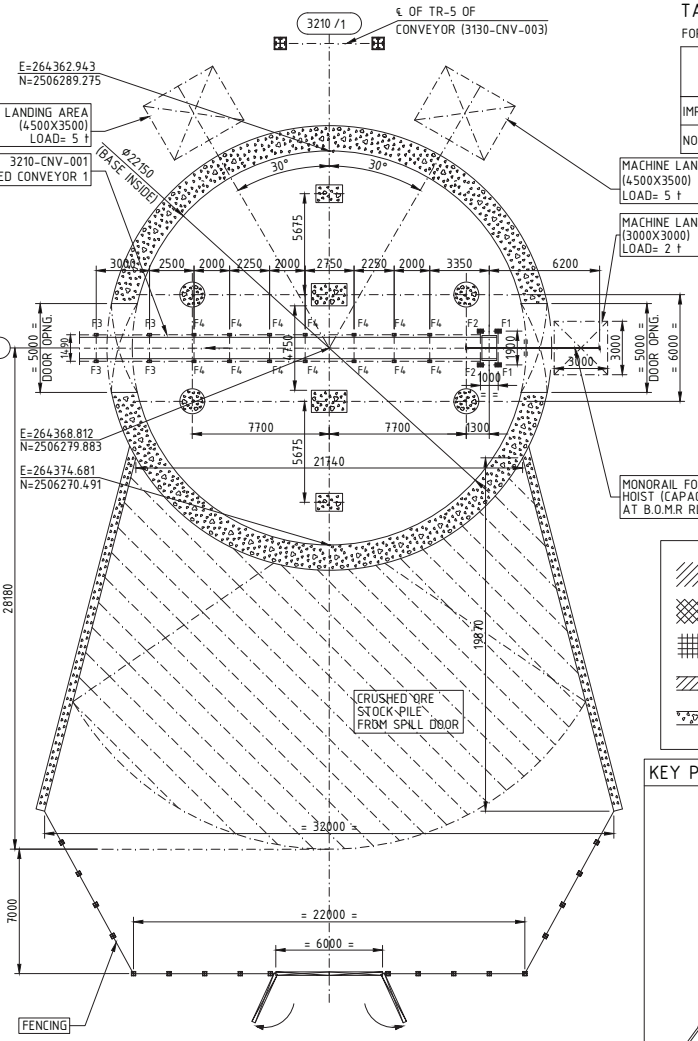
DETAIL-D1
SCALE 1:20



VIEW V1-V1
SCALE 1:20



SECTION B-B
SCALE 1:125



PLAN AT RL(+) 912.750 (F.F.L)
SCALE 1:150

NOTE:
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5. ALL MAINTENANCE FACILITIES, HANDLING PROVISIONS, ACCESS WAYS, PLATFORMS FOR MAINTENANCE ARE SHOWN TENTATIVE.

TABLE- 2 : LOAD TABLE FOR SHORT POST

FOR CONV. 3210-CNV-001

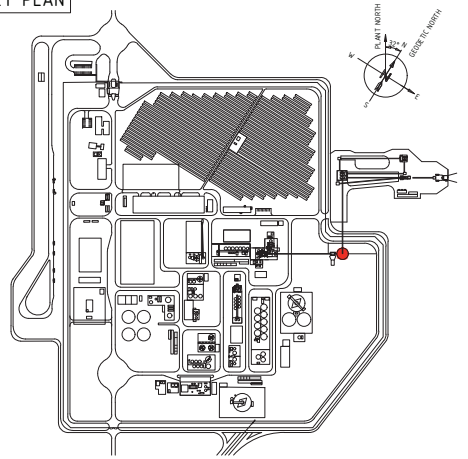
ZONE	POINT	VERTICAL LOAD (IN "t")		HORIZONTAL LOAD (IN "t")	
		DEAD LOAD	LIVE LOAD	ALONG	ACROSS
IMPACT ZONE	F4	(+)1.2	(+)0.55	(±)0.17	(±)0.09
NORMAL ZONE	F3	(+)0.98	(+)0.55	(±)0.17	(±)0.09

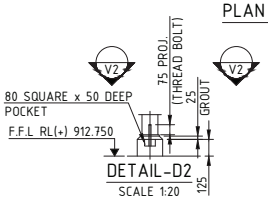
CONCRETE FLOOR
CHECKERED PLATE
GRATING
BRICK WALL
CONCRETE WALL

ABBREVIATION

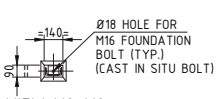
W.P.:	WORKING POINT
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CRS.:	CENTERS
thk.:	THICKNESS
RCC:	REINFORCED CEMENT CONCRETE
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B.O.M.R:	BOTTOM OF MONORAIL
OPNG.:	OPENING
TR:	TRESTLE
TYP.:	TYPICAL
PROJ:	PROJECTED
W:	WIDTH
H:	HEIGHT

KEY PLAN





DETAIL-D2
SCALE 1:20



VIEW V2-V2
SCALE 1:20

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-3210-EG-ME-LPX-1006	CRUSHED ORE STORAGE BIN & TRANSFER - PLAN								
MD-216-3210-EG-ME-LPX-1005	CRUSHED ORE STORAGE BIN & TRANSFER - PLAN								
MD-216-3210-EG-ME-LPX-1004	CRUSHED ORE STORAGE BIN & TRANSFER - PLAN								
MD-216-3210-EG-ME-ELV-1009	CRUSHED ORE STORAGE BIN & TRANSFER - ELEVATION								
MD-216-3210-EG-ME-LPX-1003	CRUSHED ORE STORAGE BIN & TRANSFER - PLAN	A01	21/02/2020	ISSUED FOR APPROVAL	HKD	BRD	BD	BGR	

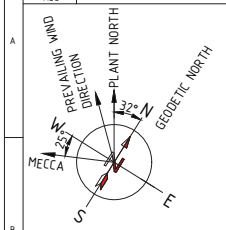
RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

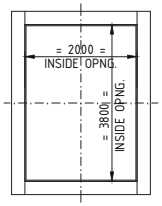
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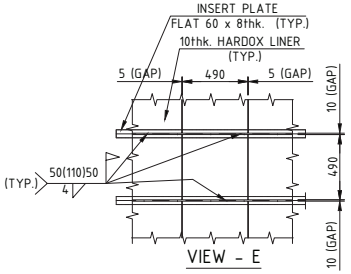
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LARSEN & TOUBRO DRAWING NUMBER			REVISION
O19045_100_H_G_16126_003			0
OUTOTEC DRAWING NUMBER		DOCUMENT ID	REVISION
900707 _GAU01 _ZC01 _US01 _32115		OU100669134	0
DRAWN	HKD	18/02/2020	MANSOURAH - MASSARAH GOLD PROJECT
CHECKED	BRD	20/02/2020	GENERAL ARRANGEMENT DRAWING
ENGINEER	BD	20/02/2020	GRINDING & CLASSIFICATION
APPROVED	BGR	21/02/2020	CRUSHED ORE STORAGE BIN & TRANSFER - SECTION
EMPLOYER			
SCALE	1:150, 1:125, 1:20	A1	DRAWING NUMBER: MD-216-3210-EG-ME-ELV-1010 REV: A01

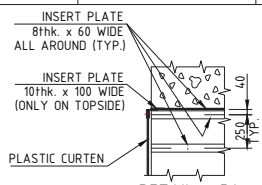




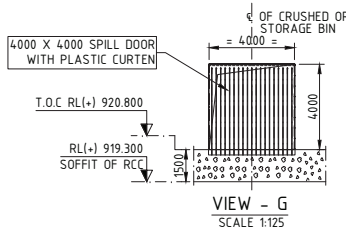
VIEW C-C
SCALE 1:40



VIEW - E
(TYP. LINER FIXING DETAILS
ON VERTICAL RCC WALL)
SCALE 1:20



DETAIL - D4
SCALE 1:20

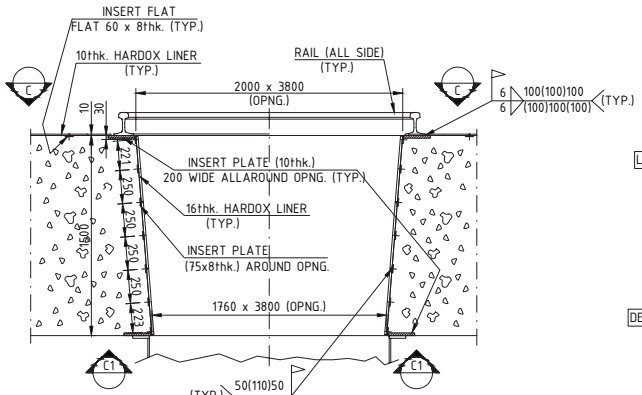


VIEW - G
SCALE 1:125

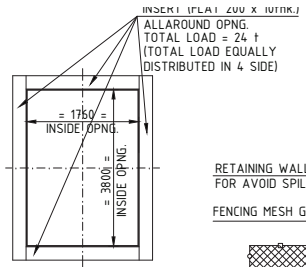
ABBREVIATION	
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thk.:	THICKNESS
RCC:	REINFORCED CEMENT CONCRETE
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B.O.M.R:	BOTTOM OF MONORAIL
OPNG.:	OPENING
TR:	TRESTLE
TYP.:	TYPICAL
PROJ:	PROJECTED
W:	WIDTH
H:	HEIGHT

STORAGE MATERIAL CHARACTERISTICS		
PARTICULARS		SPECIFICATION
STORAGE MATERIAL		CRUSHED GOLD ORE
LUMP SIZE	(mm)	(-)10
BULK DENSITY	(t/m³)	1.6 (FOR VOLUME CALCULATION)
		1.9 (FOR STRUCTURAL CALCULATION)
MOISTURE CONTENT	(%)	2
ANGLE OF REPOSE	(deg.)	37
STORAGE PERIOD	(day)	1
STORAGE CAPACITY	(m³)	~9600 (INCLUDING DEAD VOLUME)
CHEMICAL PROPERTIES OF GOLD ORE :		REFER PROCESS DESIGN CRITERIA DOC. NO. MD-216-3000-PR-CRT-0001
DEAD VOLUME TO BE CLEARED PERIODICALLY TO AVOID CAKE FORMATION		

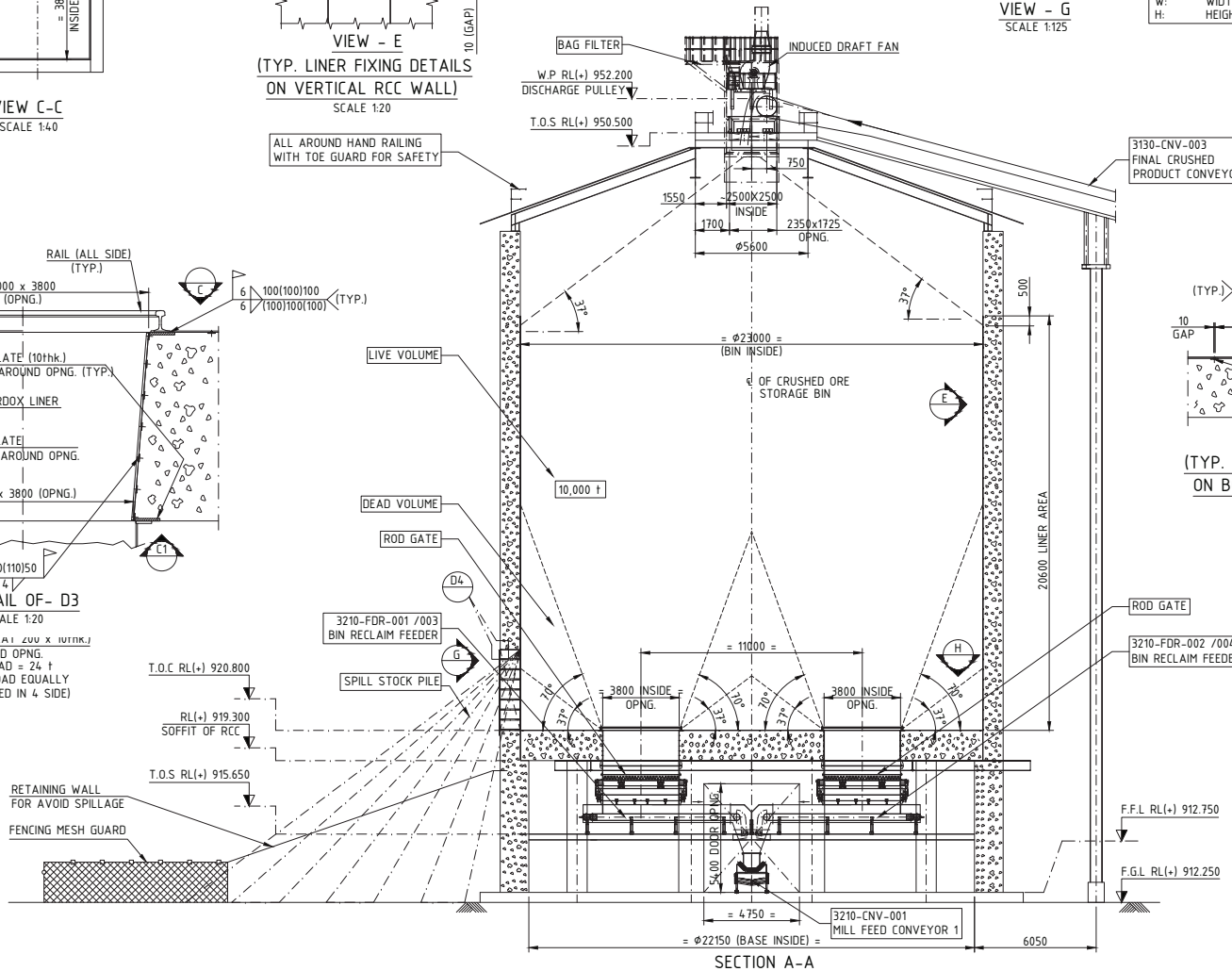
NOTE:
1. ALL DIMENSIONS ARE IN MILLIMETER, CO-ORDINATES AND ELEVATION ARE IN METER, UNLESS OTHERWISE STATED.
2. FINISHED FLOOR LEVEL (F.F.L) FOR CRUSHING AREA = RL(+)913.000 m
3. FINISHED GROUND LEVEL (F.G.L) FOR CRUSHING AREA = RL(+)912.500 m
4. F.F.L, F.G.L, T.O.C, = FINISH FLOOR LEVEL, FINISH GROUND LEVEL, TOP OF CONCRETE, T.O.S, B.O.S, B.O.B= TOP OF STEEL, BOTTOM OF STEEL, BOTTOM OF BASE PLATE
5. ALL MAINTENANCE FACILITIES, HANDLING PROVISIONS, ACCESS WAYS, PLATFORMS FOR MAINTENANCE ARE SHOWN TENTATIVE.



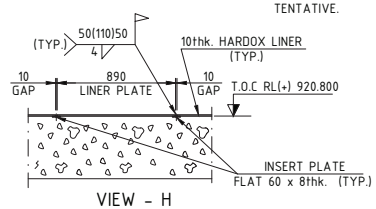
DETAIL OF- D3
SCALE 1:20



VIEW C1-C1
SCALE 1:40



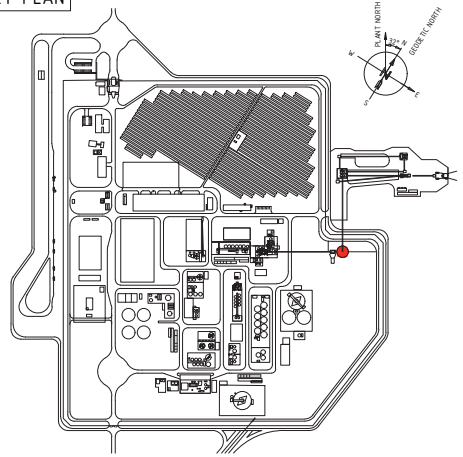
SECTION A-A
SCALE 1:125



VIEW - H
(TYP. LINER FIXING DETAILS
ON BOTTOM OF RCC BIN)
SCALE 1:20

CONCRETE FLOOR
CHECKERED PLATE
GRATING
BRICK WALL
CONCRETE WALL

KEY PLAN



DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-3210-EG-ME-LPX-1006	CRUSHED ORE STORAGE BIN & TRANSFER - PLAN								
MD-216-3210-EG-ME-LPX-1005	CRUSHED ORE STORAGE BIN & TRANSFER - PLAN								
MD-216-3210-EG-ME-LPX-1004	CRUSHED ORE STORAGE BIN & TRANSFER - PLAN								
MD-216-3210-EG-ME-ELV-1010	CRUSHED ORE STORAGE BIN & TRANSFER - SECTION								
MD-216-3210-EG-ME-LPX-1003	CRUSHED ORE STORAGE BIN & TRANSFER - PLAN	A01	21/02/2020	ISSUED FOR APPROVAL	HKD	BRD	BD	BGR	

RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

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Gold Mining Company شركة المعادن للتعدين الذهبي

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LARSEN & TOUBRO DRAWING NUMBER	O19045_100_H_G_16126_002	REVISION 0
OUTOTEC DRAWING NUMBER	900707 _GAU01 _US01_ MTL01 _32114	DOCUMENT ID OU100669133 / REVISION 0

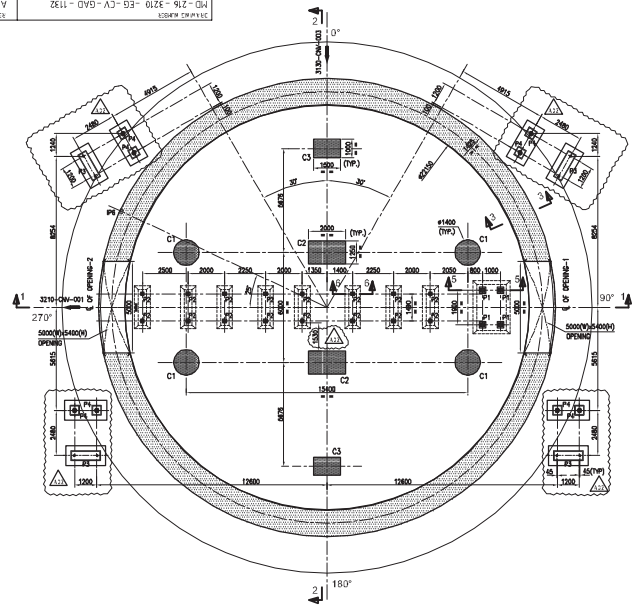
DRAWN	HKD	18/02/2020	MANSOURAH - MASSARAH GOLD PROJECT
CHECKED	BRD	20/02/2020	GENERAL ARRANGEMENT DRAWING
ENGINEER	BD	20/02/2020	GRINDING & CLASSIFICATION
APPROVED	BGR	21/02/2020	CRUSHED ORE STORAGE BIN & TRANSFER - ELEVATION
EMPLOYER			
SCALE	1:125, 1:40, 1:20	A1	DRAWING NUMBER MD-216-3210-EG-ME-ELV-1009 / REV A01

ATTACHMENT 4 CIVIL DRAWINGS

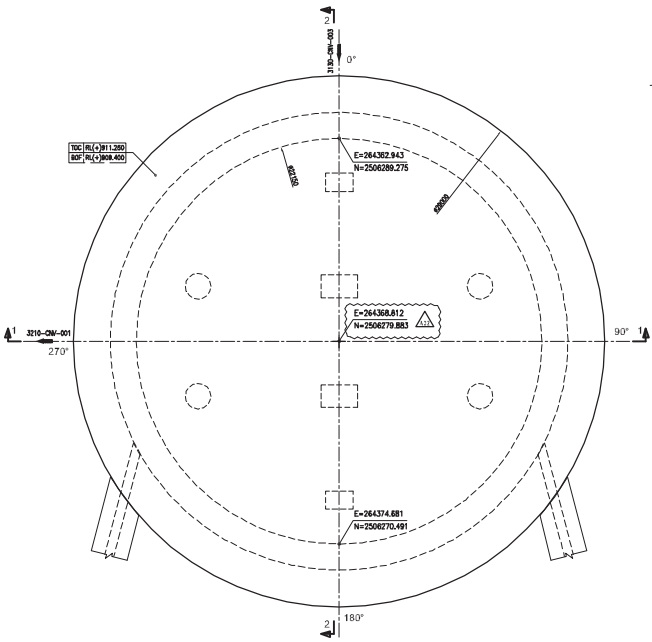
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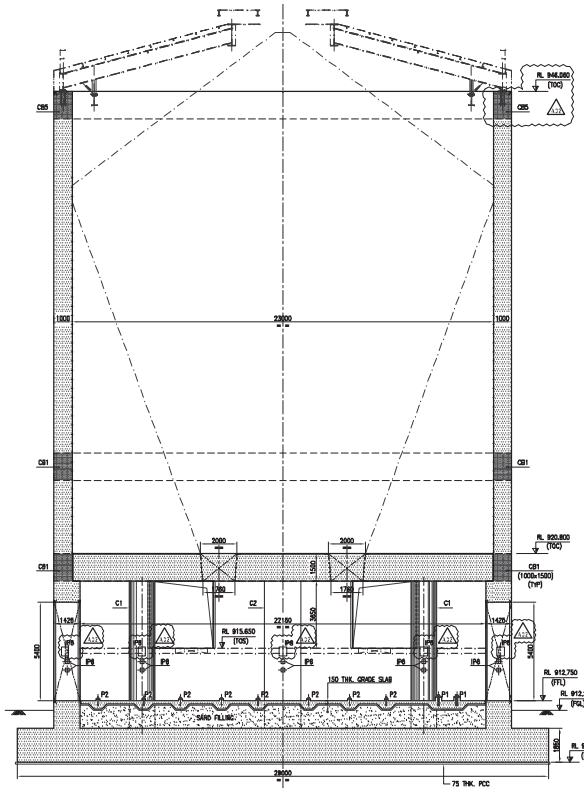
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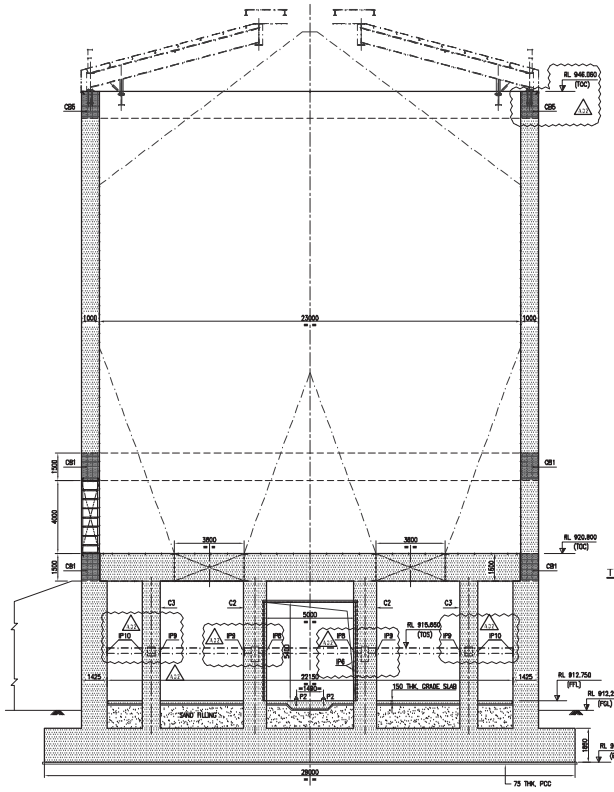
PLAN AT RL 912.750 (FFL)
SCALE-(1:100)



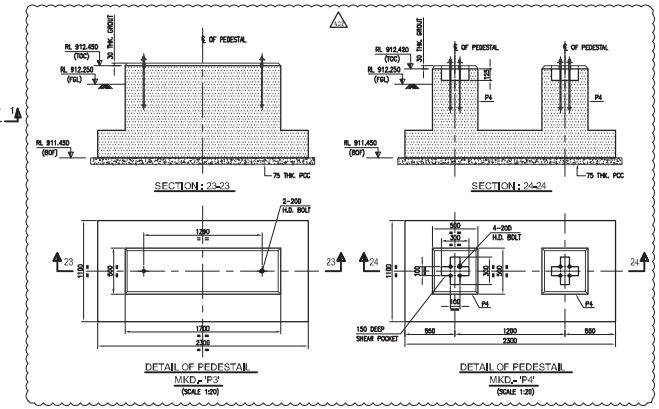
PLAN AT RL 909.400 (BOF)
SCALE-(1:100)

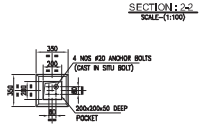


SECTION : 1-1
SCALE-(1:100)

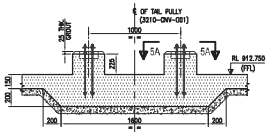


SECTION : 2-2
SCALE-(1:100)

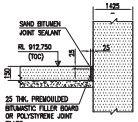




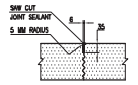
VIEW : 5A-5A



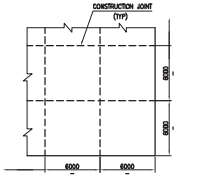
SECTION : 5-5
DETAIL OF PEDESTAL
MKD.- 'P1'
(SCALE 1:20)



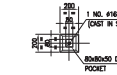
SECTION : 3-3
(TYP)



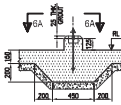
DETAIL OF CONSTRUCTION JOINT
(SCALE 1:10)



TYPICAL PLAN OF CONSTRUCTION JOINT
(CONCRETE GRADE SLAB AREA)



VIEW : 6A-6A



SECTION : 6-6
DETAIL OF PEDESTAL
MKD.- 'P2'
(SCALE 1:20)

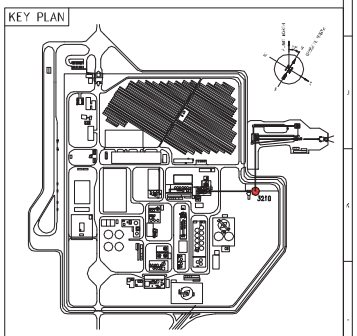
LEGEND :-

TOC = TOP OF CONCRETE
BOC = BOTTOM OF CONCRETE
TOS = TOP OF STRUCTURE
FGL = FINISHED GROUND LEVEL
BOF = BOTTOM OF FOOTING
BOM = BOTTOM OF MONORAIL
TYP = TYPICAL
MR = MONORAIL
PCC = PLAIN CEMENT CONCRETE
FL = FLAT
IP = INSERT PLATE
TOI = TOP OF INSERT
MR = MONORAIL
W = WIDTH
H = HEIGHT
℄ = CENTRE LINE
THK = THICKNESS
Φ = DIAMETER
R = RADIUS

NOTES :-

1. ALL DIMENSIONS ARE IN MILLIMETRES AND LEVELS AND CO-ORDINATES ARE IN METRES, UNLESS OTHERWISE STATED.
2. THIS DRAWING SHALL BE READ IN CONJUNCTION WITH DRG. NO. MD-216-3210-EG-CV-GAD-1133 & 1134.
3. FOR ALL OTHER NOTES REFER DRAWING NO. MD-216-1000-EG-CV-DED-1001, 1003 & 1004.

KEY PLAN



DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRW	CHK	DES	APP	DEP
MD-216-3210-EG-MC-CVA-1000	CRUSHED ORE BIN SECTION								
MD-216-3210-EG-MC-CVA-1001	CRUSHED ORE BIN ELEVATION								
MD-216-3210-EG-MC-CPA-1001	CRUSHED ORE BIN PLAN								
MD-216-3210-EG-MC-CPA-1002	CRUSHED ORE BIN PLAN	A02	23/01/2023	REVISED AS PER DMP/SITA COMMENTS	ASH	JR	SJK	AG	
MD-216-3210-EG-MC-CPA-1004	CRUSHED ORE BIN PLAN			DW 24/01/2023 & REVISED AS MARKED					
MD-216-3210-EG-MC-CPA-1003	CRUSHED ORE BIN PLAN	A01	20/01/2023	ISSUED FOR REFERENCE	ASH	JR	SJK	AG	

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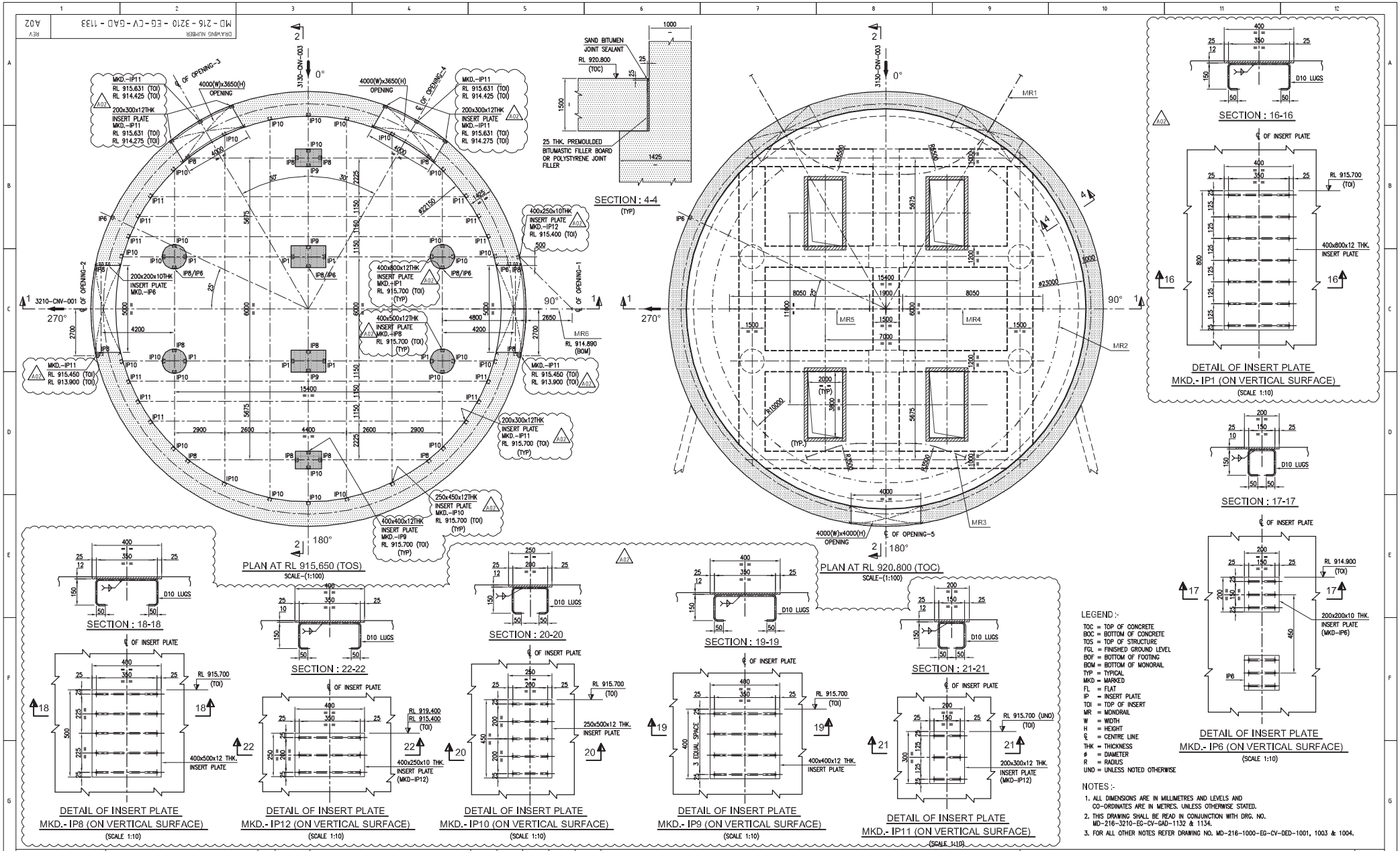
MYSTEEL Mysteel Global Consulting (Shanghai) Co., Ltd.

LARSEN & TOUBRO DRAWING NUMBER: 079045_450_C_G_24071_001 — REVISION: 1
SUBSIDIARY DRAWING NUMBER: 900707__GAU01__US01__MTL01__32121 — DOCUMENT ID: OU100673540 — REVISION: 1

DRAWN	ASH	20/01/2023
CHECKED	JR	24/01/2023
ENGINEER	SJK	23/01/2023
APPROVED	AG	20/01/2023
TRACEABILITY		
SCALE	1:100	A0

MANSOURAH - MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT DRAWING
GRINDING & CLASSIFICATION
CRUSHED ORE STORAGE BIN - LAYOUT & DETAIL

DRAWING NUMBER: MD-216-3210-EG-CV-GAD-1132 — REV: A02



LEGEND :-

TOC = TOP OF CONCRETE
BOC = BOTTOM OF CONCRETE
TOS = TOP OF STRUCTURE
FGL = FINISHED GROUND LEVEL
BOF = BOTTOM OF FOOTING
BOM = BOTTOM OF MONORAIL
TYP = TYPICAL
MKD = MARKED
FL = FLAT
IP = INSERT PLATE
TOI = TOP OF INSERT
MR = MONORAIL
W = WIDTH
H = HEIGHT
℄ = CENTRE LINE
THK = THICKNESS
ø = DIAMETER
R = RADIUS
UNO = UNLESS NOTED OTHERWISE

NOTES :-

1. ALL DIMENSIONS ARE IN MILLIMETRES AND LEVELS AND CO-ORDINATES ARE IN METRES, UNLESS OTHERWISE STATED.
2. THIS DRAWING SHALL BE READ IN CONJUNCTION WITH DRG. NO. MD-216-3210-EG-CV-GAD-1132 & 1134.
3. FOR ALL OTHER NOTES REFER DRAWING NO. MD-216-1000-EG-CV-DED-1001, 1003 & 1004.

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-3210-EG-ME-ELV-1010	CRUSHED ORE BIN-SECTION								
MD-216-3210-EG-ME-ELV-1009	CRUSHED ORE BIN-ELEVATION								
MD-216-3210-EG-ME-LPX-1006	CRUSHED ORE BIN-PLAN								
MD-216-3210-EG-ME-LPX-1005	CRUSHED ORE BIN-PLAN	A02	30/04/2020	REVISED AS PER EMPLOYER COMMENTS	ABH	UR	SVK	AG	
MD-216-3210-EG-ME-LPX-1004	CRUSHED ORE BIN-PLAN			ON 04/04/2020 & REVISED AS MARKED A02					
MD-216-3210-EG-ME-LPX-1003	CRUSHED ORE BIN-PLAN	A01	05/03/2020	ISSUED FOR REFERENCE	ABH	UR	SVK	AG	

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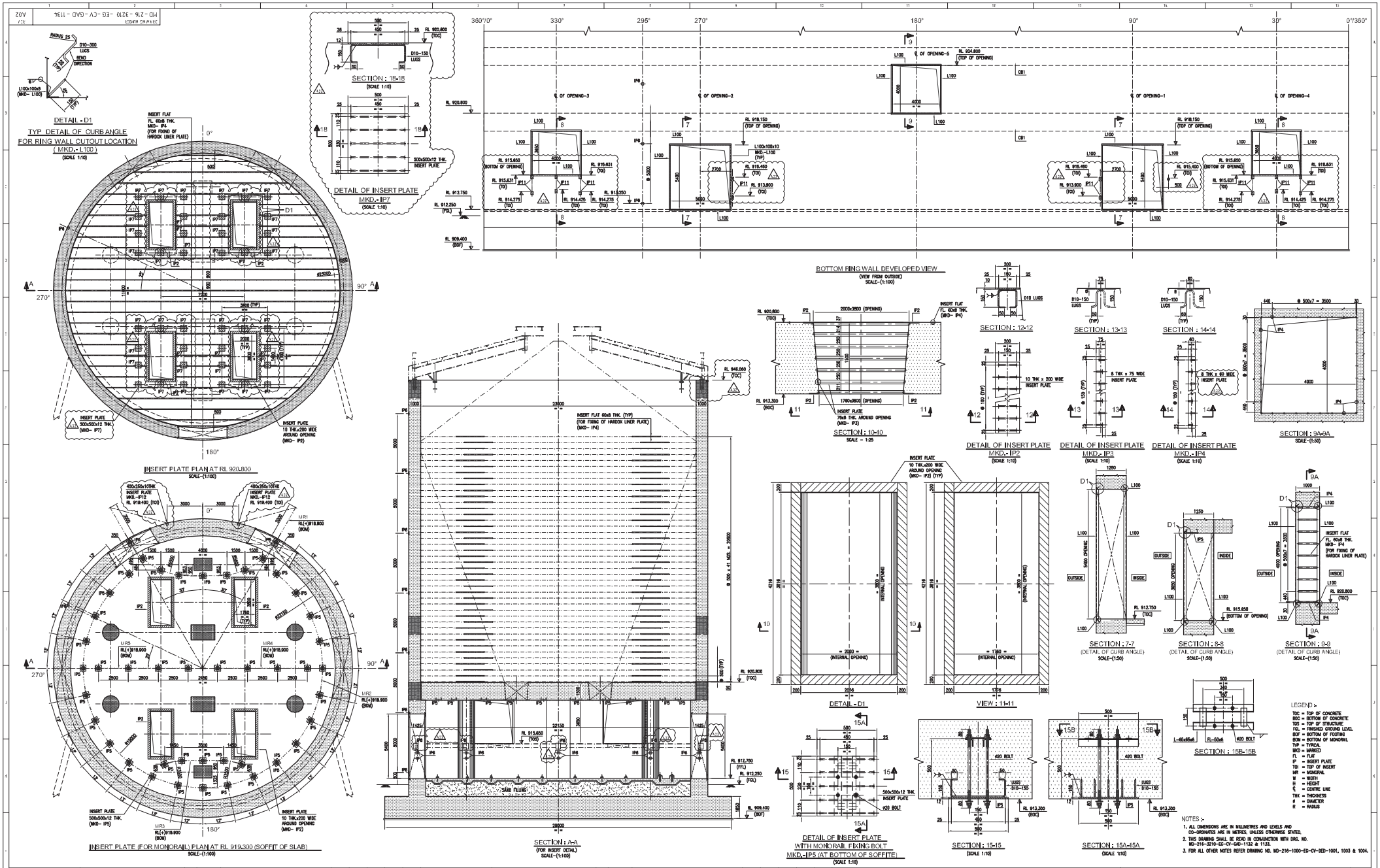
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LARSEN & TOUBRO DRAWING NUMBER: 019045_450_C_G_24071_002 — REVISION: 1
OUTOTEC DRAWING NUMBER: 900707 _GAU01 _US01 _MTL01 _32122 — DOCUMENT ID: OU100673541 — REVISION: 1

DRAWN	ABH	03/03/2020
CHECKED	UR	04/03/2020
ENGINEER	SVK	05/03/2020
APPROVED	AG	05/03/2020
EMPLOYER		
SCALE	1:100	A1

MANSOURAH - MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT DRAWING
GRINDING & CLASSIFICATION
CRUSHED ORE STORAGE BIN - LAYOUT & DETAIL

DRAWING NUMBER: MD - 216 - 3210 - EG - CV - GAD - 1133 — REV: A02



DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRW	CHK	DES	APP	DEP
MD-216-3210-EG-MS-1134/200	CRUSHED ORE BIN PLAN								
MD-216-3210-EG-MS-1134/201	CRUSHED ORE BIN ELEVATION								
MD-216-3210-EG-MS-1134/A00	CRUSHED ORE BIN P.A.								
MD-216-3210-EG-MS-1134/A01	CRUSHED ORE BIN P.A.	A02	22/24/2022	REVISED AS PER EMPLOYER COMMENTS	ASH	JR	SVK	AG	
MD-216-3210-EG-MS-1134/A02	CRUSHED ORE BIN P.A.			SM 24/24/2022 & REVISED AS MARKED					
MD-216-3210-EG-MS-1134/A03	CRUSHED ORE BIN P.A.	A01	22/23/2022	ISSUED FOR REFERENCE	ASH	JR	SVK	AG	

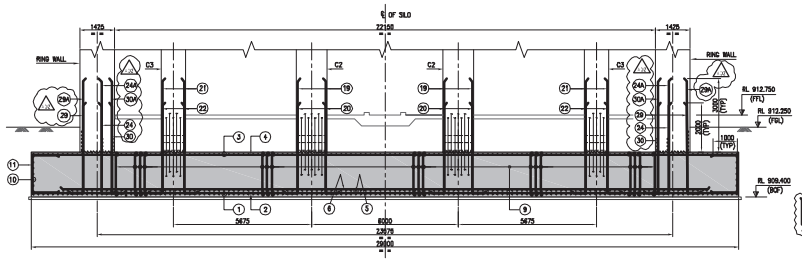
RESERVED FOR VENDOR JOB NO AND DRAWING NUMBER

LARSEN & TOUBRO JOB NUMBER: 079045_650_C_G_24071_003
SUBSTRUCTURE DRAWING NUMBER: 900707__GAU01__US01__MTL01__32123
DOCUMENT ID: OU100673542
REVISION: 1

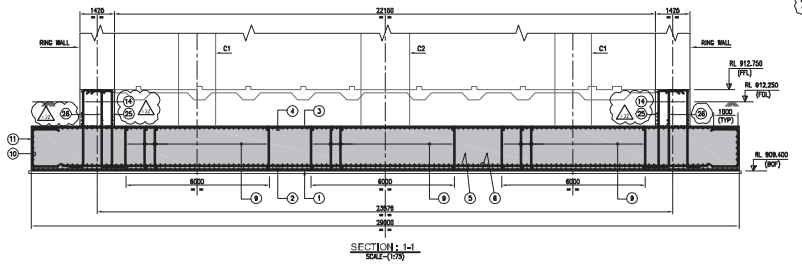
DRAWN	ASH	22/23/2022
CHECKED	JR	24/23/2022
DESIGNER	SVK	22/23/2022
APPROVED	AG	22/23/2022
FINAL VIEW		
SCALE	1:100	A0

MANSOURAH - MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT DRAWING
GRINDING & CLASSIFICATION
CRUSHED ORE STORAGE BIN - LAYOUT & DETAIL

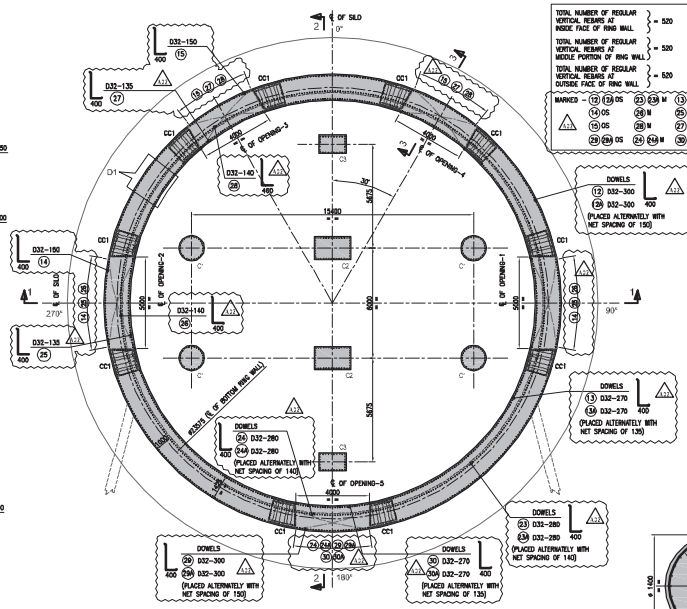
DRAWING NUMBER: MD - 216 - 3210 - EG - CV - GAD - 1134
REV: A02



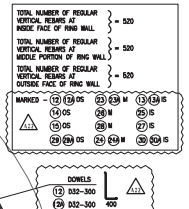
SECTION : 2-2
SCALE-(1:75)

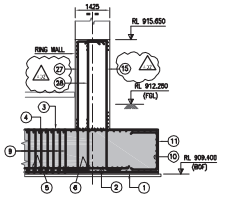


SECTION : 1-1
SCALE-(1:75)

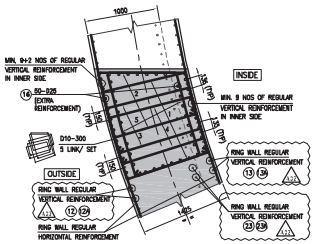


PLAN SHOWING DOWELS FOR RING WALL AND INTERNAL COLUMN REINFORCEMENT
SCALE-(1:100)

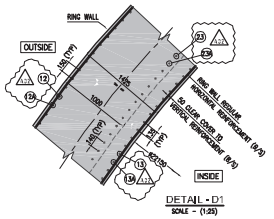




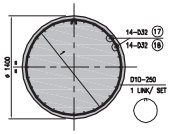
SECTION : 3-3
SCALE-(1:75)



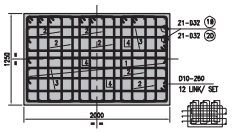
SECTIONAL DETAIL OF COLUMN (MKD. CC1)
("e"-EXTRA VERTICAL REINFORCEMENT AROUND OPENING)
("r"-REGULAR VERTICAL REINFORCEMENT)
SCALE 1:20



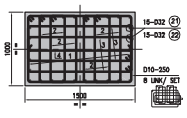
DETAIL - D1
SCALE - (1:25)



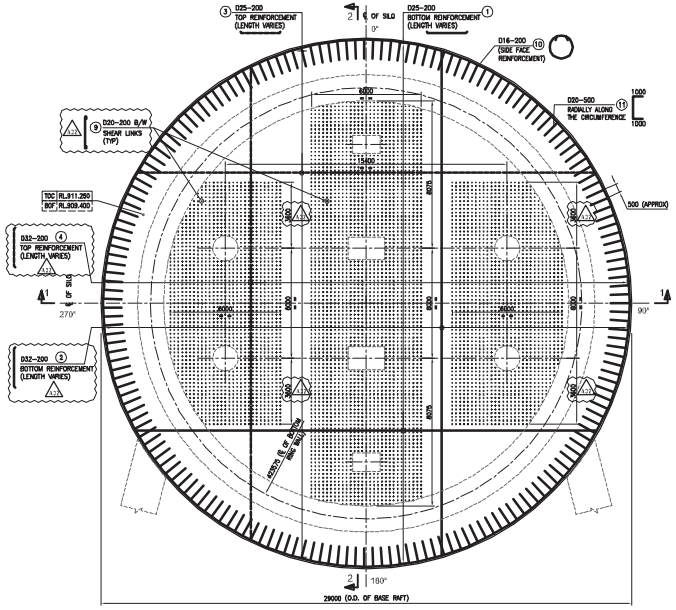
SECTIONAL DETAIL OF COLUMN (MKD. C1)
SCALE 1:25



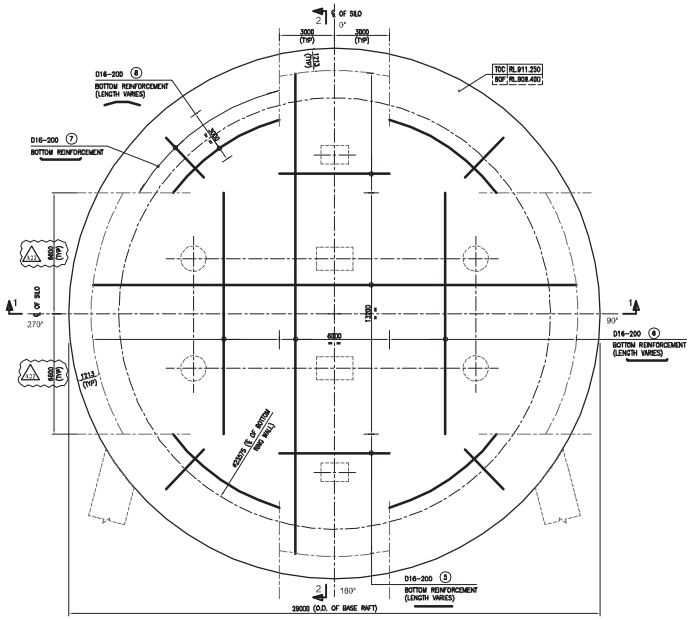
SECTIONAL DETAIL OF COLUMN (MKD. C2)
SCALE 1:25



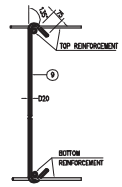
SECTIONAL DETAIL OF COLUMN (MKD. C3)
SCALE 1:25



PLAN AT RL 909.400 (BOF)
(SHOWING TOP AND BOTTOM REINFORCEMENT)
SCALE-(1:100)



PLAN AT RL 909.400 (BOF)
(SHOWING BOTTOM EXTRA REINFORCEMENT)
SCALE-(1:100)



TYP DETAIL OF VERTICAL SHEAR LINK (IN RAFT)
(ELEVATION)

LEGEND:-
TOC = TOP OF CONCRETE
FGL = FINISHED GROUND LEVEL
BOF = BOTTOM OF FOOTING
TYP = TYPICAL
MKD = MARKED
℄ = CENTRE LINE
ø = DIAMETER
R = RADIUS
B/S = BOTH SIDE
OS = OUT SIDE
M = MIDDLE
I/S = IN SIDE
= TOP REINFORCEMENT
= BOTTOM REINFORCEMENT

NOTES:-
1. ALL DIMENSIONS ARE IN MILLIMETRES AND LEVELS AND CO-ORDINATES ARE IN METRES, UNLESS OTHERWISE STATED.
2. THIS DRAWING SHALL BE READ IN CONJUNCTION WITH DRAWING NUMBER : MD-216-3210-EG-CV-DAD-1132, 1133 & 1134.
3. FOR ALL OTHER NOTES REFER DRAWING NO. MD-216-1006-EG-CV-DED-1001, 1003 & 1004.
4. GRADE OF CONCRETE SHALL BE C35 AND GRADE OF CONCRETE SHALL BE C30 FOR GRADE SLAB.
5. GRADE OF REINFORCEMENT SHALL BE DEFORMED BAR ASTM A615-GRADE 60 (fy=420 MPa)
6. GRADE OF WELDED WIRE FABRIC – ASTM A185(PLAIN) / A497(DEFORMED).
7. NET SAFE BEARING CAPACITY OF SOIL AT BOTTOM OF FOUNDATION CONSIDER 400KN/ SQ. M
8. IF ANY PART OF FOUNDATION COMES ON BACKFILL ZONE OF ADJACENT FOUNDATION OR ANY LOOSE SOIL ARE ENCOUNTERED BELOW FOUNDATION LEVEL THE SAME SHALL BE REMOVED AND FILLED WITH PCC.
9. REINFORCEMENT SHALL BE SUITABLY ADJUSTED AT SITE TO CLEAR POCKETS, BOLTS ETC.
10. SUFFICIENT AND SUITABLE NOS OF CHAIRS SHALL BE PROVIDED AT APPROPRIATE LOCATIONS TO MAINTAIN CORRECT POSITION OF REINFORCEMENT.
11. CLEAR COVER TO MAIN REINFORCEMENT SHALL BE AS FOLLOWS :
    a) RAFT : 75 mm (BOTTOM & SIDE)
    b) RAFT : 60 mm (TOP)
    c) WALL : 50 mm
    d) SUB : 40 mm
    e) PEDESTAL & COLUMN : 50 mm
12. MINIMUM LAP LENGTHS OF REINFORCEMENT SHALL BE 40xBAR DIAMETER.
13. THIS DRAWING SHALL BE READ IN CONJUNCTION WITH DRG. NO. MD-216-3210-EG-CV-DED-1137, 1138 & 1139.

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRW	CHK	DES	APP	DEP
		A02	22/06/2023	REVISED AS PER EMPLOYER COMMENTS DW 26/04/2023 & REVISED AS MARKED	ASH	JR	SJK	AG	
		A01	22/03/2023	ISSUED FOR REFERENCE	ASH	JR	SJK	AG	

RESERVED FOR VENDOR CODE AND DRAWING NUMBER

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L&T/O&M & PROJECT DRAWING NUMBER	019045_450_C_D_24072_001
SUBSITE DRAWING NUMBER	900707__GAU01__US01__CTB17__32101
DOCUMENT ID	OU100673544
REVISION	1
DRAWN	ASH 22/03/2023
CHECKED	JR 24/03/2023
ENGINEER	SJK 22/03/2023
APPROVED	AG 22/03/2023
FINAL VIEW	
SCALE	1:100
SIZE	A0

MANSOURAH – MASSARAH GOLD PROJECT
DETAIL DRAWING
GRINDING & CLASSIFICATION
CRUSHED ORE STORAGE BIN – R.C. DETAIL

DRAWING NUMBER: MD - 216 - 3210 - EG - CV - DED - 1136
REV: A02